Smarter Regular Expression Matching

From Theory to Tools

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1 Introduction

1.1 Motivation

Regular expressions are a fundamental formalism for describing regular languages and are among the most widely used abstractions in computer science. They underpin applications including lexical analysis, input validation, log processing, search tools, and network security. Across these applications, the problem is regular expression matching. Given a regular expression and an input string, determine whether the string belongs to the language denoted by the expression. Formally, this is the *membership problem*.

For example, the expression `(ab|a)*` matches `"aba"`, as it decomposes into `"ab"` followed by `"a"`, but does not match `"abb"`, since no sequence of subexpressions can produce it.

For many applications, a Boolean answer is enough. An input validator may only need to accept or reject a password or URL. Others require more. For instance, a lexer must determine how input decomposes into tokens. Capturing groups, schema validation, and information extraction require identifying which input parts correspond to parts of the expression. Recognition alone is not enough; the matcher must also recover the *structure*.

Efficiently solving these problems remains an active research area. The choice of algorithm influences performance, memory consumption, and security. Once matching becomes parsing, the way in which match structure is recovered also influences correctness and raises the question of how to handle ambiguity when multiple matches are possible.

Existing approaches include backtracking and finite-automata-based techniques, such as nondeterministic (NFA) and deterministic (DFA) automata. These approaches have different trade-offs in implementation complexity, memory consumption, and worst-case performance. Backtracking engines are flexible and widely used, but may exhibit exponential running time on constructed inputs, leading to regular expression denial-of-service (ReDoS) vulnerabilities [Cox, 2007]. Automata-based approaches provide stronger worst-case guarantees but may require preprocessing or suffer from state-space explosion.

Derivative-based algorithms provide an alternative by transforming the regular expression as input symbols are consumed. Each derivative represents the language remaining after an input symbol has been consumed. By maintaining a correspondence between an expression and its successive residuals, derivative-based methods provide an elegant foundation for regular expression matching and, with suitable extensions, structured parsing.

However, derivative-based techniques are less common in practical regular expression tooling. Their use for practical matching, parsing, and diagnostic tools motivates this thesis: "From Theory to Tools".

1.2 Problem Statement

Derivative-based approaches provide a principled foundation for regular expression matching and parsing. Brzozowski introduced regular expression derivatives as a way of transforming a regular expression after consuming an input symbol [Brzozowski, 1964]. Antimirov later introduced partial derivatives, which represent sets of residual expressions and are closely related to nondeterministic automata [Antimirov, 1996].

Building on Brzozowski's derivatives, Sulzmann and Lu developed a derivative-based algorithm for POSIX regular expression parsing [Sulzmann and Lu, 2014]. Their work also presents a bit-coded variant that represents parse information as a compact bit sequence, which can be decoded after a successful match rather than maintaining an explicit parse tree throughout the computation. A Haskell implementation of the POSIX parser is available in Sulzmann's *Programming Paradigms* teaching material [Sulzmann, 2026].

Sulzmann and Lu also studied partial derivatives in the context of regular expression sub-matching [Sulzmann and Lu, 2012]. While sub-matching is not the focus of this thesis, their work provides relevant background for the partial-derivative approach. The *Programming Paradigms* teaching material also presents a bit-coded partial-derivative-based parser in Haskell.

These constructions have primarily been presented in a theoretical and functional programming setting, leaving open practical questions about execution time, memory consumption, and data representation.

This thesis investigates these questions through implementations in Rust. As a systems programming language, Rust provides fine-grained control over memory allocation and data representation, allowing the derivative-based and partial-derivative-based parsers, including their bit-coded variants, to be evaluated in practice.

The implementation also provides traced versions of the parsing algorithms that record intermediate derivative computations during matching. These traces show the transformations performed by the algorithms and help explain successful and unsuccessful matches.

1.3 Contributions

The following contributions are being made:

- **Rust implementation of derivative-based parsers.** The thesis implements derivative-based and partial-derivative-based parsing constructions in Rust, including direct and bit-coded variants. The implementations are based on the constructions described in the literature and the corresponding Haskell material, resulting in four parser variants in total.
- **Empirical evaluation of the parser variants.** The four implementations are evaluated with respect to execution time and memory consumption using synthetic benchmarks and a corpus of real-world regular expressions from Snort/Suricata, SpamAssassin, and RegexLib, following the source categories investigated by Mamouras et al. [Mamouras et al., 2024]. They are also compared with Rust's `regex` crate and Google's RE2.

1.4 Thesis Outline

The remainder of this thesis follows the parsing constructions from their formal concepts through their Rust implementation to their evaluation. Theory and implementation are presented together.

- **Chapter 2 - Regular Expressions.** This chapter introduces the syntax and semantics of regular expressions and the Rust representation used throughout the implementation.
- **Chapter 3 - Derivatives.** This chapter introduces Brzozowski's derivatives and Antimirov's partial derivatives, together with their basic matching algorithms and Rust implementations.
- **Chapter 4 - Matching vs. Parsing.** This chapter distinguishes regular expression matching from parsing and introduces the disambiguation problem.
- **Chapter 5 - Derivative Parser.** This chapter presents Sulzmann and Lu's derivative-based POSIX parser and its Rust implementation in direct and bit-coded variants.
- **Chapter 6 - Partial-Derivative Parser.** This chapter presents the partial-derivative-based parser and its Rust implementation in direct and bit-coded variants. It also discusses the differences between the two derivative-based approaches.
- **Chapter 7 - External versus Internal Syntax.** This chapter describes the parsing of regular expression strings into an external AST and their translation into the internal representation used by the parsers.
- **Chapter 8 - Evaluation.** This chapter evaluates the four parser variants with respect to execution time, memory consumption, and correctness. The evaluation uses synthetic benchmarks and a corpus of real-world regular expressions, and compares the implementations with Rust's `regex` crate and Google's RE2. It also examines optimization challenges and the insights gained from the evaluation, including the derivative traces produced by the implementations.
- **Chapter 9 - Conclusion.** Finally, this chapter summarizes the findings of the thesis and discusses the practical implications of the results and directions for future work.

2 Regular Expressions

2.1 Syntax

Let Σ be an alphabet. Regular expressions over Σ are defined inductively by the grammar:

$$r, s ::= x \mid \varepsilon \mid \emptyset \mid r + s \mid r \cdot s \mid r^* \qquad (x \in \Sigma). \tag{2.1}$$

In the grammar, r and s range over regular expressions, while x ranges over the alphabet Σ and denotes an arbitrary symbol, such as a, b, or c. The constants ε and $\emptyset$ represent the *empty-word* and *empty-language* expressions. The operators $+$, $\cdot$, and * denote alternation, concatenation, and Kleene star, respectively.

Concatenation is written implicitly, so rs denotes $r \cdot s$. Given this notation, we adopt the precedence and associativity conventions. Unless parentheses are shown explicitly, Kleene star binds most tightly, followed by concatenation, followed by alternation. For example,

$$ab^*c = (a \cdot (b^*)) \cdot c,$$
$$a + bc^* = a + (b \cdot (c^*)).$$

For repeated operators we use left association, so

$$a + b + c = (a + b) + c,$$
$$abc = (ab)c.$$

The grammar above is the core syntax. Most regular expression libraries also offer one-or-more, optional, and bounded repetition (r^+, $r?$, $r^{\{m,n\}}$), but these add no expressive power since each can be defined by the core constructs: $r^+ = r \cdot r^*$, $r? = r + \varepsilon$, and

$$r^{\{m,n\}} = \underbrace{r \cdot \ldots \cdot r}_{m \text{ times}} \cdot \underbrace{(r + \varepsilon) \cdot \ldots \cdot (r + \varepsilon)}_{n-m \text{ times}}.$$

In the implementation, expressions using such operators are translated into the core constructs before they reach the parsing algorithms.

As a running example, consider the regular expression over $\Sigma = \{0, 1\}$

$$r_{01} = (0 + 1)^* \cdot 0 \cdot 1. \tag{2.2}$$

By the conventions above, its syntax tree is $((0 + 1)^* \cdot 0) \cdot 1$. This tree represents the hierarchical structure of the expression: a Kleene star applied to the alternation $0 + 1$, followed by concatenation with the literals 0 and 1.

The syntactic structure must be given a semantic interpretation.

2.2 Semantics

A semantic interpretation assigns a language to each regular expression. A *word* over Σ is a finite sequence of symbols from Σ. The set of all words over Σ, including the empty word ε, is denoted by Σ^*. A *language* over Σ is a subset of Σ^*.

For a regular expression r, we write $L(r)$ for the language it denotes, where $L(r) \subseteq \Sigma^*$. The definition follows the structure of the syntax tree by *structural recursion*. Each case corresponds to one production of the grammar in (2.1):

$$\begin{aligned} L(x) &= \{x\}, \\ L(\varepsilon) &= \{\varepsilon\}, \\ L(\emptyset) &= \emptyset, \\ L(r + s) &= L(r) \cup L(s), \\ L(r \cdot s) &= \{vw \mid v \in L(r),\ w \in L(s)\}, \\ L(r^*) &= \{\varepsilon\} \cup \{w_1 \cdots w_n \mid n \geq 1,\ w_1, \ldots, w_n \in L(r)\}. \end{aligned} \tag{2.3}$$

Table 2.1 applies these definitions to small expressions. The literal a denotes the language containing only the word a. The expression ε denotes the language containing only the empty word, while $\emptyset$ denotes the language containing no words. The alternation $a + b$ combines the languages of a and b to give $\{a, b\}$, and concatenation ab combines words from the languages of a and b to give $\{ab\}$. The Kleene star a^* allows zero or more concatenations of words from the language of a, giving $\{\varepsilon, a, aa, aaa, \ldots\}$.

Expression r	Language $L(r)$
a	$\{a\}$
ε	$\{\varepsilon\}$
$\emptyset$	$\emptyset$
$a + b$	$\{a, b\}$
ab	$\{ab\}$
a^*	$\{\varepsilon, a, aa, aaa, \ldots\}$

Table 2.1: Languages denoted by basic expressions.

For example $r_{01} = (0 + 1)^* \cdot 0 \cdot 1$ from (2.2), we compute its language step by step:

$$\begin{aligned} L(0) &= \{0\}, \\ L(1) &= \{1\}, \\ L(0 + 1) &= L(0) \cup L(1) = \{0, 1\}, \\ L((0 + 1)^*) &= \{0, 1\}^*, \\ L(r_{01}) &= L((0 + 1)^*) \cdot L(0) \cdot L(1) = \{\, w \cdot 0 \cdot 1 \mid w \in \{0, 1\}^* \,\}. \end{aligned} \tag{2.4}$$

This is the set of binary words ending in 01. For example, the words 01, 001, and 1101 belong to $L(r_{01})$, whereas ε, 0, and 010 do not.

This is the question of whether a word *matches* a regular expression: w *matches* r if $w \in L(r)$. Determining this is the *membership problem*.

2.3 Rust Representation

The grammar from (2.1) is represented directly as a Rust enum with one variant for each syntactic construct, as shown in Figure 2.1:

```
#[derive(Debug, Clone, PartialEq, Eq, Hash)]
pub enum Regex {
    Phi, Eps, Lit(char),
    Seq(Box<Regex>, Box<Regex>),  // r . s
    Alt(Box<Regex>, Box<Regex>),  // r + s
    Star(Box<Regex>),              // r*
}

impl Regex {
    pub fn seq(r: Regex, s: Regex) -> Regex {
        Regex::Seq(Box::new(r), Box::new(s))
    }
    pub fn alt(r: Regex, s: Regex) -> Regex {
        Regex::Alt(Box::new(r), Box::new(s))
    }
    pub fn star(r: Regex) -> Regex { Regex::Star(Box::new(r)) }
    pub fn lit(c: char) -> Regex { Regex::Lit(c) }
}
```

Figure 2.1: The `Regex` enum and constructors. Derived traits support operations needed by parsing: `Clone` for duplication, `PartialEq`/`Eq` for comparison, `Hash` for collections.

The type carries no precedence or associativity information, as these are determined by the syntax tree. Rust requires explicit indirection because enum sizes must be known at compile time. Recursive variants therefore use `Box<Regex>`. A direct definition such as `Seq(Regex, Regex)` would have infinite size, whereas `Box` stores recursive values separately, providing finite size [The Rust Project Developers, 2026a,b]. This differs from Haskell's heap-allocated data types.

The running example $r_{01} = (0 + 1)^* \cdot 0 \cdot 1$ from (2.2) is represented by the syntax tree:

$$\mathrm{Seq}\Big(\mathrm{Seq}(\mathrm{Star}(\mathrm{Alt}(0, 1)), 0), 1\Big). \tag{2.5}$$

In Rust, this is represented as shown in Figure 2.2, with the outermost `Seq` containing `Lit('1')` and `Seq(Star(Alt(0,1)), 0)` as its right and left children.

```
Regex::seq(
    Regex::seq(
        Regex::star(Regex::alt(Regex::lit('0'), Regex::lit('1'))),
        Regex::lit('0')),
    Regex::lit('1'))
```

Figure 2.2: Rust representation of r_{01}.

3 Derivatives and Partial Derivatives

Chapter 2 introduced the syntax and semantics of regular expressions, and defined the *membership problem*: Decide whether a word w belongs to $L(r)$. This chapter introduces three algorithms for solving it, each returning a Boolean answer: match or no match.

The first is a naive recursive matcher that follows the structure of the regular-expression semantics. The other two transform the regular expression as input symbols are consumed. Brzozowski derivatives transform it into a single residual expression, while Antimirov's partial derivatives represent the possible residual expressions as a set.

3.1 Naive Matching

A direct way to decide whether $w \in L(r)$ is to interpret the regular-expression semantics as recursive matching rules, by following the structure of the expression and searching for possible decompositions of the input. Let $w[i:j]$ denote the substring from position i to j. A recursive procedure $\mathrm{match}(w, i, j, r)$ determines whether $w[i:j] \in L(r)$:

$$\begin{aligned}
\mathrm{match}(w,i,j,\varepsilon) &\iff i = j,\\
\mathrm{match}(w,i,j,x) &\iff j = i+1 \wedge w[i] = x,\\
\mathrm{match}(w,i,j,r+s) &\iff \mathrm{match}(w,i,j,r) \vee \mathrm{match}(w,i,j,s),\\
\mathrm{match}(w,i,j,r\cdot s) &\iff \exists k \in [i,j].\ \mathrm{match}(w,i,k,r) \wedge \mathrm{match}(w,k,j,s),\\
\mathrm{match}(w,i,j,r^*) &\iff i = j \vee \exists k \in (i,j].\ \mathrm{match}(w,i,k,r) \wedge \mathrm{match}(w,k,j,r^*).
\end{aligned} \tag{3.1}$$

Each rule corresponds directly to the semantics. Concatenation does not specify where to split, so the matcher must consider all possible split points k.

For matching $(a \cdot a) \cdot a$ against `aaa`, the split points $k \in \{0,1,2,3\}$ are:

$$\begin{aligned}
&\underbrace{\mathrm{match}(w,0,0,a\cdot a)}_{a\cdot a \text{ against } w[0:0]=\epsilon} \wedge \underbrace{\mathrm{match}(w,0,3,a)}_{a \text{ against } w[0:3]=\texttt{aaa}}\\
\vee\ &\underbrace{\mathrm{match}(w,0,1,a\cdot a)}_{a\cdot a \text{ against } w[0:1]=\texttt{a}} \wedge \underbrace{\mathrm{match}(w,1,3,a)}_{a \text{ against } w[1:3]=\texttt{aa}}\\
\vee\ &\underbrace{\mathrm{match}(w,0,2,a\cdot a)}_{a\cdot a \text{ against } w[0:2]=\texttt{aa}} \wedge \underbrace{\mathrm{match}(w,2,3,a)}_{a \text{ against } w[2:3]=\texttt{a}}\\
\vee\ &\underbrace{\mathrm{match}(w,0,3,a\cdot a)}_{a\cdot a \text{ against } w[0:3]=\texttt{aaa}} \wedge \underbrace{\mathrm{match}(w,3,3,a)}_{a \text{ against } w[3:3]=\epsilon}.
\end{aligned} \tag{3.2}$$

At $k = 2$, both operands match, triggering a recursive search on $(a \cdot a)$ against `aa` with split points $k \in \{0, 1, 2\}$, where $k = 1$ succeeds:

$$\begin{aligned} & \underbrace{\text{match}(w, 0, 0, a)}_{a \text{ against } w[0:0]=\epsilon} \wedge \underbrace{\text{match}(w, 0, 2, a)}_{a \text{ against } w[0:2]=\texttt{aa}} \\ \vee\ & \underbrace{\text{match}(w, 0, 1, a)}_{a \text{ against } w[0:1]=\texttt{a}} \wedge \underbrace{\text{match}(w, 1, 2, a)}_{a \text{ against } w[1:2]=\texttt{a}} \\ \vee\ & \underbrace{\text{match}(w, 0, 2, a)}_{a \text{ against } w[0:2]=\texttt{aa}} \wedge \underbrace{\text{match}(w, 2, 2, a)}_{a \text{ against } w[2:2]=\epsilon}. \end{aligned} \tag{3.3}$$

This completes the original match through the nested split points $k = 2$ and $k = 1$.

In Rust. The recursive rules translate directly into the corresponding Rust match arms, shown in Figure 3.1. Alternation tries both branches on the same input, while concatenation and star search possible split points. The star arm clones the body to construct the recursive `Star` expression. This is necessary because the recursive call needs to own the newly constructed expression. The range-based search short-circuits at the first successful split. Nested concatenations repeatedly search smaller substrings, exploring multiple paths before succeeding. This leads to exponential backtracking in the worst case.

```
Regex::Alt(r, s) => {
    match_naive_range(word, i, j, r)
        || match_naive_range(word, i, j, s)
}
Regex::Seq(r, s) => {
    (i..=j).any(|k| {
        match_naive_range(word, i, k, r)
            && match_naive_range(word, k, j, s)
    })
}
Regex::Star(r) => {
    i == j || (i + 1..=j).any(|k| {
        match_naive_range(word, i, k, r)
            && match_naive_range(word, k, j,
                &Regex::star(*r.clone()))
    })
}
```

Figure 3.1: Alternation, concatenation, and star arms of the naive matcher.

Although inefficient, this matcher is useful as a reference implementation because it follows the regular-expression semantics directly without any intermediate transformation. The cost is visible in the worked examples: nested concatenations repeatedly search smaller substrings, each exploring all possible split points before finding a match.

The derivative approach takes a different approach. Rather than searching for a decomposition of w while keeping r fixed, it transforms r as each character of w is consumed. The resulting expression describes the language that can still match the remaining suffix.

3.2 Brzozowski Derivatives

Brzozowski's derivative of r with respect to a symbol x, written $r\backslash x$, is the expression describing exactly what remains of r's language after consuming the leading symbol x of a word $w \in L(r)$ [Brzozowski, 1964]:

$$L(r\backslash x) \;=\; \{\, w \mid xw \in L(r) \,\}. \tag{3.4}$$

For $w = x_1 x_2 \cdots x_n$, consume one symbol at a time by repeatedly taking derivatives:

$$r, \quad r\backslash x_1, \quad (r\backslash x_1)\backslash x_2, \quad \ldots$$

Nullability. After all symbols have been consumed, the remaining input is ε. Thus, the resulting expression must accept ε for the original word to belong to $L(r)$. This is the question of whether the expression is *nullable*. Let $n(r)$ be true exactly when $\varepsilon \in L(r)$:

$$\begin{array}{llll} n(\emptyset) = \text{false} & (\mathsf{N}_\emptyset) & n(r+s) = n(r) \vee n(s) & (\mathsf{N}_+) \\ n(\varepsilon) = \text{true} & (\mathsf{N}_\varepsilon) & n(r \cdot s) = n(r) \wedge n(s) & (\mathsf{N}_{\text{seq}}) \\ n(x) = \text{false} & (\mathsf{N}_{\text{lit}}) & n(r^*) = \text{true} & (\mathsf{N}_*) \end{array} \tag{3.5}$$

The empty language $\emptyset$ is not nullable because it contains no words, including ε, while a literal cannot match the empty word. ε and r^* always match ε. An alternation is nullable when either branch is nullable, and a concatenation only when both components are.

In Rust. The definition is transcribed one `Regex` variant at a time, using the enum of Figure 2.1. The case split of (3.5) becomes the arms of a `match`, shown in Figure 3.2.

```
pub fn nullable(r: &Regex) -> bool {
    match r {
        Regex::Phi => false,
        Regex::Eps => true,
        Regex::Lit(_) => false,
        Regex::Alt(r, s) => nullable(r) || nullable(s),
        Regex::Seq(r, s) => nullable(r) && nullable(s),
        Regex::Star(_) => true,
    }
}
```

Figure 3.2: `nullable` implementation and its six cases: $\mathsf{N}_\emptyset$, N_ε, N_{lit}, N_+, N_{seq} and N_*.

The expression is passed by reference because `nullable` only inspects r and does not modify or consume it. The underscores in N_{lit} and N_* indicate that the corresponding operands are irrelevant. A literal is never nullable, regardless of its symbol, while a Kleene-star expression is always nullable, regardless of its body. The short-circuiting operators `||` and `&&` avoid evaluating the second operand when the result is already determined, while preserving the corresponding logical semantics.

Derivative Rules. The rules specify the expression that remains after consuming x:

$$\begin{array}{llll} \emptyset\backslash x = \emptyset & (\mathsf{D}_\emptyset) & (r+s)\backslash x = (r\backslash x) + (s\backslash x) & (\mathsf{D}_+) \\ \varepsilon\backslash x = \emptyset & (\mathsf{D}_\varepsilon) & r^*\backslash x = (r\backslash x)\cdot r^* & (\mathsf{D}_*) \\ y\backslash x = \begin{cases} \varepsilon & x = y \\ \emptyset & x \neq y \end{cases} & (\mathsf{D}_{\text{lit}}) & (r\cdot s)\backslash x = \begin{cases} (r\backslash x)\cdot s + s\backslash x & n(r) = \text{true} \\ (r\backslash x)\cdot s & n(r) = \text{false} \end{cases} & \begin{array}{l}(\mathsf{D}^n_{\text{seq}}) \\ (\mathsf{D}_{\text{seq}})\end{array} \end{array} \tag{3.6}$$

The empty language and ε cannot consume x. A literal y succeeds exactly when $x = y$, leaving ε. Alternation retains both derivatives and for r^*, one occurrence of r consumes x while further occurrences remain available.

Concatenation is the only rule that depends on nullability. If r is not nullable, r must consume the leading x, leaving $r\backslash x$ as the remainder of r while s remains untouched. If r is nullable, it may either consume x giving $(r\backslash x)\cdot s$, or r matches ε and s consumes x leaving $s\backslash x$. Hence the nullable case produces the alternation $(r\backslash x)\cdot s + s\backslash x$.

In Rust. Figure 3.3 implements (3.6) similarly, one variant at a time.

```
pub fn deriv(r: &Regex, x: char) -> Regex {
    match r {
        Regex::Phi => Regex::Phi,
        Regex::Eps => Regex::Phi,
        Regex::Lit(c) => if *c == x { Regex::Eps } else { Regex::Phi },
        Regex::Alt(r1, r2) =>
            Regex::alt(deriv(r1, x), deriv(r2, x)),
        Regex::Seq(r1, r2) => {
            let dr1 = Regex::seq(deriv(r1, x), *r2.clone());
            if nullable(r1) {
                Regex::alt(dr1, deriv(r2, x))
            } else { dr1 }
        }
        Regex::Star(r1) =>
            Regex::seq(deriv(r1, x), Regex::star(*r1.clone())),
    }
}
```

Figure 3.3: `deriv` implementation with six cases.

`deriv` differs from `nullable` in that it borrows its input but returns an owned `Regex`, since a derivative is a new expression. The `if nullable(r1)` implements the case split of (3.6), with branches in order $\mathsf{D}^n_{\text{seq}}$ then D_{seq}. This is the only place where a derivative rule consults nullability and where the output can be larger than the input, since the nullable branch produces an alternation while the other produces a concatenation.

The cloning in `*r2.clone()` is necessary because s appears untouched in both branches of the concatenation case, and the star arm must preserve r^*'s body both derived and underived. This copying mechanism drives expression growth, which justifies simplification after every derivative step.

Example: Growth without simplification. Consider $(x+y)^*$ against the word $\mathtt{xx}$: The expression is a star, so D_* applies with body $r = x + y$:

$$\begin{aligned} (x+y)^*\backslash\mathtt{x} &= ((x+y)\backslash\mathtt{x}) \cdot (x+y)^* && \mathsf{D}_*,\ r = x+y \\ &= (\varepsilon + \emptyset) \cdot (x+y)^* && \mathsf{D}_+,\ \mathsf{D}_{\text{lit}} \end{aligned} \tag{3.7}$$

The expression is now the concatenation $(\varepsilon + \emptyset) \cdot (x+y)^*$, and deriving it consumes the second $\mathtt{x}$. The result is an alternation because $\mathsf{D}^n_{\text{seq}}$ applies with $r = \varepsilon + \emptyset$ and $s = (x+y)^*$, its nullable case being selected since $n(r) = n(\varepsilon + \emptyset) = \text{true}$ by N_+ and N_ε:

$$\begin{aligned} &((\varepsilon + \emptyset) \cdot (x+y)^*)\backslash\mathtt{x} \\ &= ((\varepsilon + \emptyset)\backslash\mathtt{x}) \cdot (x+y)^* + (x+y)^*\backslash\mathtt{x} && \mathsf{D}^n_{\text{seq}} \\ &= (\emptyset + \emptyset) \cdot (x+y)^* + (x+y)^*\backslash\mathtt{x} && \mathsf{D}_\varepsilon \\ &= (\emptyset + \emptyset) \cdot (x+y)^* + (\varepsilon + \emptyset) \cdot (x+y)^* && \mathsf{D}_*,\ \mathsf{D}_+,\ \mathsf{D}_{\text{lit}} \end{aligned} \tag{3.8}$$

After both symbols are consumed, the result is checked for nullability:

$$\begin{aligned} &n((\emptyset + \emptyset) \cdot (x+y)^* + (\varepsilon + \emptyset) \cdot (x+y)^*) \\ &= n((\emptyset + \emptyset) \cdot (x+y)^*) \vee n((\varepsilon + \emptyset) \cdot (x+y)^*) && \mathsf{N}_+ \\ &= (n(\emptyset + \emptyset) \wedge n((x+y)^*)) \vee (n(\varepsilon + \emptyset) \wedge n((x+y)^*)) && \mathsf{N}_{\text{seq}} \\ &= (\text{false} \wedge \text{true}) \vee (\text{true} \wedge \text{true}) && \mathsf{N}_+,\ \mathsf{N}_\emptyset,\ \mathsf{N}_\varepsilon,\ \mathsf{N}_* \\ &= \text{true}. \end{aligned} \tag{3.9}$$

Thus $(x+y)^*$ matches $\mathtt{xx}$.

Simplification. Derivatives can grow unboundedly. Repeatedly deriving $(x+y)^*$ by the same symbol produces expressions of increasing size. This growth results from the nullable branch of $\mathsf{D}^n_{\text{seq}}$, which introduces an alternation as a redundant structure that is carried into subsequent derivatives. ε and $\emptyset$ accumulate without bound:

$$\begin{aligned} & (x+y)^* && 4 \\ \xrightarrow{\mathtt{x}}\ & (\varepsilon + \emptyset) \cdot (x+y)^* && 8 \\ \xrightarrow{\mathtt{x}}\ & (\emptyset + \emptyset) \cdot (x+y)^* \ +\ (\varepsilon + \emptyset) \cdot (x+y)^* && 17 \\ \xrightarrow{\mathtt{x}}\ & (\emptyset + \emptyset) \cdot (x+y)^* \ +\ \left((\emptyset + \emptyset) \cdot (x+y)^* \ +\ (\varepsilon + \emptyset) \cdot (x+y)^*\right) && 26 \end{aligned} \tag{3.10}$$

To counter the growth, expressions are simplified after each derivative step, using identities that preserve the denoted language:

$$\begin{aligned} \emptyset \cdot r &= \emptyset && (\mathsf{S}_1) \\ \varepsilon \cdot r &= r && (\mathsf{S}_2) \\ \emptyset + r &= r && (\mathsf{S}_3) \end{aligned} \tag{3.11}$$

S_1 eliminates dead concatenations with $\emptyset$, S_2 eliminates identity concatenations with ε, and S_3 eliminates $\emptyset$ from alternations. Each holds equally with the unit on the right.

Applying them to (3.8), after each step, returns the expression back to $(x+y)^*$, keeping the size bounded regardless of input length:

$$\begin{aligned}
&(\emptyset + \emptyset) \cdot (x+y)^* + (\varepsilon + \emptyset) \cdot (x+y)^* \\
&= \emptyset \cdot (x+y)^* + \varepsilon \cdot (x+y)^* && \mathsf{S}_3,\ \text{twice} \\
&= \emptyset + (x+y)^* && \mathsf{S}_1,\ \mathsf{S}_2 \\
&= (x+y)^* && \mathsf{S}_3
\end{aligned} \tag{3.12}$$

In Rust. The identities are implemented recursively, applying the rules bottom-up over the expression tree. Each arm of the inner match implements an identity from (3.11).

`simplify` takes ownership of its input and returns simplified subexpressions directly, unlike `deriv`, which borrows and constructs. This model is visible where the two compose:

```
current = simplify(deriv(&current, c));
```

`deriv` borrows `current`, constructs a new expression, and `simplify` takes ownership of that result to return it without additional copying. The recursive calls ensure that all dead code is eliminated before an expression is returned, keeping size bounded at each step. Together, derive-then-simplify forms the core loop of the matcher, transforming and cleaning the expression at every input symbol.

```
pub fn simplify(r: Regex) -> Regex {
    match r {
        Regex::Seq(r, s) => {
            let r = simplify(*r);
            let s = simplify(*s);
            match (&r, &s) {
                (Regex::Phi, _) | (_, Regex::Phi) => Regex::Phi,
                (Regex::Eps, _) => s,
                (_, Regex::Eps) => r,
                _ => Regex::seq(r, s),
            }
        }
        Regex::Alt(r, s) => {
            let r = simplify(*r);
            let s = simplify(*s);
            match (&r, &s) {
                (Regex::Phi, _) => s,
                (_, Regex::Phi) => r,
                _ if r == s => r,
                _ => Regex::alt(r, s),
            }
        }
        Regex::Star(r) => match simplify(*r) {
            Regex::Eps | Regex::Phi => Regex::Eps,
            r => Regex::star(r),
        },
        other => other,
    }
}
```

Figure 3.4: Simplification rules: S_1 and S_2 in concatenation, S_3 in both branches.

Running example: $r_{01} = (0+1)^* \cdot 0 \cdot 1$. Recall from (2.2) that it is left-associated as $\Big((0+1)^* \cdot 0\Big) \cdot 1$, so the outer operands are $r = (0+1)^* \cdot 0$ and $s = 1$. We derive against $w = \mathtt{101}$ one symbol at a time.

1. **Derive by $\mathtt{1}$.** At the outer node, $r = (0+1)^* \cdot 0$ and $s = 1$. This r is not nullable:

$$n\Big((0+1)^* \cdot 0\Big) = \text{true} \wedge \text{false} = \text{false},$$

by N_{seq}, N_* and N_{lit}. Thus D_{seq} applies without a second summand:

$$r_{01}\backslash\mathtt{1} = \Big((0+1)^* \cdot 0)\backslash\mathtt{1}\Big) \cdot 1.$$

At the middle node, the left operand $(0+1)^*$ is nullable, so $\mathsf{D}^n_{\text{seq}}$ applies. First,

$$\begin{aligned}
(0+1)^*\backslash\mathtt{1} &= \Big((0+1)\backslash\mathtt{1}\Big) \cdot (0+1)^* && \mathsf{D}_* \\
&= \Big((0\backslash\mathtt{1}) + (1\backslash\mathtt{1})\Big) \cdot (0+1)^* && \mathsf{D}_+ \\
&= (\emptyset + \varepsilon) \cdot (0+1)^* && \mathsf{D}_{\text{lit}} \\
&= (0+1)^* && \mathsf{S}_3,\ \mathsf{S}_2.
\end{aligned}$$

Hence

$$\begin{aligned}
((0+1)^* \cdot 0)\backslash\mathtt{1} &= (0+1)^* \cdot 0 + (0\backslash\mathtt{1}) && \mathsf{D}^n_{\text{seq}} \\
&= (0+1)^* \cdot 0 + \emptyset && \mathsf{D}_{\text{lit}} \\
&= (0+1)^* \cdot 0 && \mathsf{S}_3.
\end{aligned}$$

Substituting back into the outer node gives

$$r_{01}\backslash\mathtt{1} = \Big((0+1)^* \cdot 0\Big) \cdot 1 = r_{01}. \tag{3.13}$$

Thus a $\mathtt{1}$ not yet part of the final $\mathtt{01}$ is absorbed by the leading star.

2. **Derive by $\mathtt{0}$.** The expression is unchanged. Again, $(0+1)^*\backslash\mathtt{0}$ simplifies to $(0+1)^*$, but now the second summand at the middle node survives:

$$\begin{aligned}
((0+1)^* \cdot 0)\backslash\mathtt{0} &= (0+1)^* \cdot 0 + (0\backslash\mathtt{0}) && \mathsf{D}^n_{\text{seq}} \\
&= (0+1)^* \cdot 0 + \varepsilon && \mathsf{D}_{\text{lit}}.
\end{aligned}$$

No simplification applies. The outer node is still non-nullable, so

$$r_{01}\backslash\mathtt{1}\backslash\mathtt{0} = \Big((0+1)^* \cdot 0 + \varepsilon\Big) \cdot 1. \tag{3.14}$$

After $\mathtt{10}$, either the trailing $\mathtt{01}$ is still owed in full, represented by $(0+1)^* \cdot 0$, or the consumed $\mathtt{0}$ was already its first symbol, represented by ε.

3. **Derive by $\mathtt{1}$.** The outer binding has now changed. The left operand is $(0+1)^* \cdot 0 + \varepsilon$, which is nullable by N_+ and N_ε. So $\mathsf{D}^n_{\text{seq}}$ applies and contributes the second summand $1\backslash\mathtt{1} = \varepsilon$. For the first summand, D_+ applies, reusing the result from step 1:

$$\begin{aligned}
((0+1)^* \cdot 0 + \varepsilon)\backslash\mathtt{1} &= ((0+1)^* \cdot 0)\backslash\mathtt{1} + (\varepsilon\backslash\mathtt{1}) && \mathsf{D}_+ \\
&= (0+1)^* \cdot 0 + \emptyset && \text{step 1; } \mathsf{D}_\varepsilon \\
&= (0+1)^* \cdot 0 && \mathsf{S}_3.
\end{aligned}$$

Therefore

$$r_{01}\backslash\mathtt{1}\backslash\mathtt{0}\backslash\mathtt{1} = \Big((0+1)^* \cdot 0\Big) \cdot 1 + \varepsilon = r_{01} + \varepsilon. \tag{3.15}$$

The final expression is nullable by N_+ and N_ε, so $\texttt{101} \in L(r_{01})$. The ε summand witnesses that the match is complete. For comparison, $w = \texttt{10}$ leaves $\Big((0+1)^* \cdot 0 + \varepsilon\Big) \cdot 1$. Neither branch is nullable: $(0 + 1)^* \cdot 0$ still requires a 0, while $\varepsilon \cdot 1$ still requires a 1. Hence $\texttt{10} \notin L(r_{01})$.

The matcher. The implementation can be summarized by the sequence of residual expressions produced while consuming the input. For $w = x_1 x_2 \cdots x_n$, the initial residual is $r_0 = r$, and each subsequent residual is obtained by deriving with respect to the next symbol and simplifying the result:

$$r_0 = r, \qquad r_i = \text{simplify}(r_{i-1} \backslash x_i) \quad (1 \le i \le n). \tag{3.16}$$

The residual r_i denotes exactly the set of suffixes that complete the consumed prefix $x_1 \cdots x_i$ to a word in $L(r)$. Since simplification preserves the represented language, each r_i is equivalent to the unsimplified derivative after consuming that prefix. After the final symbol has been consumed, the matcher accepts when the remaining residual is nullable:

$$w \in L(r) \iff n(r_n) = \text{true}.$$

So $r_0 \xrightarrow{x_1} r_1 \xrightarrow{x_2} \cdots \xrightarrow{x_n} r_n$ can be viewed as a path through a lazily constructed DFA. Each residual expression represents a state, and $\xrightarrow{x}$ denotes the transition obtained by taking the derivative with respect to x. Unlike an explicitly constructed DFA, only the states reached by the input are constructed. A residual is an accepting state exactly when it is nullable, so acceptance is determined by whether the final residual is nullable.

In Rust. Figure 3.5 implements (3.16) by composing `deriv`, `simplify`, and `nullable`. The implementation follows the mathematical definition directly. The current residual is transformed at each step, and acceptance is determined after the complete input has been consumed.

```
pub fn match_deriv(input: &str, r: &Regex) -> bool {
    let mut current = r.clone();
    for c in input.chars() {
        current = simplify(deriv(&current, c));
    }
    nullable(&current)
}
```

Figure 3.5: `match_deriv`: the core matcher loop. For each input character, derive the current residual expression and simplify it in place. Accept if the final residual is nullable.

The loop implements the residual sequence. At each iteration, `current` contains the current residual, which is then derived and simplified before processing the next input symbol. Only the current residual is retained, since the previous residual is dropped when `current` is reassigned. In contrast to parsing, where intermediate results may be required to reconstruct a parse tree, Boolean matching requires only the final residual.

The derivative-based matcher avoids the exponential backtracking of naive matching by transforming the expression rather than searching for decompositions. Simplification keeps the resulting expressions compact, preventing the growth shown in (3.10).

3.3 Antimirov Partial Derivatives

The derivative of Section 3.2 combines alternative residuals into a single expression using $+$. Antimirov's partial derivatives instead keep them as a *set* of expressions, one per strand of nondeterminism [Antimirov, 1996]. Writing $pd(r, x)$ for this set:

$$
\begin{aligned}
pd(\emptyset, x) &= \{\} && (\mathsf{P}_\emptyset)\\
pd(\varepsilon, x) &= \{\} && (\mathsf{P}_\varepsilon)\\
pd(y, x) &= \begin{cases} \{\varepsilon\} & x = y \\ \{\} & x \neq y \end{cases} && (\mathsf{P}_{\text{lit}})\\
pd(r + s, x) &= pd(r, x) \cup pd(s, x) && (\mathsf{P}_+)\\
pd(r^*, x) &= \{\, smartC(r', r^*) \mid r' \in pd(r, x) \,\} && (\mathsf{P}_*)\\
pd(r \cdot s, x) &= \begin{cases} \{\, smartC(r', s) \mid r' \in pd(r, x) \,\} \cup pd(s, x) & n(r) = \text{true} \\ \{\, smartC(r', s) \mid r' \in pd(r, x) \,\} & n(r) = \text{false} \end{cases} && \begin{matrix}(\mathsf{P}^n_{\text{seq}})\\ (\mathsf{P}_{\text{seq}})\end{matrix}
\end{aligned}
\tag{3.17}
$$

The empty language and ε produce the empty set, since neither has a residual with respect to x: where $\mathsf{D}_\emptyset$ and D_ε returned $\emptyset$ as a node in the result, $\mathsf{P}_\emptyset$ and P_ε return no residual at all. A literal y gives $\{\varepsilon\}$ when $x = y$ and $\{\}$ otherwise. Alternation unions the two residual sets, and for r^* one iteration of r consumes x, with each resulting residual placed in front of r^* so the star may continue. None of these alternatives are combined with $+$; each stays a separate element of the set.

Concatenation depends on nullability for the same reason as D_{seq}. If r is not nullable it must consume x, and each of its residuals is concatenated with the untouched s. If r is nullable, r may instead match ε and let s consume x, in which case $pd(s, x)$ joins the set. Where $\mathsf{D}^n_{\text{seq}}$ combined the two cases with $+$ inside one expression, $\mathsf{P}^n_{\text{seq}}$ keeps them as two elements of one set.

In Rust. Figure 3.6 presents the implementation of (3.17), with each `match` arm corresponding to one of its cases. The `Phi`, `Eps`, and `Lit` variants implement $\mathsf{P}_\emptyset$, P_ε, and P_{lit} directly, returning `Vec::new()` where the equation gives $\{\}$. `Alt` concatenates the two child vectors, implementing the union of P_+. `Star` maps `smart_seq` over r's own residuals, matching P_*; `Regex::star(*r.clone())` clones because `Regex::star` needs an owned `Regex` and `r` is only borrowed, once per residual since the closure runs once for each.

`Seq` implements the case split directly. The `map` applies `smart_seq` to each residual of the left operand, matching P_{seq}, and the guarded `out.extend(pderiv(s, x))` adds $pd(s, x)$ for $\mathsf{P}^n_{\text{seq}}$. Although the mathematical definition uses sets, the implementation uses `Vec<Regex>` throughout the recursion and defers deduplication to a single `HashSet` insertion at the top, rather than deduplicating at every `Alt` or `Seq` node on the way down.

Every union in `pderiv` is therefore a vector `extend`, not a set operation. Duplicate residuals reached by different strands are only merged once inserted into the `HashSet` used by the matcher below to keep the residual generation separate from deduplication.

```
fn pderiv(r: &Regex, x: char) -> Vec<Regex> {
    match r {
        Regex::Phi => Vec::new(),
        Regex::Eps => Vec::new(),
        Regex::Lit(c) => {
            if *c == x { vec![Regex::Eps] } else { Vec::new() }
        }
        Regex::Alt(r, s) => {
            let mut out = pderiv(r, x);
            out.extend(pderiv(s, x));
            out
        }
        Regex::Seq(r, s) => {
            let mut out: Vec<Regex> = pderiv(r, x)
                .into_iter()
                .map(|r_prime| smart_seq(r_prime, s))
                .collect();
            if nullable(r) {
                out.extend(pderiv(s, x));
            }
            out
        }
        Regex::Star(r) => {
            pderiv(r, x)
                .into_iter()
                .map(|r_prime| smart_seq(r_prime, &Regex::star(*r.clone())))
                .collect()
        }
    }
}
```

Figure 3.6: $pd(r, x)$ in full, corresponding arm for arm to (3.17).

The smart constructor. The concatenation in P_{seq} is constructed using the auxiliary constructor *smartC*, denoted by SC:

$$smartC(r', s) = \begin{cases} s & r' = \varepsilon \\ r' \cdot s & \text{otherwise} \end{cases} \tag{3.18}$$

If $r' = \varepsilon$, the consumed symbol completed the left operand and there is nothing left to concatenate with s, so the residual is s alone. Otherwise r' stays in front of s. *smartC* therefore states the concatenation rule without ever constructing $\varepsilon \cdot s$.

A set element is one possible residual, not one branch of a sum. When several alternatives can consume the same symbol, they stay as separate elements, and *smartC* is applied to each independently. One strand may finish the left operand while another still carries a non-empty residual in front of s. The constructor is therefore applied independently to each residual produced by $pd(r, x)$, preserving the distinct continuations of the concatenation without introducing redundant $\varepsilon \cdot s$ expressions.

In Rust. Figure 3.7 is *smartC*. `smart_seq` takes `r` by value and `s` by reference, matching (3.18). The residual r' comes from the recursive derivative and is consumed, while s must survive into the result. The $r' = \varepsilon$ branch clones only `s` and drops ε, so $\varepsilon \cdot s$ is never built.

```
pub fn smart_seq(r: Regex, s: &Regex) -> Regex {
    match r {
        Regex::Eps => s.clone(),
        r => Regex::seq(r, s.clone()),
    }
}
```

Figure 3.7: `smart_seq`, the smart constructor *smartC*

Example: No growth under *pd*. Consider $(x+y)^*$ against `xx` under *pd*.

P_* applies with star body $r = x + y$:

$$\begin{aligned} pd((x+y)^*, \mathtt{x}) &= \{\, smartC(r', (x+y)^*) \mid r' \in pd(x+y, \mathtt{x}) \,\} && \mathsf{P}_* \\ &= \{\, smartC(r', (x+y)^*) \mid r' \in \{\varepsilon\} \,\} && \mathsf{P}_+,\ \mathsf{P}_{\text{lit}} \text{ twice} \\ &= \{(x+y)^*\} && \mathsf{SC},\ r' = \varepsilon. \end{aligned} \tag{3.19}$$

$pd(x+y, \mathtt{x}) = \{\varepsilon\} \cup \{\} = \{\varepsilon\}$ by P_+, since `x` matches the left literal and misses the right, and *smartC* absorbs the ε straight back into $(x+y)^*$.

The set is $\{(x+y)^*\}$, so deriving by the second `x` repeats the computation:

$$pd((x+y)^*, \mathtt{x}, \mathtt{x}) = \{(x+y)^*\}.$$

Since $(x+y)^*$ is nullable by N_*, the final partial derivative accepts ε, so $\mathtt{xx} \in L((x+y)^*)$.

For comparison, applying the Brzozowski derivative to the same expression gives the intermediate alternation in (3.8), $(\emptyset+\emptyset)\cdot(x+y)^* + (\varepsilon+\emptyset)\cdot(x+y)^*$. Only after simplification in (3.12) does this reduce to $(x+y)^*$. Under *pd*, the residual set is $\{(x+y)^*\}$ after both steps, since $\mathsf{P}_\emptyset$ and P_ε produce no residual at all. So redundant terms are avoided at construction time rather than built and simplified afterwards

Running example: $r_{01} = (0+1)^* \cdot 0 \cdot 1$. Derive r_{01} against $w = \mathtt{101}$, tracking the residual set. The concatenations bind as in Section 3.2, with $(r, s) = ((0+1)^* \cdot 0, 1)$ at the outer node and $(r, s) = ((0+1)^*, 0)$ at the inner node; the star body is $0 + 1$. The derivation proceeds from the inside outward.

1. **Derive by `1`.** At the star body, P_+ distributes the derivative over two alternatives:

$$\begin{aligned} pd(0{+}1,\ \mathtt{1}) &= pd(0, \mathtt{1}) \cup pd(1, \mathtt{1}) && \mathsf{P}_+,\ r = 0,\ s = 1 \\ &= \{\} \cup \{\varepsilon\} && \mathsf{P}_{\text{lit}},\ \text{both branches} \\ &= \{\varepsilon\}. \end{aligned} \tag{3.20}$$

   The star rule then applies *smartC* to the resulting residual:

$$\begin{aligned} pd\Big((0+1)^*,\ \mathtt{1}\Big) &= \{\, smartC(r', (0+1)^*) \mid r' \in \{\varepsilon\} \,\} && \mathsf{P}_*,\ r = 0+1 \\ &= \{(0+1)^*\} && \mathsf{SC},\ r' = \varepsilon. \end{aligned} \tag{3.21}$$

At the middle concatenation, the left operand is nullable by N_*, so the nullable case contributes both the derivatives through the left and right operand:

$$\begin{aligned} pd\Big((0+1)^* \cdot 0,\ \mathtt{1}\Big) &= \{\, smartC(r', 0) \mid r' \in pd\Big((0+1)^*, \mathtt{1}\Big)\,\} \cup pd(0, \mathtt{1}) && \mathsf{P}^n_{\text{seq}} \\ &= \{smartC((0+1)^*, 0)\} \cup \{\} && \text{above; } \mathsf{P}_{\text{lit}} \\ &= \{(0+1)^* \cdot 0\} && \mathsf{SC},\ r' \neq \varepsilon. \end{aligned} \tag{3.22}$$

At the outer node, the left operand is not nullable, so only one branch contributes:

$$\begin{aligned} pd(r_{01},\ \mathtt{1}) &= \{\, smartC(r', 1) \mid r' \in \{(0+1)^* \cdot 0\}\,\} && \mathsf{P}_{\text{seq}},\ r = (0+1)^* \cdot 0,\ s = 1 \\ &= \{(0+1)^* \cdot 0 \cdot 1\} = \{r_{01}\} && \mathsf{SC}. \end{aligned} \tag{3.23}$$

The first symbol leaves $\{r_{01}\}$, returning to the starting expression since the consumed `1` belongs to the leading star.

2. **Derive by `0`.** The current set is still $\{r_{01}\}$. As before, $pd\Big((0+1)^*, \mathtt{0}\Big) = \{(0+1)^*\}$. At the middle concatenation, however, the right operand is now the matching literal 0, so P_{lit} contributes $\{\varepsilon\}$:

$$\begin{aligned} pd\Big((0+1)^* \cdot 0,\ \mathtt{0}\Big) &= \{smartC((0+1)^*, 0)\} \cup pd(0, \mathtt{0}) && \mathsf{P}^n_{\text{seq}},\ r = (0+1)^*,\ s = 0 \\ &= \{(0+1)^* \cdot 0\} \cup \{\varepsilon\} && \mathsf{P}_{\text{lit}},\ \text{since } \mathtt{0} = 0 \\ &= \{(0+1)^* \cdot 0,\ \varepsilon\}. \end{aligned} \tag{3.24}$$

The outer concatenation applies $smartC$ separately to both residuals:

$$\begin{aligned} pd(r_{01}, \mathtt{1}, \mathtt{0}) &= \{\, smartC(r', 1) \mid r' \in \{(0+1)^* \cdot 0, \varepsilon\}\,\} && \mathsf{P}_{\text{seq}},\ s = 1 \\ &= \{(0+1)^* \cdot 0 \cdot 1,\ 1\} && \mathsf{SC} \\ &= \{r_{01},\ 1\}. \end{aligned} \tag{3.25}$$

The two residuals are the possible strands after `10`: r_{01} keeps both symbols in the leading star, while 1 means the `0` began the final `01`. Since $smartC(\varepsilon, 1) = 1$, the latter is represented directly as 1.

3. **Derive by `1`.** The two residuals are derived independently and their results are unioned:

$$\begin{aligned} pd(r_{01}, \mathtt{1}, \mathtt{0}, \mathtt{1}) &= pd(r_{01}, \mathtt{1}) \cup pd(1, \mathtt{1}) && \text{union over the set} \\ &= \{r_{01}\} \cup \{\varepsilon\} && \text{step 1; } \mathsf{P}_{\text{lit}} \\ &= \{r_{01},\ \varepsilon\}. \end{aligned} \tag{3.26}$$

The final set contains ε, so one surviving residual is nullable and $\mathtt{101} \in L(r_{01})$. This agrees with the conclusion reached by the Brzozowski derivative in (3.15).

For $w = \mathtt{10}$, the first two steps leave $\{r_{01}, 1\}$, which represents the same alternatives as $\Big((0+1)^* \cdot 0 + \varepsilon\Big) \cdot 1$ in (3.14). Either the trailing `01` remains to be matched, or the consumed `0` was its first symbol. Neither residual is nullable, so $\mathtt{10} \notin L(r_{01})$.

Finiteness. For a fixed regular expression r, the set of reachable partial derivatives is finite and bounded by $\|r\| + 1$, where $\|r\|$ denotes the number of literal occurrences in r [Antimirov, 1996, Theorem 3.4, p. 309]. For r_{01}, $\|r_{01}\| = 4$ (the four literals: $0, 1, 0, 1$), so the bound is $4 + 1 = 5$ distinct residuals across all possible inputs. The example with $w = \texttt{101}$ reaches only three of them: $\{r_{01}\}$, $\{r_{01}, 1\}$, and $\{r_{01}, \varepsilon\}$. Other residuals in the theoretical set are reachable only by different input sequences.

For ordinary derivatives, finiteness requires identifying equivalent expressions [Brzozowski, 1964, Theorem 4.3], yielding an exponential bound $d_R \leq 2^{d_P} - 1$. Simplification rules eliminate only units and redundant $\emptyset$ expressions, not all equivalent forms. The linear bound for partial derivatives applies directly without simplification.

The matcher. A *frontier* of residuals is updated one symbol at a time by taking pd of every element and unioning the results. The frontier F_i represents all residuals reachable after consuming the first i input symbols—every possible state the computation could be in. For frontier F_i after i symbols,

$$F_0 = \{r\}, \qquad F_i = \bigcup_{r' \in F_{i-1}} pd(r', x_i) \quad (1 \leq i \leq n). \tag{3.27}$$

The matcher accepts when some residual in the final frontier is nullable. Where Brzozowski's $r_0 \to r_1 \to \cdots \to r_n$ traces one path through a lazily constructed DFA, $F_0 \to F_1 \to \cdots \to F_n$ traces all states an NFA explores simultaneously, holding each state at once rather than committing to a single residual expression.

In Rust. Figure 3.8 implements (3.27) using a `HashSet<Regex>` frontier, where `Regex`'s `Hash` and `Eq` implementations allow residuals from different paths to collapse. Where the Brzozowski matcher holds one `Regex`, this holds a set. The early return when the frontier empties follows from construction: no later derivative produces residuals from an empty set. The final `any(nullable)` checks whether some residual is nullable. The matcher trades a single growing expression for a set of small ones that never need simplification.

```
pub fn match_pderiv(input: &str, r: &Regex) -> bool {
    let mut states: HashSet<Regex> = HashSet::new();
    states.insert(r.clone());
    for c in input.chars() {
        let mut next: HashSet<Regex> = HashSet::new();
        for state in &states {
            next.extend(pderiv(state, c));
        }
        if next.is_empty() { return false; }
        states = next;
    }
    states.iter().any(nullable)
}
```

Figure 3.8: The partial-derivative matcher.

4 Matching and Parsing

Chapter 3 established matching with a Boolean answer: $w \in L(r)$. A matcher only answers whether the input is accepted by the regular expression. Parsing identifies the evidence for the match and *parse trees* act as their structured witnesses.

4.1 Matching: A Boolean Answer

Consider $r = (a + ab)(b + \varepsilon)$ and $w = \texttt{ab}$. The matcher can determine $w \in L(r)$ using the algorithms of Chapter 3. There are, however, two ways in which r can match w:

$$\underbrace{a}_{a+ab}\ \underbrace{b}_{b+\varepsilon} \quad \text{and} \quad \underbrace{\texttt{ab}}_{a+ab}\ \underbrace{\varepsilon}_{b+\varepsilon} \tag{4.1}$$

Both choices establish the same membership result, but assign the input differently to the subexpressions of r. The Boolean result alone cannot distinguish these cases, so the matcher cannot reveal which decomposition was chosen.

Parsing makes this additional structure explicit, by showing which alternatives were chosen and which parts of the input were consumed by the corresponding subexpressions.

4.2 Parsing: Parse Trees as Matching Evidence

Parse Tree. A *parse tree* provides a structured witness for such a derivation. Sulzmann and Lu view regular expressions as types and *parse trees* as their values [Sulzmann and Lu, 2014]. The judgment $\vdash v : r$ states that v is a valid parse tree for r, recording which alternatives were chosen and how each part of the input was consumed during matching:

$$\frac{\vdash v_1 : r_1}{\vdash \mathit{Left}\ v_1 : r_1 + r_2}\ (\mathsf{T}^L_+) \qquad \vdash () : \varepsilon\ (\mathsf{T}_\varepsilon)$$

$$\frac{\vdash v_2 : r_2}{\vdash \mathit{Right}\ v_2 : r_1 + r_2}\ (\mathsf{T}^R_+) \qquad \frac{l \in \Sigma}{\vdash l : l}\ (\mathsf{T}_{\text{lit}}) \tag{4.2}$$

$$\frac{}{\vdash [\,] : r^*}\ (\mathsf{T}^0_*) \qquad \frac{\vdash v_1 : r_1 \qquad \vdash v_2 : r_2}{\vdash (v_1, v_2) : r_1 r_2}\ (\mathsf{T}_{\text{seq}})$$

$$\frac{\vdash v : r \qquad \vdash vs : r^*}{\vdash (v : vs) : r^*}\ (\mathsf{T}_*)$$

In Rust. Figure 4.1 shows the parse-tree type, with one variant for each constructor.

```
#[derive(Debug, Clone, PartialEq, Eq)]
pub enum ParseTree {
    Empty,                                  // ()
    Char(char),                             // l
    Pair(Box<ParseTree>, Box<ParseTree>),   // (v1, v2)
    Left(Box<ParseTree>),                   // Left v
    Right(Box<ParseTree>),                  // Right v
    Star(Vec<ParseTree>),                   // [v1, ..., vn]
}
```

Figure 4.1: The parse tree type, representing the structure of a parse.

The definitions above translate directly to Rust, except for the star case. $\mathsf{T}\varepsilon$ corresponds to `Empty` for the empty parse, T_{lit} to `Char` for a literal, T_{seq} to `Pair` for a concatenation of parse trees, and T_{+}^{L} and T_{+}^{R} to `Left` and `Right`, recording the chosen branch of an alternation. Both star rules are represented by `Star(vec![])`.

Flattening. The typing judgment describes the structure of a match, but a parse tree must also be related to the word that it matches. For this purpose, they define a *flattening* function $|v|$ that maps a parse tree to the word represented by its structure:

$$\begin{aligned} |()| &= \varepsilon, & |(v_1, v_2)| &= |v_1||v_2|, \\ |l| &= l, & |\mathit{Left}\ v| &= |v|, \\ |[]| &= \varepsilon & |\mathit{Right}\ v| &= |v|, \\ & & |v : vs| &= |v||vs|. \end{aligned} \tag{4.3}$$

The flattening function directly connects parse trees to $L(r)$, and thus validates that a parse tree witnesses membership in $L(r)$, serving as an correctness check:

$$L(r) = \{\, |v| \mid\mid\!\vdash v : r \,\}. \tag{4.4}$$

In Rust. Figure 4.2 shows how `flatten` recursively traverses the parse tree and concatenates its leaf characters. The function borrows the tree to allow subsequent inspection and comparison. The `Pair` case concatenates the results of flattening both children, while `Left` and `Right` delegate to their child, and `Star` concatenates the results for each element.

```
pub fn flatten(v: &ParseTree) -> String {
    match v {
        ParseTree::Empty        => String::new(),
        ParseTree::Char(c)      => c.to_string(),
        ParseTree::Pair(v1, v2) => format!("{}{}", flatten(v1), flatten(v2)),
        ParseTree::Left(v)      => flatten(v),
        ParseTree::Right(v)     => flatten(v),
        ParseTree::Star(vs)     => vs.iter().map(flatten).collect(),
    }
}
```

Figure 4.2: The `flatten` function extracts the matched word from a parse tree.

Unambiguous example. Consider $r = ab$ and $w = \texttt{ab}$. Since each subexpression is a literal, the rule T_{lit} gives each literal a unique witness. Then T_{seq} requires the parse tree for a to be followed by the parse tree for b, combining them into a parse tree for ab:

$$\frac{a \in \Sigma}{\vdash a : a}\,(\mathsf{T}_{\text{lit}}) \qquad \frac{b \in \Sigma}{\vdash b : b}\,(\mathsf{T}_{\text{lit}}) \qquad \frac{\vdash a : a \qquad \vdash b : b}{\vdash (a, b) : ab}\,(\mathsf{T}_{\text{seq}}) \tag{4.5}$$

The parse tree $v = (a, b)$ flattens to $|(a, b)| = \texttt{ab}$.

Ambiguous Example. Recall $r = (a + ab)(b + \varepsilon)$ and $w = \texttt{ab}$ from Section 4.1. The word admits two distinct decompositions, corresponding to different parse trees.

The first decomposition uses the right alternatives of both subexpressions:

$$\underbrace{\frac{\dfrac{a \in \Sigma}{\vdash a : a}\,(\mathsf{T}_{\text{lit}}) \qquad \dfrac{b \in \Sigma}{\vdash b : b}\,(\mathsf{T}_{\text{lit}})}{\vdash (a, b) : ab}\,(\mathsf{T}_{\text{seq}})}{\vdash \mathit{Right}(a, b) : a + ab}\,(\mathsf{T}^R_+)}_{a + ab} \qquad \underbrace{\frac{\dfrac{b \in \Sigma}{\vdash b : b}\,(\mathsf{T}_{\text{lit}}) \qquad \vdash () : \varepsilon\,(\mathsf{T}_\varepsilon)}{\vdash \mathit{Right}() : b + \varepsilon}\,(\mathsf{T}^R_+)}_{b + \varepsilon}$$

By T_{seq}, these derivations combine into the parse tree:

$$\frac{\dfrac{\vdash a : a \qquad \vdash b : b}{\vdash (a, b) : ab}}{\vdash \mathit{Right}(a, b) : a + ab} \qquad \frac{\vdash () : \varepsilon}{\vdash \mathit{Right}() : b + \varepsilon}$$

$$v_1 = \Big(\mathit{Right}(a, b), \mathit{Right}()\Big), \qquad |v_1| = \texttt{ab}. \tag{4.6}$$

The second decomposition uses the left alternatives of both subexpressions:

$$\underbrace{\frac{\dfrac{a \in \Sigma}{\vdash a : a}\,(\mathsf{T}_{\text{lit}})}{\vdash \mathit{Left}\, a : a + ab}\,(\mathsf{T}^L_+)}_{a + ab} \qquad \underbrace{\frac{\dfrac{b \in \Sigma}{\vdash b : b}\,(\mathsf{T}_{\text{lit}})}{\vdash \mathit{Left}\, b : b + \varepsilon}\,(\mathsf{T}^L_+)}_{b + \varepsilon}$$

By T_{seq}, these derivations combine into the second parse tree:

$$\frac{\vdash a : a}{\vdash \mathit{Left}\, a : a + ab} \qquad \frac{\vdash b : b}{\vdash \mathit{Left}\, b : b + \varepsilon}$$

$$v_2 = \Big(\mathit{Left}\, a, \mathit{Left}\, b\Big), \qquad |v_2| = \texttt{ab}. \tag{4.7}$$

Both trees are valid parse trees for the same expression and word, yet they differ in structure. The typing judgment alone does not determine which parse tree should be selected. A *disambiguation policy* specifies this choice.

5 Derivative-Based Parser

Chapter 3 showed how a Brzozowski-derivative matcher determines whether $w \in L(r)$ and Chapter 4 showed that a matching word may admit multiple valid parse trees. A parser must therefore select a single parse tree according to a disambiguation policy.

Following Sulzmann and Lu [Sulzmann and Lu, 2014], this chapter presents two derivative-based methods for constructing POSIX parse trees. The direct method performs a forward derivative pass followed by a backward pass for tree reconstruction. The bit-coded method records parse-tree information during the forward pass, avoiding a separate backward pass.

5.1 The Standard Derivative-Based Parser

Consider the ambiguous example from Chapter 4, $r = (a + ab)(b + \varepsilon)$ with input $w = \texttt{ab}$, and ask whether a parse tree can be recovered from the final derivative.

$$\begin{aligned} r &\xrightarrow{a} (\varepsilon + \varepsilon b)(b + \varepsilon) \\ &\xrightarrow{b} (\phi + (\phi b + \varepsilon))(b + \varepsilon) + (\varepsilon + \phi). \end{aligned} \tag{5.1}$$

The final derivative is nullable, so $w \in L(r)$ and both disjuncts accept the empty word. Its first disjunct, $(\phi + (\phi b + \varepsilon))(b + \varepsilon)$, reflects $ab \cdot \varepsilon$, and its second, $(\varepsilon + \phi)$, reflects $a \cdot b$.

Since a derivative step only records the remaining input language and not how consumed characters were matched, the final derivative alone does not indicate which decomposition was selected, and therefore cannot be used to recover the parse tree.

However, the derivation sequence itself contains this information. Each step consumed one character and is invertible given that character, so reversing the steps recovers the tree. Figure 5.1 shows the forward pass as the derivation sequence and the backward pass as the reconstruction.

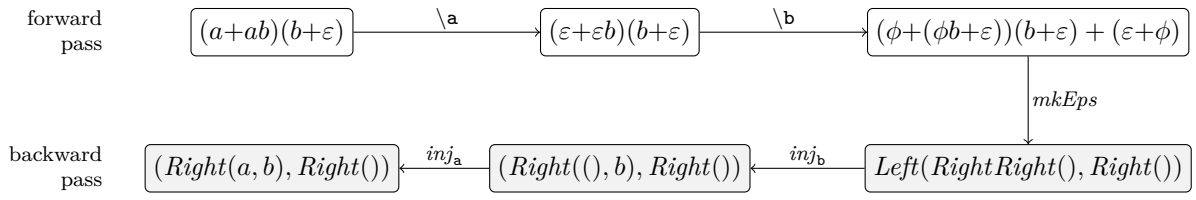


Figure 5.1: Overview of the standard derivative-based parser on the ambiguous example. The forward pass computes derivatives, while the backward pass uses *mkEps* to construct a parse tree for the final nullable derivative and *inj* to reconstruct the original parse tree.

The two-pass pipeline. The direct construction uses two passes. The forward pass applies `deriv` (Figure 3.3) and `nullable` (Figure 3.2) to process the input.

The backward pass applies two new functions to walk the derivation sequence in reverse and reconstruct the parse tree. First *mkEps* builds a parse tree for the empty word from the final nullable expression. Then injection steps are applied, with *inj* reinserting one consumed character into the parse tree, moving in w from right to left. Each injection step takes the current parse tree, the corresponding derivative step, and its consumed character, and returns the parse tree for the preceding expression.

As shown in Figure 5.1 after the first injection of b, the tree is $(\mathit{Right}((), b), \mathit{Right}())$, where $()$ marks the position still to be filled. After the second injection of a, that position is replaced by the subtree $\mathit{Right}(a, b)$, giving the complete parse tree $(\mathit{Right}(a, b), \mathit{Right}())$.

In the following the operations for the standard derivative parser are shown in more detail.

5.1.1 Building the Empty-Word Tree

The backward pass begins with *mkEps*, which reads a parse tree for the empty word off the final nullable expression. The function follows [Sulzmann and Lu, 2014, Fig. 3] and is defined recursively:

$$\begin{aligned}
\mathit{mkEps}_{\varepsilon} &= () && (\mathsf{E}_{\varepsilon}) \\
\mathit{mkEps}_{r^*} &= [\,] && (\mathsf{E}_{*}) \\
\mathit{mkEps}_{r_1 r_2} &= (\mathit{mkEps}_{r_1}, \mathit{mkEps}_{r_2}) && (\mathsf{E}_{\mathrm{seq}}) \\
\mathit{mkEps}_{r_1 + r_2} &= \begin{cases} \mathit{Left}\ \mathit{mkEps}_{r_1} & \text{if } n(r_1) \quad (\mathsf{E}_{+}^{L}) \\ \mathit{Right}\ \mathit{mkEps}_{r_2} & \text{otherwise} \quad (\mathsf{E}_{+}^{R}) \end{cases}
\end{aligned} \tag{5.2}$$

The first three rules follow from the structure of the expression: $(\mathsf{E}_{\varepsilon})$ gives ε an empty parse tree, (E_{*}) treats a star matching the empty word as zero iterations, and $(\mathsf{E}_{\mathrm{seq}})$ matches a concatenation against ε only when both factors do. No rules are needed for $\emptyset$ or literals, since neither can match the empty word. The alternation rule is where disambiguation occurs. If exactly one of r_1 and r_2 is nullable, that branch is the only possible witness. If both are, (E_{+}^{L}) breaks the tie by preferring the left one, following POSIX's leftmost-longest policy [Sulzmann and Lu, 2014]. In the example, the final expression is:

$$(\phi + (\phi b + \varepsilon))(b + \varepsilon) + (\varepsilon + \phi).$$

At top-level $+$, both summands are nullable, so (E_{+}^{L}) applies and picks the left one. Within the left summand's first factor $\phi + (\phi b + \varepsilon)$, only the right alternative is nullable, so (E_{+}^{R}) applies, and within it only the right sub-alternative ε is nullable, so (E_{+}^{R}) applies again, contributing $\mathit{RightRight}()$. The second factor $b + \varepsilon$ contributes $\mathit{Right}()$ via (E_{+}^{R}), giving:

$$\mathit{Left}(\mathit{RightRight}(), \mathit{Right}()).$$

Because disambiguation is deferred to *mkEps*, the forward pass cannot simplify. The second injection recovers $\mathit{Right}(a, b)$ only because the forward pass preserved the branch that consumed a. Had it been simplified away, the backward pass would have had no node to reinsert a into. For example, simplifying at each step collapses the derivation to:

$$r \xrightarrow{a} (\varepsilon + b)(b + \varepsilon) \xrightarrow{b} \varepsilon,$$

This still witnesses $w \in L(r)$ but keeps no record of which alternative consumed a or b. The matcher accepts, but a parser cannot recover $\mathit{Right}(a, b)$ from ε.

In Rust. The recursive definition of *mkEps* (5.2) is implemented directly in Rust, as shown in Figure 5.2, following the same cases for each expression form.

```
pub fn mk_eps(r: &Regex) -> ParseTree {
    match r {
        Regex::Eps          => ParseTree::Empty,
        Regex::Star(_)      => ParseTree::Star(Vec::new()),
        Regex::Seq(r1, r2) => ParseTree::Pair(
                               Box::new(mk_eps(r1)), Box::new(mk_eps(r2))),
        Regex::Alt(r1, r2) =>
            if nullable(r1)   {ParseTree::Left(Box::new(mk_eps(r1)))}
            else              {ParseTree::Right(Box::new(mk_eps(r2)))},
        Regex::Lit(c)       => panic!("mk_eps called on Lit('{}')", c),
        Regex::Phi          => panic!("mk_eps called on Phi"),
    }
}
```

Figure 5.2: `mk_eps`: construct a parse tree for the empty word.

The expression r is borrowed rather than consumed because the backward pass needs it again during injection. The alternation arm encodes disambiguation as a conditional, examining `r2` only when `nullable(r1)` is false. The panic arms enforce the precondition that *mkEps* applies only to nullable expressions, rejecting the non-nullable ϕ and literals.

5.1.2 Reinserting Characters via Injection

After *mkEps* constructed the parse tree v for the empty input, the reconstruction starts from this initial tree and proceeds backwards through the derivation sequence. Each application of *inj* takes the current tree, the corresponding derivative step, and its consumed character, returning a tree for the preceding expression. *inj* recursively follows the structure of the residual until it reaches the subexpression containing the literal that consumed the character. It reinserts the character there and then unwinds to rebuild the surrounding tree. After all characters have been reinserted, the parse tree for the original expression and input is recovered. The injection function is defined by the following rules [Sulzmann and Lu, 2014, Fig. 3]:

$$
\begin{aligned}
\mathit{inj}_{x\backslash x}() &= x && (\mathsf{I}_{\mathrm{lit}}) \\
\mathit{inj}_{(r_1+r_2)\backslash x}(\mathit{Left}\ v_1) &= \mathit{Left}\ (\mathit{inj}_{r_1\backslash x}\ v_1) && (\mathsf{I}^{L}_{+}) \\
\mathit{inj}_{(r_1+r_2)\backslash x}(\mathit{Right}\ v_2) &= \mathit{Right}\ (\mathit{inj}_{r_2\backslash x}\ v_2) && (\mathsf{I}^{R}_{+}) \\
\mathit{inj}_{r^*\backslash x}(v, vs) &= (\mathit{inj}_{r\backslash x}\ v) : vs && (\mathsf{I}_{*}) \\
\mathit{inj}_{(r_1 r_2)\backslash x}(v_1, v_2) &= (\mathit{inj}_{r_1\backslash x}\ v_1,\ v_2) && (\mathsf{I}_{\mathrm{seq}}),\ n(r_1) = \text{false} \\
\mathit{inj}_{(r_1 r_2)\backslash x}(\mathit{Left}(v_1, v_2)) &= (\mathit{inj}_{r_1\backslash x}\ v_1,\ v_2) && (\mathsf{I}^{L}_{\mathrm{seq}}),\ n(r_1) = \text{true} \\
\mathit{inj}_{(r_1 r_2)\backslash x}(\mathit{Right}\ v_2) &= (\mathit{mkEps}_{r_1},\ \mathit{inj}_{r_2\backslash x}\ v_2) && (\mathsf{I}^{R}_{\mathrm{seq}}),\ n(r_1) = \text{true}
\end{aligned}
\tag{5.3}
$$

Rules (I_{lit}) and (I_{*}) invert their derivative rules directly, reconstructing the consumed character and the current iteration, the latter prepended to the remaining ones.

Since the consumed character lies in exactly one branch of an alternation, the incoming tree is either *Left* v_1 or *Right* v_2. Rule (I_{+}^{L}) recurses into v_1 and rebuilds the *Left* node, and (I_{+}^{R}) recurses into v_2 and rebuilds the *Right* node.

The concatenation rules cover the three cases of the sequence derivative. If $n(r_1) = \text{false}$, r_1 cannot match ε and must consume x, so (I_{seq}) recurses into v_1. If $n(r_1) = \text{true}$, the derivative recorded which component consumed x as a tag: ($\mathsf{I}_{\text{seq}}^{L}$) handles an incoming $\mathit{Left}(v_1, v_2)$ tag and recurses into v_1, while ($\mathsf{I}_{\text{seq}}^{R}$) handles a *Right* v_2 tag by reconstructing r_1's empty match with mkEps_{r_1} and recursing into v_2.

Since ϕ consumes no characters, *inj* is never applied with ϕ as the residual.

```
pub fn inject(r: &Regex, l: char, v: ParseTree) -> ParseTree {
    match r {
        Regex::Lit(c) => { assert!(*c == l);
            ParseTree::Char(l) }
        Regex::Alt(r1, r2) => match v {
            ParseTree::Left(v1) =>
                ParseTree::Left(Box::new(inject(r1, l, *v1))),
            ParseTree::Right(v2) =>
                ParseTree::Right(Box::new(inject(r2, l, *v2))),
            _ => panic!("inject/Alt: want Left|Right"),
        },
        Regex::Star(r1) => match v {
            ParseTree::Pair(v1, vs) => {
                let mut it = vec![inject(r1, l, *v1)];
                if let ParseTree::Star(rest) = *vs { it.extend(rest); }
                ParseTree::Star(it)
            }
            _ => panic!("inject/Star: want Pair"),
        },
        Regex::Seq(r1, r2) => if !nullable(r1) {
            match v {
                ParseTree::Pair(v1, v2) => ParseTree::Pair(
                    Box::new(inject(r1, l, *v1)), v2),
                _ => panic!("inject/Seq: want Pair"),
            }
        } else {
            match v {
                ParseTree::Left(vp) => match *vp {
                    ParseTree::Pair(v1, v2) => ParseTree::Pair(
                        Box::new(inject(r1, l, *v1)), v2),
                    o => panic!("inject/SeqL: want Pair, got {:?}", o),
                },
                ParseTree::Right(v2) => ParseTree::Pair(
                    Box::new(mk_eps(r1)),
                    Box::new(inject(r2, l, *v2))),
                _ => panic!("inject/Seq: want Left|Right"),
            }
        },
        Regex::Eps | Regex::Phi => panic!("inject called on Eps/Phi"),
    }
}
```

Figure 5.3: `inject`: reinsert a character into a parse tree.

In Rust. `inject(r, l, v)` takes three arguments because *inj* is indexed by both an expression and a character. Figure 5.3 shows the complete implementation.

The function matches on the expression `r` first and on the tree `v` second, following the order in which (5.3) indexes its rules. This order matters because a tree tag is interpreted relative to the expression that produced it. Under an alternation, `Left` selects the left disjunct, while under a nullable concatenation it indicates that the derivative continued inside the left operand. The `Lit` arm reconstructs a leaf after asserting that the consumed

character matches, and the `Star` arm inverts (I_*), reinserting the character into the current iteration and prepending it to the remaining ones.

The expression `r` is borrowed and read only, while the tree `v` is consumed and decomposed. In the concatenation arms where the right operand was not derived, `v2` moves directly into the new pair, since (5.3) leaves it unchanged. The `Right` arm of concatenation is the only concatenation case that synthesizes content, calling `mk_eps(r1)` because the derivative skipped `r1` and the forward pass produced no tree for that half of the pair.

Calling `nullable(r1)` in the `Seq` arm recomputes a decision the forward pass already made when it chose between D_{seq} and $\mathsf{D}^n_{\text{seq}}$. However, carrying this as a flag at every step would require additional storage, while `nullable` is cheap to evaluate on demand.

5.1.3 Parser

The parser composes `mkEps` and `inject` by storing every forward-pass expression and reconstructing the tree in reverse. Since `inject` takes the preceding expression as an explicit argument, each call needs r_{i-1}. Re-deriving it from r through the same prefix would duplicate the forward pass, so storing it avoids computing the derivation again.

Figure 5.4 implements this iteratively, using a `Vec` that grows dynamically on the heap.

```
pub fn parse_deriv_std_loop(input: &str, r: &Regex) -> Option<ParseTree>
{
    let chars: Vec<char> = input.chars().collect();
    let mut expressions = Vec::with_capacity(chars.len() + 1);
    expressions.push(r.clone());
    for &c in &chars {
        expressions.push(deriv(expressions.last().unwrap(), c));
    }
    if !nullable(expressions.last().unwrap()) {
      return None;
      }
    let mut tree = mk_eps(expressions.last().unwrap());
    for i in (0..chars.len()).rev() {
        tree = inject(&expressions[i], chars[i], tree);
    }
    Some(tree)
}
```

Figure 5.4: The standard derivative-based parser. The forward pass stores the residual after each character in `expressions`, entry i being the residual after i characters, and returns `None` if the final entry is not nullable. The backward pass starts from `mk_eps` of the final entry and applies `inject` to the preceding entries in reverse, yielding the parse tree for r.

5.1.4 Worked Example: Ambiguous Expression

Recall the ambiguous example of Section 4.2, $r_0 = (a \mid ab)(b \mid \varepsilon)$ on input `ab`.

Forward pass. There are two derivative steps, one per input symbol. After deriving $\mathtt{a}$, the left factor $\varepsilon + \varepsilon b$ is nullable, so the second derivation on $\mathtt{b}$ uses $\mathsf{D}^n_{\text{seq}}$ rather than D_{seq}:

$$r_0 = (a + ab)(b + \varepsilon) \xrightarrow[\mathsf{D}_{\text{seq}}]{a} r_1 = (\varepsilon + \varepsilon b)(b + \varepsilon)$$
$$\xrightarrow[\mathsf{D}^n_{\text{seq}}]{b} r_2 = \underbrace{(\phi + (\phi b + \varepsilon))(b + \varepsilon)}_{X} + \underbrace{(\varepsilon + \phi)}_{Y}.$$

The resulting final expression is r_2, both X and Y are nullable. The disjunct X is the residual of the ab alternative in the first factor, and Y is the residual of the a alternative.

Backward pass: mkEps. The backward pass starts by building the parse tree for the empty word: $mkEps(r_2)$. Since $r_2 = X + Y$ and both disjuncts are nullable, E^L_+ selects X. As X is a concatenation, E_{seq} combines the trees for its two factors:

$$mkEps(r_2) = mkEps(X + Y) \xrightarrow{\mathsf{E}^L_+} Left\, mkEps(X)$$

$$mkEps(X) \xrightarrow{\mathsf{E}_{\text{seq}}} \Big(mkEps(\phi + (\phi b + \varepsilon)), mkEps(b + \varepsilon)\Big).$$

Since ϕ and ϕb are not nullable, the first factor selects ε by applying the rule E^R_+ twice, while the second factor selects ε through E^R_+ since b is not nullable, giving v_2:

$$mkEps(\phi + (\phi b + \varepsilon)) \xrightarrow{\mathsf{E}^R_+} Right\, mkEps(\phi b + \varepsilon) \xrightarrow{\mathsf{E}^R_+} Right\, Right\, mkEps(\varepsilon) \xrightarrow{\mathsf{E}_\varepsilon} Right\, Right\, ()$$

$$mkEps(b + \varepsilon) \xrightarrow{\mathsf{E}^R_+} Right\, mkEps(\varepsilon) \xrightarrow{\mathsf{E}_\varepsilon} Right\, ()$$

$$v_2 = mkEps(r_2) = Left\Big(Right\, Right\, (), Right\, ()\Big).$$

Backward pass: Inj. Reinsertion begins with $\mathtt{b}$ and maps v_2 to a tree for $r_1 = (\varepsilon + \varepsilon b)(b + \varepsilon)$. Since v_2 carries a *Left* tag, $\mathsf{I}^L_{\text{seq}}$ recurses into $\varepsilon + \varepsilon b$, leaving the second factor's tree $Right\, ()$ unchanged, as all input characters belong to the first factor under the $ab \cdot \varepsilon$ decomposition. Inside $\varepsilon + \varepsilon b$, I^R_+ selects εb, the residual of the ab alternative. Since ε is nullable, $\mathsf{I}^R_{\text{seq}}$ supplies $mkEps(\varepsilon) = ()$ and recurses into b, where I_{lit} reinserts $\mathtt{b}$.

$$inj_{r_1 \backslash \mathtt{b}}(v_2) = inj_{r_1 \backslash \mathtt{b}}\Big(\underbrace{Left(Right\, Right\, (),\ Right\, ())}_{v_2}\Big)$$
$$\xrightarrow{\mathsf{I}^L_{\text{seq}}} \Big(inj_{(\varepsilon+\varepsilon b)\backslash \mathtt{b}}(Right\, Right\, ()),\ Right\, ()\Big)$$
$$\xrightarrow{\mathsf{I}^R_+} \Big(Right\, inj_{\varepsilon b \backslash \mathtt{b}}(Right\, ()),\ Right\, ()\Big)$$
$$\xrightarrow{\mathsf{I}^R_{\text{seq}}} \Big(Right\, (mkEps(\varepsilon),\ inj_{b\backslash \mathtt{b}}()),\ Right\, ()\Big)$$
$$\xrightarrow{\mathsf{E}_\varepsilon,\ \mathsf{I}_{\text{lit}}} \underbrace{\Big(Right((), b),\ Right\, ()\Big)}_{v_1}$$

Reinsertion of $\mathtt{a}$ maps v_1 to a parse tree for $r_0 = (a + ab)(b + \varepsilon)$. The outer sequence's first factor $a + ab$ is not nullable, so I_{seq} fires directly on v_1's first component, again leaving the

second component $Right\,()$ untouched. That first component, $Right((), b)$, carries a $Right$ tag, so I_{+}^{R} selects ab. Inside $ab = Seq(a, b)$ the first factor a is again not nullable, so I_{seq} fires once more and recurses into a, where I_{lit} reinserts the character:

$$\begin{aligned}
inj_{r_0\backslash \mathtt{a}}(v_1) &= inj_{r_0\backslash \mathtt{a}}\Big(\underbrace{(Right((), b),\ Right\,())}_{v_1}\Big) \\
&\xrightarrow{\mathsf{I}_{\text{seq}}} \Big(inj_{(a+ab)\backslash \mathtt{a}}(Right((), b)),\ Right\,()\Big) \\
&\xrightarrow{\mathsf{I}_{+}^{R}} \Big(Right\, inj_{ab\backslash \mathtt{a}}(((), b)),\ Right\,()\Big) \\
&\xrightarrow{\mathsf{I}_{\text{seq}}} \Big(Right\,(inj_{a\backslash \mathtt{a}}(()),\ b),\ Right\,()\Big) \\
&\xrightarrow{\mathsf{I}_{\text{lit}}} \underbrace{\Big(Right(a, b),\ Right\,()\Big)}_{v_0}
\end{aligned}$$

This implementation stores three intermediate expressions, r_0, r_1, and r_2, one per input character plus the initial expression, for the backward pass to reconstruct from.

5.2 The Bit-Coded Derivative-Based Parser

The direct construction of Sulzmann and Lu [Sulzmann and Lu, 2014] reconstructs the tree only after the full input, since each injection step needs the preceding expression. Their bit-coded optimization records this information during a single pass instead.

Each subexpression carries the bit-sequence it has committed to so far, written $bs@r$. P and Q denote the annotated factors for $r_0 = (a + ab)(b + \varepsilon)$. Deriving $w = \mathtt{ab}$ gives:

$$\begin{aligned}
ri_0 &= []@\Big(\underbrace{[0]@a \oplus [1]@(a\,b)}_{P} \cdot \underbrace{[0]@b \oplus [1]@\varepsilon}_{Q}\Big) \\
&\xrightarrow{a} ri_1 = []@\Big(\underbrace{[0]@\varepsilon \oplus [1]@b}_{P'} \cdot Q\Big) \\
&\xrightarrow{b} ri_2 = []@\Big([1]@Q \ \oplus\ [0, 0]@\varepsilon\Big).
\end{aligned} \tag{5.4}$$

Figure 5.5 shows this derivation together with the extraction step it feeds into:

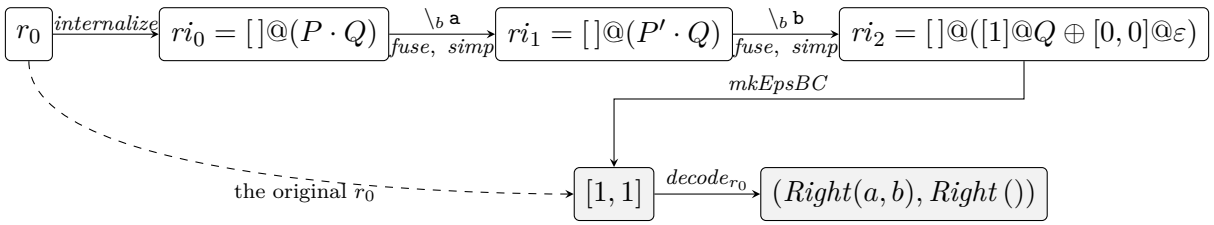


Figure 5.5: Overview of the bit-coded derivative parser on the ambiguous example. *internalize* introduces the bits, each derivative step threads them forward, so consumed factors' bits transfer to survivors. *mkEpsBC* extracts them and $decode_{r_0}$ reads them against the original pattern.

The bit-coded pipeline. *internalize* tags each alternation branch with a bit that records its preference, producing $ri_0 = []@(P \cdot Q)$ with $P = [0]@a \oplus [1]@(a\,b)$ and $Q = [0]@b \oplus [1]@\varepsilon$, where 0 selects the left branch and 1 the right.

Deriving by `a` consumes P's leading a, giving P', while Q remains untouched. Deriving by `b` consumes the rest. Since P''s ε branch is nullable, both decompositions survive. In the first, the derivative continues inside P', which finishes and folds its bit 1 onto Q. In the second, Q matches `b` instead, and P''s completion bit fuses with Q's match into $[0,0]@\varepsilon$. This yields $ri_2 = []@([1]@Q \oplus [0,0]@\varepsilon)$. The helper *fuse* performs this folding and fusing of bits onto surviving nodes, prepending a bit-sequence to a node's annotation without descending into its children. Both $deriv_{bc}$ and *simp* rely on it.

Unlike the direct construction, the bit-coded construction requires simplification to prevent unbounded growth of annotated expressions. The function *simp* discards dead branches and collapses finished subexpressions, while *fuse* preserves their bits on surviving nodes, as when P collapses into P' and then onto Q.

Once `ab` is exhausted, *mkEpsBC* walks ri_2 the way *mkEps* walks a plain expression, taking the leftmost nullable branch at each remaining choice: $[1]@Q$ is nullable, so it contributes its 1, and inside Q the ε branch is nullable, contributing a second 1, giving $bits = [1,1]$. $decode_{r_0}$ then replays these two bits against the original r_0. The first 1 selects $a+ab$'s right branch and the second selects $b+\varepsilon$'s right branch, giving the tree $(Right(a,b), Right\,())$.

This shows that the bit-coded construction reconstructs the parse tree in a single step, while the direct construction reverses residuals step by step.

5.2.1 Annotated Expressions

Building parse-tree information during the forward pass extends the derivative computation with bit-sequence annotations, propagated by the bit-threaded derivative $ri\backslash_b l$. An annotated expression ri extends r by associating each non-$\emptyset$ node with a bit-sequence bs recording the reconstruction choices made during the forward pass.

$$ri ::= \emptyset \mid (bs@\varepsilon) \mid (bs@x) \mid (bs@ri_1 \oplus ri_2) \mid (bs@ri_1\,ri_2) \mid (bs@ri^*). \tag{5.5}$$

In Rust. Figure 5.6 defines `ARegex` as the annotated counterpart of `Regex`. Every variant except `Phi` carries a leading `Vec<bool>` corresponding to (5.5), with one `bool` per bit. `Phi` has no annotation, since it cannot contribute to a successful parse.

```
pub enum ARegex {
    Phi,                        // no annotation slot
    Eps(Vec<bool>), Lit(Vec<bool>, char),
    Alt(Vec<bool>, Box<ARegex>, Box<ARegex>),
    Seq(Vec<bool>, Box<ARegex>, Box<ARegex>),
    Star(Vec<bool>, Box<ARegex>),
}
```

Figure 5.6: The annotated-expression type extends `Regex` with a leading bit-list.

All following operations act on this `ARegex`. Since derivative and simplification steps remove nodes while the bits attached to them must survive, each operation needs a way to transfer those bits onto the node that remains. The helper *fuse* provides this transfer.

5.2.2 Fusing Bit-Lists

fuse, defined in [Sulzmann and Lu, 2014, Fig. 5], transfers bits from a node that is about to be removed onto the node that survives in its place. It prepends bs to the top node's annotation bs' without descending into its children, so outer reconstruction choices precede those already recorded in the subtree.

$$\begin{aligned} \mathit{fuse}(bs, \emptyset) &= \emptyset \\ \mathit{fuse}(bs, (bs'@\varepsilon)) &= (bs +\!\!+ bs')@\varepsilon \\ \mathit{fuse}(bs, (bs'@x)) &= (bs +\!\!+ bs')@x \\ \mathit{fuse}(bs, (bs'@ri_1 \oplus ri_2)) &= (bs +\!\!+ bs')@(ri_1 \oplus ri_2) \\ \mathit{fuse}(bs, (bs'@ri_1\, ri_2)) &= (bs +\!\!+ bs')@(ri_1\, ri_2) \\ \mathit{fuse}(bs, (bs'@ri^*)) &= (bs +\!\!+ bs')@ri^*. \end{aligned} \tag{5.6}$$

In Rust. Figure 5.7 implements the six rules of (5.6), one per variant. The helper `combine` corresponds to the concatenation $+\!\!+$ and places `bs` before `p`, so the newly prepended bits precede those already recorded:

```
pub fn fuse(bs: &[bool], ri: ARegex) -> ARegex {
    if bs.is_empty() { return ri; }
    match ri {
        ARegex::Phi => ARegex::Phi,
        ARegex::Eps(p) => ARegex::Eps(combine(bs, p)),
        ARegex::Lit(p, c) => ARegex::Lit(combine(bs, p), c),
        ARegex::Alt(p, r1, r2) => ARegex::Alt(combine(bs, p), r1, r2),
        ARegex::Seq(p, r1, r2) => ARegex::Seq(combine(bs, p), r1, r2),
        ARegex::Star(p, r) => ARegex::Star(combine(bs, p), r),
    }
}

fn combine(bs: &[bool], p: Vec<bool>) -> Vec<bool> {
    let mut new = bs.to_vec();
    new.extend(p);
    new
}
```

Figure 5.7: `fuse`: prepend a bit-list to the annotation of the top node.

The `Phi` arm is the only case that returns without calling `combine`, because `Phi` represents the empty language and carries no annotation slot, so there is no bit-list for `fuse` to prepend to. Every other arm rebuilds the top node with a combined annotation and moves its two children, or single child in the star case, unchanged. Since `fuse` prepends `bs` only to the top node's annotation, the bits carried by its children are not touched, and the operation rewrites exactly one node regardless of how large the subtree below it is.

The function is therefore non-recursive, and no subtree needs to be copied. The implementation realizes this directly, since `ri` is consumed by value and its untouched children move into the rebuilt node without a clone. The helper `combine` allocates a fresh `Vec`, at a cost proportional to the combined length of the two bit-lists rather than to the size of the subtree below. This allocation happens on every call with a non-empty `bs`. An empty

`bs` is the one case that skips it, since combining an empty prefix with `p` gives `p` and the rebuilt node would be identical to the input.

5.2.3 Internalizing Expressions

The forward pass operates on annotated expressions, but the input is a plain expression, so r must first be converted into its annotated counterpart ri. The function *internalize*, following [Sulzmann and Lu, 2014, Fig. 5], performs this conversion recursively, and it is the only step that introduces new bits:

$$\begin{aligned}
\mathit{internalize}(\emptyset) &= \emptyset \\
\mathit{internalize}(\varepsilon) &= [\,]@\varepsilon \\
\mathit{internalize}(x) &= [\,]@x \\
\mathit{internalize}(r_1 + r_2) &= [\,]@\Big(\mathit{fuse}([0], \mathit{internalize}(r_1)) \oplus \mathit{fuse}([1], \mathit{internalize}(r_2))\Big) \\
\mathit{internalize}(r_1\, r_2) &= [\,]@\Big(\mathit{internalize}(r_1)\; \mathit{internalize}(r_2)\Big) \\
\mathit{internalize}(r^*) &= [\,]@\mathit{internalize}(r)^*.
\end{aligned} \tag{5.7}$$

The tags [0] and [1] on the two branches of $r_1 + r_2$ mark the left and right branch respectively, so the decoder can later recover which one was taken. Every other case introduces no new bit and uses only the empty list $[\,]$. The two calls to *fuse* in (5.7) are the only points in the construction where new bits are introduced.

In Rust. Figure 5.8 directly maps the constructors of `Regex` to their annotated counterparts. Each recursive call applies `internalize` to the corresponding subexpression, preserving the structure of the original expression in the annotated representation.

```
pub fn internalize(r: &Regex) -> ARegex {
    match r {
        Regex::Phi => ARegex::Phi,
        Regex::Eps => ARegex::Eps(vec![]),
        Regex::Lit(c) => ARegex::Lit(vec![], *c),
        Regex::Alt(r1, r2) => {
            let ri1 = fuse(&[false], internalize(r1));
            let ri2 = fuse(&[true], internalize(r2));
            ARegex::Alt(vec![], Box::new(ri1), Box::new(ri2))
        }
        Regex::Seq(r1, r2) => ARegex::Seq(vec![],
            Box::new(internalize(r1)), Box::new(internalize(r2))),
        Regex::Star(r1) => ARegex::Star(vec![], Box::new(internalize(r1))),
    }
}
```

Figure 5.8: `internalize` converts a `Regex` into an `ARegex`. Each alternation branch receives a tag recording its side, and the two `fuse` calls are the only points where new bits are introduced.

The `Alt` case calls `fuse` with a non-empty bit-list, tagging the left operand with `[false]` and the right with `[true]` before wrapping them in a `Alt` with an empty outer bit-list. Every later bit originates from one of these two tags and is propagated during derivation.

The `Phi`, `Eps`, `Lit`, `Seq`, and `Star` cases use an empty bit-list, since they introduce no disambiguation choice at this stage. Star iteration choices are instead determined during derivative computation.

5.2.4 Extracting Bits

mkEpsBC, from [Sulzmann and Lu, 2014, Fig. 5], mirrors *mkEps* but extracts the corresponding bit-list, using the same left preference at alternation and a closing 1 for a star. Unlike *mkEps*, it constructs no parse-tree nodes. Each rule prepends the current node's bits to those returned recursively.

$$
\begin{aligned}
mkEpsBC_{(bs@\varepsilon)} &= bs \\
mkEpsBC_{(bs@ri^*)} &= bs +\!\!+ [1] \\
mkEpsBC_{(bs@ri_1\, ri_2)} &= bs +\!\!+ mkEpsBC_{ri_1} +\!\!+ mkEpsBC_{ri_2} \\
mkEpsBC_{(bs@ri_1 \oplus ri_2)} &= bs +\!\!+ \begin{cases} mkEpsBC_{ri_1} & \varepsilon \in L(ri_1) \\ mkEpsBC_{ri_2} & \text{otherwise.} \end{cases}
\end{aligned}
\tag{5.8}
$$

The rules cover the four expression forms that can be nullable, as in *mkEps*. No rules are needed for $\emptyset$ or literals, since neither can match the empty word, and *mkEpsBC* is called only on nullable expressions, so both are handled as preconditions.

In Rust. Figure 5.9 mirrors `mk_eps` (Figure 5.2), with one difference. Where `mk_eps` builds a `ParseTree`, `mk_eps_bc` concatenates bit-lists. The `Star` arm also does not recurse, since a star matching the empty word uses zero iterations, so it only appends the closing `true` and never examines the child expression:

```
pub fn mk_eps_bc(ri: &ARegex) -> Vec<bool> {
    match ri {
        ARegex::Phi => panic!("mk_eps_bc called on Phi"),
        ARegex::Lit(_, c) => panic!("mk_eps_bc called on Lit('{}')", c),
        ARegex::Eps(bs) => bs.clone(),
        ARegex::Alt(bs, r1, r2) => {
            let mut result = bs.clone();
            if nullable_bc(r1) {
                result.extend(mk_eps_bc(r1));
            } else {
                result.extend(mk_eps_bc(r2));
            }
            result
        }
        ARegex::Seq(bs, r1, r2) => {
            let mut result = bs.clone();
            result.extend(mk_eps_bc(r1));
            result.extend(mk_eps_bc(r2));
            result
        }
        ARegex::Star(bs, _) => {
            let mut result = bs.clone();
            result.push(true);
            result
        }
    }
}
```

Figure 5.9: `mk_eps_bc` implements (5.8). `Phi` and `Lit` panic, since neither can be nullable.

5.2.5 Annotated Derivatives

$ri\backslash_b l$ has the same case structure as the plain derivative (3.6), but extended with bits threaded through the rules. Each rule corresponds to its unannotated counterpart:

$$
\begin{aligned}
\emptyset\backslash_b x &= \emptyset && (\mathsf{B}_\emptyset)\\
(bs@\varepsilon)\backslash_b x &= \emptyset && (\mathsf{B}_\varepsilon)\\
(bs@c)\backslash_b x &= \begin{cases} bs@\varepsilon & c = x \\ \emptyset & \text{otherwise} \end{cases} && (\mathsf{B}_{\text{lit}})\\
(bs@ri_1 \oplus ri_2)\backslash_b x &= (bs@(ri_1\backslash_b x) \oplus (ri_2\backslash_b x)) && (\mathsf{B}_+)\\
(bs@ri_1\ ri_2)\backslash_b x &= (bs@(ri_1\backslash_b x)\ ri_2) && (\mathsf{B}_{\text{seq}}),\ \varepsilon \notin L(ri_1)\\
(bs@ri_1\ ri_2)\backslash_b x &= bs@\Big(([\,]@(ri_1\backslash_b x)\ ri_2) \oplus \mathit{fuse}(\mathit{mkEpsBC}_{ri_1}, ri_2\backslash_b x)\Big) && (\mathsf{B}^n_{\text{seq}}),\ \varepsilon \in L(ri_1)\\
(bs@ri^*)\backslash_b x &= (bs@\mathit{fuse}([0], ri\backslash_b x)\ ([\,]@ri^*)) && (\mathsf{B}_*).
\end{aligned}
\tag{5.9}
$$

The rules B_+, B_{seq}, and B_* preserve the outer bs, since derivation changes the residual but not the choices already recorded above it.

The nullable sequence case $\mathsf{B}^n_{\text{seq}}$ is the exception. Since ri_1 is nullable, x may be matched

by either factor, so the rule introduces a new alternation. Its left branch continues into ri_1, while its right branch recurses into ri_2, with the empty match's bits extracted by *mkEpsBC* and fused onto ri_2's derivative, so the choice inside ri_1 survives. This performs during the forward pass what $\mathsf{I}^R_{\text{seq}}$ performs during injection. The outer *bs* attaches to the new alternation, since the alternation as a whole is what $ri_1\ ri_2$ derived into.

In Rust. Figure 5.10 implements (5.9), following the case structure of plain `deriv` with bits threaded through the constructors. The base cases `Phi`, `Eps`, and `Lit` match plain `deriv`, except that `Lit` carries its bit-list onto the resulting `Eps`.

The `Alt` arm derives both operands and rebuilds the alternation with the outer `bs` on the wrapper, not pushed into either branch. This matches B_+, where the recorded choice belongs to the alternation itself and derivation does not change it, so the bit stays on the outer constructor until the decoder reads it.

The `Seq` arm reflects B_{seq} and $\mathsf{B}^n_{\text{seq}}$. In the non-nullable case, `r1` must consume the character, so only its derivative is needed and `r2` moves directly into the rebuilt sequence. In the nullable case, both alternatives are possible. The bits of `r1`'s empty match are extracted by `mk_eps_bc` *before* `r1` is consumed by `deriv_bc`, since after that call the original `r1` is gone. Those bits are then fused onto the derivative of `r2`, so the choice made inside `r1` reaches the surviving branch. The left branch is wrapped in a fresh `Seq` with an empty bit-list, since the outer `bs` moves to the new `Alt` wrapper, matching $\mathsf{B}^n_{\text{seq}}$. Two clones are therefore needed. `r2.clone()` provides one copy for the left branch and one for its own derivative, and `nullable_bc` borrows `r1` before `deriv_bc` consumes it.

The `Star` arm follows the same pattern as the nullable case of `Seq`. The body is needed both for its derivative and for the reconstructed star, so `r.clone()` provides the second copy, and the reconstructed star is wrapped in a fresh `Star` with an empty bit-list for the same reason as the left branch above. The leading `[false]` prepended to the derivative marks one more iteration, matching the outermost bit in B_*

```
pub fn deriv_bc(ri: ARegex, l: char) -> ARegex {
    match ri {
        // ...
        ARegex::Alt(bs, r1, r2) => {
            let d1 = deriv_bc(*r1, l);
            let d2 = deriv_bc(*r2, l);
            ARegex::Alt(bs, Box::new(d1), Box::new(d2))
        }
        ARegex::Seq(bs, r1, r2) => {
            if nullable_bc(&r1) {
                let eps_bits = mk_eps_bc(&r1);
                let d1 = deriv_bc(*r1, l);
                let d2 = deriv_bc(*r2.clone(), l);
                let left_branch = ARegex::Seq(vec![], Box::new(d1), r2);
                let right_branch = fuse(&eps_bits, d2);
                ARegex::Alt(bs, Box::new(left_branch), Box::new(right_branch))
            } else {
                let d1 = deriv_bc(*r1, l);
                ARegex::Seq(bs, Box::new(d1), r2)
            }
        }
        ARegex::Star(bs, r) => {
            let d = deriv_bc(*r.clone(), l);
            let fused = fuse(&[false], d);
            let new_star = ARegex::Star(vec![], r);
            ARegex::Seq(bs, Box::new(fused), Box::new(new_star))
        }
    }
}
```

Figure 5.10: Annotated derivative `deriv_bc`, implementing $\mathsf{B}_{\emptyset}$, B_{ε}, B_{lit}, B_{+}, $\mathsf{B}_{\text{seq}}/\mathsf{B}^{n}_{\text{seq}}$, B_{*}.

5.2.6 Decoding Bits

$decode_r$ reconstructs a `ParseTree` from the bit sequence produced by $mkEpsBC$. It walks the original, unannotated r and uses one bit per alternation to choose a branch:

$$decode_r(bs) = v \quad \text{if } decodeInner_r(bs) = (v, [\,])$$

$$\begin{aligned}
decodeInner_{\varepsilon}(bs) &= (Empty,\ bs)\\
decodeInner_{x}(bs) &= (Char(x),\ bs)\\
decodeInner_{r_1+r_2}(0{:}bs) &= (Left(v),\ bs_1) \quad \text{if } decodeInner_{r_1}(bs) = (v, bs_1)\\
decodeInner_{r_1+r_2}(1{:}bs) &= (Right(v),\ bs_1) \quad \text{if } decodeInner_{r_2}(bs) = (v, bs_1)\\
decodeInner_{r_1 r_2}(bs) &= (Pair(v_1, v_2),\ bs_2)\\
&\qquad \text{if } decodeInner_{r_1}(bs) = (v_1, bs_1) \text{ and } decodeInner_{r_2}(bs_1) = (v_2, bs_2)\\
decodeInner_{r^*}(1{:}bs) &= (Star([\,]),\ bs)\\
decodeInner_{r^*}(0{:}bs) &= (Star(v{:}vs),\ bs_2)\\
&\qquad \text{if } decodeInner_{r}(bs) = (v, bs_1) \text{ and } decodeInner_{r^*}(bs_1) = (Star(vs), bs_2)
\end{aligned} \tag{5.10}$$

The annotated expression has been reduced to a bare bit sequence, so $decode_r$ walks the original r and uses the bits only to choose between alternation branches. Its worker *decodeInner* returns both the tree for the current subexpression and the unconsumed bits, so the recursion can continue at the correct position.

In Rust. Figure 5.11 shows `decode` and the compound arms of its worker `decode_inner`. The worker returns a pair, `(ParseTree, &[bool])`, holding the tree for the current subexpression together with whatever bits remain, which is what lets the recursion continue at the correct position in the bit-string. Rust's slice patterns make the leading-bit test direct, since `[false, rest @ ..]` and `[true, rest @ ..]` both check the bit and name the remainder in one pattern.

```
pub fn decode(r: &Regex, bs: &[bool]) -> ParseTree {
    let (v, rest) = decode_inner(r, bs);
    assert!(rest.is_empty(),
        "decode: {} leftover bit(s) after decoding {:?}", rest.len(), r);
    v
}

fn decode_inner<'a>(r: &Regex, bs: &'a [bool]) -> (ParseTree, &'a [bool]) {
    match r {
        // ...
        Regex::Alt(r1, r2) => match bs {
            [false, rest @ ..] => {
                let (v, remaining) = decode_inner(r1, rest);
                (ParseTree::Left(Box::new(v)), remaining)
            }
            [true, rest @ ..] => {
                let (v, remaining) = decode_inner(r2, rest);
                (ParseTree::Right(Box::new(v)), remaining)
            }
            // ...
        },
        Regex::Seq(r1, r2) => {
            let (v1, after_r1) = decode_inner(r1, bs);
            let (v2, after_r2) = decode_inner(r2, after_r1);
            (ParseTree::Pair(Box::new(v1), Box::new(v2)), after_r2)
        }
        Regex::Star(r1) => {
            let mut items = Vec::new();
            let mut remaining = bs;
            loop {
                match remaining {
                    [true, rest @ ..] => { remaining = rest; break; }
                    [false, rest @ ..] => {
                        let (v, after) = decode_inner(r1, rest);
                        items.push(v);
                        remaining = after;
                    }
                    // ...
                }
            }
            (ParseTree::Star(items), remaining)
        }
    }
}
```

Figure 5.11: `decode` and the `Alt`, `Seq`, and `Star` arms of `decode_inner`.

`decode` calls the worker and asserts that no bits remain, which is exactly the side condition that $decode_r$ be defined only when the worker consumes the entire bit-string. The `Alt` arm tests the leading bit with a slice pattern and recurses into the branch it names, returning the remaining bits unchanged, since only the leading bit belongs to the alternation.

The `Seq` arm shows why the worker returns a pair. It recurses into the first factor, receives both a tree and the bits the first factor left unconsumed, and passes those bits to the second factor. The second factor then returns its own tree and the bits that follow it, which are the bits for any enclosing expression. Returning only the tree would leave the second recursion with no way to know where to start.

The `Star` arm loops, since the number of iterations is not known in advance and is itself encoded in the bits. Each leading `false` corresponds to one iteration, so the arm recurses into the body and then continues the loop from whatever bits the body left behind. The loop ends at the leading `true` that `mk_eps_bc` appended. The `Eps`, `Lit`, and `Phi` arms are elided, since the first two consume no bits and the third is unreachable.

5.2.7 Parser

Figure 5.12 implements the bit-coded driver. It internalizes r, applies `deriv_bc` and `simp` to each input character, checks nullability and, if successful, calls `mk_eps_bc` and `decode`.

```
pub fn parse_bitcoded_loop(input: &str, r: &Regex) -> Option<ParseTree> {
    let mut ri = internalize(r);
    for l in input.chars() { ri = simp(deriv_bc(ri, l)); }
    if !nullable_bc(&ri) { return None; }
    Some(decode(r, &mk_eps_bc(&ri)))
}
```

Figure 5.12: `parse_bitcoded_loop`: the bit-coded parser implementation.

Unlike the direct implementation, which has a growing `Vec<Regex>` for backward reconstruction, the bit-coded implementation needs only the current `ri`. The recursive entry point has the same three-step computation but differs only in how intermediate expressions are stored.

5.2.8 Worked Example: Ambigious Expression

Consider the same input `ab` and regular expression $r_0 = (a \mid ab)(b \mid \varepsilon)$.

Internalize. Before consuming input, *internalize* tags each alternative with [0] for the left branch and [1] for the right branch:

$$\mathit{internalize}(r_0) = ri_0 = [\,]@\Big([\,]@([0]@a \oplus [1]@(a\,b)) \cdot [\,]@([0]@b \oplus [1]@\varepsilon)\Big). \tag{5.11}$$

Forward pass. The two input symbols are consumed by successive bit-threaded derivatives with *simp* applied after each step. The first derivative uses B_{seq}. After consuming `a`, the first factor becomes nullable, so the second derivative uses $\mathsf{B}^{n}_{\text{seq}}$, which propagates bits from the empty match of the first factor into the derivative of the second:

$$\begin{aligned} ri_0 &\xrightarrow[\mathsf{B}_{\text{seq}}]{a} ri_1 = [\,]@\Big([\,]@([0]@\varepsilon \oplus [1]@b) \cdot [\,]@([0]@b \oplus [1]@\varepsilon)\Big) \\ ri_1 &\xrightarrow[\mathsf{B}^{n}_{\text{seq}}]{b} ri_2 = [\,]@\Big([1]@([0]@b \oplus [1]@\varepsilon) \oplus [0,0]@\varepsilon\Big). \end{aligned} \tag{5.12}$$

The [1] records the right alternative ab of the first factor. The $[0, 0]$ on the second branch results from fusing bits during the derivative. Unlike the direct construction, where *inj* recovers this information later, the bit-coded method propagates it during the forward pass.

Extracting and decoding the bits. Both disjuncts of ri_2 are nullable, so *mkEpsBC* applies the left-preference rule and selects the left disjunct. Inside, b is not nullable while ε is, so EB_+^R contributes its [1] tag:

$$mkEpsBC(ri_2) \xrightarrow{\mathsf{EB}_+^L} mkEpsBC\Big([1]@([0]@b \oplus [1]@\varepsilon)\Big) \xrightarrow{\mathsf{EB}_+^R} [1, 1]. \tag{5.13}$$

Unlike *mkEps*, which constructs a partial parse tree, *mkEpsBC* extracts only the committed choices as a bit-string. The decoder then interprets $[1, 1]$ against the original r_0: the first bit selects the right branch of $a + ab$ (producing ab), and the second selects the right branch of $b + \varepsilon$:

$$decode_{r_0}([1, 1]) = \Big(Right(a, b), Right(\,)\Big) = v_0. \tag{5.14}$$

Flattening validates the result:

$$flatten(v_0) = a \cdot b \cdot \varepsilon = \mathtt{ab}. \tag{5.15}$$

The bit-coded construction produces the same parse tree as the direct construction but stores only the final annotated expression ri_2, not the intermediate expressions r_0 and r_1.

6 Partial-Derivative-Based Parser

Chapter 5 presented two derivative-based methods for constructing POSIX parse trees. Both keep the structure of the original expression throughout the forward pass, defer disambiguation to a final *mkEps* step and, in the standard variant, reconstruct the tree in a backward pass. This chapter presents the partial-derivative-based alternative of Sulzmann and Lu [Sulzmann and Lu, 2012]. Instead of one residual per step, the partial derivative produces a set of states, each pairing a residual expression with the reconstruction information for one path through the original expression. The parser maintains this set as an NFA state set, transitions every state on the next input character, and selects the first accepting state at the end.

The defining difference from the derivative-based parsers is the disambiguation policy. Because partial derivatives break apart the structure of the original expression—alternations are split into separate states, and concatenations are expanded without preserving the outer shape—the NFA traversal yields the greedy leftmost parse tree rather than the POSIX one. On the running ambiguous example from Chapter 5, $r_0 = (a+ab)(b+\varepsilon)$ with input $\mathtt{ab}$, the derivative-based parsers return $(\mathit{Right}(a, b), \mathit{Right}())$, whereas the partial-derivative parsers return $(\mathit{Left}(a), \mathit{Left}(b))$.

As in Chapter 5, two variants are presented. The standard partial-derivative parser attaches an injection closure to each configuration, composing reconstruction steps during the forward pass. The bit-coded partial-derivative parser replaces these closures with bit-sequences carried alongside the residuals. Both compute the same greedy parse tree; they differ only in how the reconstruction information is represented.

6.1 The Standard Partial-Derivative Parser

Consider again the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ with input $w = \mathtt{ab}$. The partial derivative of r_0 with respect to $\mathtt{a}$ produces two states, one for each way $\mathtt{a}$ can be consumed:

$$\begin{aligned} \mathit{pderiv}(r_0, \mathtt{a}) = \big\{ &(\; b + \varepsilon, \; \lambda(_, v_2).\ (\mathit{Left}\, a, \; v_2) \;), \\ &(\; b\,(b + \varepsilon), \; \lambda((v_1, v_2), v_2').\ (\mathit{Right}(a, v_2), \; v_2') \;) \big\}. \end{aligned} \tag{6.1}$$

The first state comes from the left branch a of the first factor: the character is matched by a, and the injection places it in the left injection of the alternation. The second state comes from the right branch ab: the character is matched by the leading a of ab, and the injection places it in the right injection, leaving the residual b to be matched later. Two notational conventions appear in the closures. The underscore in the first injection marks

a binder that is never used: the residual is ε, whose only tree carries no information, so the closure discards its argument. The leading ε inside the second state's residual has already been removed by the smart constructor, since $\varepsilon\, b$ and b describe the same language; the closure's pair-binder reflects the shape the residual would have had before the simplification.

Transitioning both states on the second character `b` gives a new state set:

$$\begin{aligned} \mathit{pderiv}(r_0, \mathtt{a}, \mathtt{b}) = \big\{ &(\ \varepsilon,\ \lambda_.\ (\mathit{Left}\, a,\ \mathit{Left}\, b)\ ), \\ &(\ b + \varepsilon,\ \lambda(_, v_2).\ (\mathit{Right}(a, b),\ v_2)\ ) \big\}. \end{aligned} \tag{6.2}$$

The first configuration has residual ε and is accepting; the second has residual $b + \varepsilon$, which is also accepting. Selection walks the state set in priority order and returns the first accepting configuration. Applying its injection to $\mathit{mkEps}_\varepsilon = ()$ yields $(\mathit{Left}\, a, \mathit{Left}\, b)$, the greedy parse tree.

Figure 6.1 shows the two transitions and the selection on the running example. Each arrow is labelled by the function that performs the step: $\mathit{pderiv}(-, \mathtt{a})$ and $\mathit{pderiv}(-, \mathtt{b})$ advance the state set by one character, *select* picks the first accepting configuration.

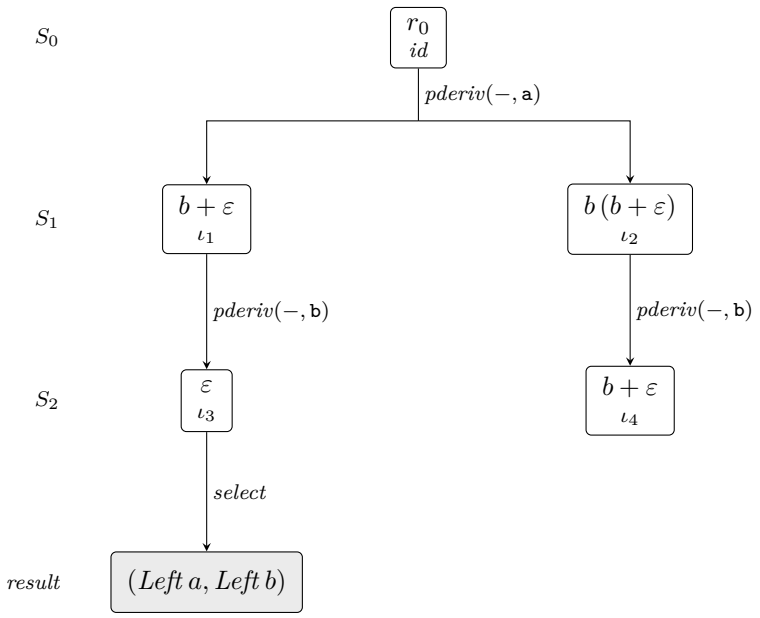


Figure 6.1: The standard partial-derivative parser on the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ with input `ab`. Each box is a configuration: a residual expression and an injection ι_i. The initial state set S_0 contains (r_0, id). Applying $\mathit{pderiv}(-, \mathtt{a})$ advances every configuration by the character `a`, producing S_1 with two configurations. Applying $\mathit{pderiv}(-, \mathtt{b})$ advances S_1 to S_2, in which the first configuration has residual ε and is accepting. Selecting that configuration's injection and applying it to $\mathit{mkEps}_\varepsilon = ()$ yields the greedy parse tree $(\mathit{Left}\, a, \mathit{Left}\, b)$.

The order of configurations in the state set encodes the greedy preference. When a state produces several successors, they are appended in the order the rules generate them, with the left alternative first. When two configurations share a residual, *nub* keeps the first, discarding the later one and its injection. Both mechanisms favour the leftmost path through the expression.

6.1.1 Partial Derivatives with Injections

A partial derivative of an expression r with respect to a character x is not a single expression but a set of pairs. Each pair consists of a residual expression r' whose language is one component of the left quotient, together with an injection closure *inj* that maps a parse tree for r' back to a parse tree for r. The function *pderiv*, following [Sulzmann and Lu, 2012, Fig. 10], is defined recursively:

$$
\begin{aligned}
\mathit{pderiv}(\phi, x) &= \emptyset \\
\mathit{pderiv}(\varepsilon, x) &= \emptyset \\
\mathit{pderiv}(c, x) &= \begin{cases} \{(\varepsilon,\ \lambda_.\, c)\} & c = x \\ \emptyset & \text{otherwise} \end{cases} \\
\mathit{pderiv}(r_1 + r_2, x) &= \{\, (r_1',\ \lambda v.\, \mathit{Left}\,(\mathit{inj}_1\, v)) \mid (r_1', \mathit{inj}_1) \in \mathit{pderiv}(r_1, x) \,\} \\
&\quad \cup \{\, (r_2',\ \lambda v.\, \mathit{Right}\,(\mathit{inj}_2\, v)) \mid (r_2', \mathit{inj}_2) \in \mathit{pderiv}(r_2, x) \,\} \\
\mathit{pderiv}(r_1\, r_2, x) &= \{\, (r_1'\, r_2,\ \lambda(v_1, v_2).\, (\mathit{inj}_1\, v_1,\ v_2)) \mid (r_1', \mathit{inj}_1) \in \mathit{pderiv}(r_1, x) \,\} \\
&\quad \cup \begin{cases} \{\, (r_2',\ \lambda v.\, (\mathit{mkEps}_{r_1},\ \mathit{inj}_2\, v)) \mid (r_2', \mathit{inj}_2) \in \mathit{pderiv}(r_2, x) \,\} & n(r_1) \\ \emptyset & \text{otherwise} \end{cases} \\
\mathit{pderiv}(r^*, x) &= \{\, (r'\, r^*,\ \lambda(v, \mathit{Star}(vs)).\, \mathit{Star}([\mathit{inj}\, v] +\!\!+ vs)) \mid (r', \mathit{inj}) \in \mathit{pderiv}(r, x) \,\}.
\end{aligned}
\tag{6.3}
$$

After expansion, the list is deduplicated by residual, keeping the first occurrence. This is the step that breaks apart the original expression: where the derivative of $r_1 + r_2$ produces a single alternation, the partial derivative produces separate states for each branch, and the outer alternation disappears from the residual.

The injection in each pair records the path taken through the expression. Rule (P_+) wraps the recursive injection in a *Left* or *Right* node, reinserting the alternation choice. Rule (P_{seq}) for a non-nullable first factor passes the outer pair through unchanged, with the recursive injection applied to the first component only. When the first factor is nullable, the rule generates two kinds of states: those that continue inside the first factor, and those that skip the first factor entirely, using mkEps_{r_1} to synthesise its empty match. Rule (P_*) reconstructs the current iteration by applying the recursive injection and prepending it to the remaining iterations.

No rules are needed for ϕ or ε beyond their empty results, since neither can consume a character.

In Rust. The recursive definition (6.3) is implemented directly in Rust. The type *Inj* is a reference-counted closure, which allows a single parent injection to be shared when a transition fans out into several states.

```
pub(crate) type Inj = Rc<dyn Fn(ParseTree) -> ParseTree>;

pub(crate) fn pderiv_tree(r: &Regex, x: char) -> Vec<(Regex, Inj)> {
    match r {
        Regex::Eps | Regex::Phi => vec![],

        Regex::Lit(c) => {
            if *c == x {
                vec![(Regex::Eps,
                      Rc::new(move |_| ParseTree::Char(x))) as Inj]
            } else { vec![] }
        }

        Regex::Alt(r1, r2) => {
            let mut out: Vec<(Regex, Inj)> = pderiv_tree(r1, x)
                .into_iter()
                .map(|(r1_, inj1)| {
                    let f: Inj = Rc::new(move |v| {
                        ParseTree::Left(Box::new(inj1(v)))
                    });
                    (r1_, f)
                })
                .collect();
            out.extend(pderiv_tree(r2, x).into_iter().map(|(r2_, inj2)| {
                let f: Inj = Rc::new(move |v| {
                    ParseTree::Right(Box::new(inj2(v)))
                });
                (r2_, f)
            }));
            nub_tree(out)
        }

        Regex::Seq(r1, r2) => {
            if nullable(r1) {
                let mut out: Vec<(Regex, Inj)> = pderiv_tree(r1, x)
                    .into_iter()
                    .map(|(r1_, inj1)| {
                        let r2c = (**r2).clone();
                        let f: Inj = Rc::new(move |v| match v {
                            ParseTree::Pair(v1, v2) =>
                                ParseTree::Pair(Box::new(inj1(*v1)), v2),
                            _ => panic!("pderiv_tree/Seq: want Pair"),
                        });
                        (Regex::seq(r1_, r2c), f)
                    })
                    .collect();

                let v1_eps = mk_eps(r1);
                out.extend(pderiv_tree(r2, x).into_iter().map(|(r2_, inj2)| {
                    let v1c = v1_eps.clone();
                    let f: Inj = Rc::new(move |v2| {
                        ParseTree::Pair(Box::new(v1c.clone()),
                                        Box::new(inj2(v2)))
                    });
                    (r2_, f)
                }));
                nub_tree(out)
```

The `Alt` arm is where the partial derivative visibly breaks apart the expression: each branch produces its own states, tagged by *Left* or *Right* in the injection. The `Seq` arm's nullable case is the richest: the first factor is consumed by `deriv`-style recursion, and the first factor's empty match is computed once with `mk_eps` and captured by each state from the second factor.

The `Star` arm follows the same pattern, with the injection inserting the reconstructed iteration at the front of the star's list. Note that the residual is built with the plain `Regex::seq` constructor rather than a smart constructor. Keeping the unsimplified shape matters because the injection pattern-matches on the tree's structure; a simplified residual would lose the outer sequence node the injection expects.

6.1.2 State Sets and Selection

The parser maintains a state set of (residual, injection) configurations. For each input character, every configuration in the state set is transitioned with `pderiv_tree`, and the resulting configurations are collected into a new state set. A configuration's parent injection is composed with the transition injection, so that at the end the state set's injections map residuals all the way back to the original expression.

The composition is direct. When a parent configuration (r, inj_p) transitions to (r', inj_s), the new configuration's injection is $\lambda v.\, inj_p(inj_s(v))$.

```
pub(super) fn step_frontier(
    frontier: Vec<(Regex, Inj)>, x: char
) -> Vec<(Regex, Inj)> {
    let mut out = Vec::new();
    for (r, inj_parent) in frontier {
        for (r_next, inj_step) in pderiv_tree(&r, x) {
            let parent = inj_parent.clone();
            let combined: Inj = Rc::new(move |v| parent(inj_step(v)));
            out.push((r_next, combined));
        }
    }
    out
}
```

Figure 6.3: `step_frontier`: one transition of the NFA state set on a single character. The function name retains the older "frontier" terminology; it computes the successor state set $\delta(S, x)$.

After the input is exhausted, selection walks the state set in priority order and returns the tree for the first accepting configuration:

```
pub(super) fn select(frontier: &[(Regex, Inj)]) -> Option<ParseTree> {
    for (r, inj) in frontier {
        if nullable(r) {
            return Some(inj(mk_eps(r)));
        }
    }
    None
}
```

Figure 6.4: `select`: return the tree for the first accepting configuration in the state set.

Because the state set lists configurations in the order the rules generated them, and because `nub_tree` keeps the first of each duplicate residual, selection favours the configuration that committed earliest to the leftmost alternative. This is the greedy leftmost policy.

6.1.3 Parser

The parser composes `step_frontier` over the input and then calls `select`.

```
pub fn parse_pderiv_std(input: &str, r: &Regex) -> Option<ParseTree> {
    let mut frontier: Vec<(Regex, Inj)> =
        vec![(r.clone(), identity_inj())];
    for c in input.chars() {
        frontier = step_frontier(frontier, c);
        if frontier.is_empty() {
            return None;
        }
    }
    select(&frontier)
}
```

Figure 6.5: The standard partial-derivative parser.

The initial state set has a single configuration: the original expression with the identity injection. The early return on an empty state set corresponds to the case where no configuration can consume the current character, so the input cannot be matched. The final `select` returns `None` if no configuration is accepting, and otherwise returns the tree for the first accepting one.

6.1.4 Worked Example: Ambiguous Expression

Recall the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ on input `ab`.

Forward pass. The initial state set holds a single configuration:

$$S_0 = \big[(r_0,\ id)\big].$$

Transitioning with respect to `a`:

$$pderiv(r_0, \mathtt{a}) = pderiv\Big(Alt(a, ab), \mathtt{a}\Big) \cdot Alt(b, \varepsilon).$$

Inside the first factor:

$$\begin{aligned} \mathit{pderiv}(a, \mathtt{a}) &= \Big[(\varepsilon,\ \lambda_.\, a)\Big] \\ \mathit{pderiv}(ab, \mathtt{a}) &= \Big[(\mathit{Seq}(\varepsilon, b),\ \lambda(v_1, v_2).\, (a, v_2))\Big]. \end{aligned}$$

After the alternation rule wraps each state with *Left* or *Right*:

$$\begin{aligned} \mathit{pderiv}(a + ab, \mathtt{a}) = \Big[&(\varepsilon,\ \lambda v.\, \mathit{Left}\,(\lambda_.\, a)(v)), \\ &(\mathit{Seq}(\varepsilon, b),\ \lambda(v_1, v_2).\, \mathit{Right}\,(a, v_2))\Big]. \end{aligned}$$

The outer sequence rule combines this with the untouched second factor:

$$\begin{aligned} S_1 = \Big[&(\mathit{Seq}(\varepsilon, b + \varepsilon),\ \lambda(v_1, v_2).\, (\mathit{Left}\, a,\ v_2)), \\ &(\mathit{Seq}(\mathit{Seq}(\varepsilon, b), b + \varepsilon),\ \lambda((v_1, v_2), v_2').\, (\mathit{Right}(a, v_2),\ v_2'))\Big]. \end{aligned}$$

Transitioning with respect to `b` steps both configurations. For the first, the left factor ε is nullable, so the sequence rule produces the state that jumps directly into the second factor:

$$\mathit{pderiv}(b + \varepsilon, \mathtt{b}) = \Big[(\varepsilon,\ \lambda v.\, \mathit{Left}\, b)\Big].$$

The second factor's injection inserts the character at the leftmost position of the pair:

$$\mathit{inj}_1(\mathit{Left}\, b) = (\mathit{Left}\, a,\ \mathit{Left}\, b).$$

For the second configuration, the leading $\mathit{Seq}(\varepsilon, b)$ has a non-nullable *Seq* structure and the derivative continues inside b:

$$\mathit{pderiv}(\mathit{Seq}(\varepsilon, b), \mathtt{b}) = \Big[(\varepsilon,\ \lambda_.\, ((), b))\Big],$$

and the outer sequence rule rebuilds the pair, giving the second configuration's residual $b + \varepsilon$. The two resulting state-set entries are:

$$\begin{aligned} S_2 = \Big[&(\varepsilon,\ \lambda_.\, (\mathit{Left}\, a,\ \mathit{Left}\, b)), \\ &(b + \varepsilon,\ \lambda(_, v_2).\, (\mathit{Right}(a, b),\ v_2))\Big]. \end{aligned}$$

Selection. The first configuration's residual is ε, which is nullable, and selection returns immediately:

$$\mathit{select}(S_2) = (\lambda_.\, (\mathit{Left}\, a, \mathit{Left}\, b))(\mathit{mkEps}_\varepsilon) = (\mathit{Left}\, a, \mathit{Left}\, b).$$

Flattening confirms the result:

$$\mathit{flatten}\Big((\mathit{Left}\, a, \mathit{Left}\, b)\Big) = \mathtt{ab}.$$

This is the greedy parse tree. The derivative-based parser from Chapter 5 returns $(\mathit{Right}(a, b), \mathit{Right}())$ for the same input, the POSIX tree. The two trees agree on which word they spell out, but differ on how the input is decomposed: the greedy tree commits to the shorter first alternative a in the first factor, while the POSIX tree commits to the longer alternative ab.

6.2 The Bit-Coded Partial-Derivative Parser

The standard partial-derivative parser carries a closure per configuration, which accumulates the reconstruction path during the forward pass. The bit-coded variant replaces closures with bit-sequences: each configuration's annotation records the choices made so far, and decoding the final bit-sequence against the original expression yields the parse tree.

Transitioning $w = \texttt{ab}$ from $r_0 = (a + ab)(b + \varepsilon)$ gives:

$$\begin{aligned} S_0 &= \big[(r_0,\ [\,])\big] \\ &\xrightarrow{\texttt{a}} S_1 = \big[(b + \varepsilon,\ [\mathit{false}]),\ (b\,(b + \varepsilon),\ [\mathit{true}])\big] \\ &\xrightarrow{\texttt{b}} S_2 = \big[(\varepsilon,\ [\mathit{false}, \mathit{false}]),\ (b + \varepsilon,\ [\mathit{true}])\big]. \end{aligned} \tag{6.4}$$

The first configuration of S_1 records that the character `a` was matched by the left branch of the first factor's alternation. The second configuration records that it was matched by the right branch, leaving the residual b to be consumed later. After the second character, only the first configuration is accepting, and its bits $[\mathit{false}, \mathit{false}]$ decode to $(\mathit{Left}\, a, \mathit{Left}\, b)$.

Figure 6.6 shows this transition together with the selection and decode steps.

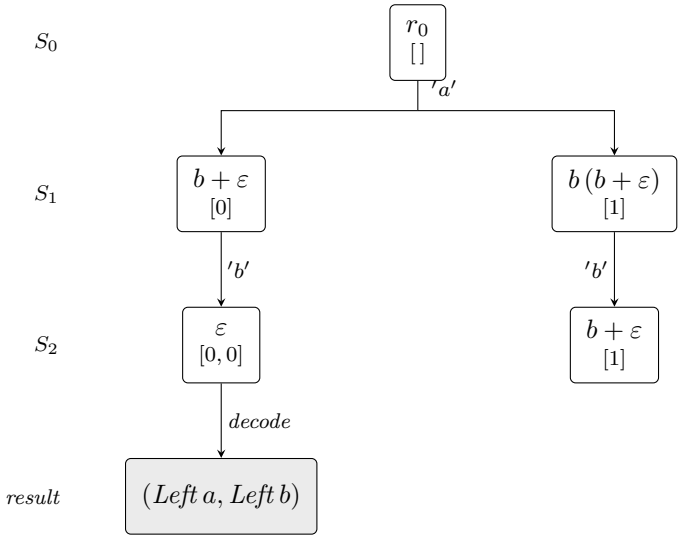


Figure 6.6: Overview of the bit-coded partial-derivative parser on the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ with input `ab`. Each box is a configuration: a residual expression and a bit-sequence, written 0 for *false* and 1 for *true*. Selection takes the first accepting configuration in S_2 and appends *mkEpsBC* of its residual, yielding the bit-string $[0, 0]$, which decode_{r_0} reads against the original expression.

The bit-coded parser keeps the same state-set structure as the standard one. Only the configuration's annotation changes, from a closure to a sequence of bits, and the final step changes from applying an injection to calling *decode*.

6.2.1 Bit-Coded Partial Derivatives

The bit-coded partial derivative $\mathit{pderivBC}(r, x)$ returns a list of (residual, bit-sequence) pairs. It follows the same case structure as (6.3), with one difference: where the standard

version builds an injection closure, the bit-coded version prepends a fixed bit-sequence to the configuration's existing bits.

The bit conventions are inherited from Chapter 5.2. Each alternation contributes a leading *false* for the left branch or a leading *true* for the right branch. Each star iteration contributes a leading *false*, and each star termination contributes a *true*. Concatenation contributes no bits of its own.

$$
\begin{aligned}
\mathit{pderivBC}(\phi, x) &= \emptyset \\
\mathit{pderivBC}(\varepsilon, x) &= \emptyset \\
\mathit{pderivBC}(c, x) &= \begin{cases} \{(\varepsilon,\, [\,])\} & c = x \\ \emptyset & \text{otherwise} \end{cases} \\
\mathit{pderivBC}(r_1 + r_2, x) &= \{\, (r_1',\, [\mathit{false}] +\!\!+\, bs) \mid (r_1', bs) \in \mathit{pderivBC}(r_1, x) \,\} \\
&\quad \cup \{\, (r_2',\, [\mathit{true}] +\!\!+\, bs) \mid (r_2', bs) \in \mathit{pderivBC}(r_2, x) \,\} \\
\mathit{pderivBC}(r_1\, r_2, x) &= \{\, (\mathit{seq}(r_1', r_2),\, bs) \mid (r_1', bs) \in \mathit{pderivBC}(r_1, x) \,\} \\
&\quad \cup \begin{cases} \{\, (r_2',\, \mathit{mkEpsBC}_{r_1} +\!\!+\, bs) \mid (r_2', bs) \in \mathit{pderivBC}(r_2, x) \,\} & n(r_1) \\ \emptyset & \text{otherwise} \end{cases} \\
\mathit{pderivBC}(r^*, x) &= \{\, (\mathit{seq}(r', r^*),\, [\mathit{false}] +\!\!+\, bs) \mid (r', bs) \in \mathit{pderivBC}(r, x) \,\}.
\end{aligned}
\tag{6.5}
$$

As in the standard version, the result is deduplicated by residual, keeping the first occurrence.

The bit-coded sequence rule is the counterpart of the injection rule in the standard version. When the first factor is nullable, the bits recorded inside its empty match are extracted with $\mathit{mkEpsBC}_{r_1}$ and prepended to the second factor's bits, so the choice made inside the first factor survives in the surviving configuration.

In Rust. Figure 6.7 implements (6.5) directly. The base cases and the alternation case mirror the standard version with bit-lists in place of closures. The sequence case inlines both the non-nullable and nullable paths. The star case prepends the leading *false* that marks one more iteration.

```
pub fn pderiv_bc(r: &Regex, x: char) -> Vec<(Regex, Vec<bool>)> {
    match r {
        Regex::Phi | Regex::Eps => vec![],

        Regex::Lit(y) => {
            if *y == x { vec![(Regex::Eps, vec![])] } else { vec![] }
        }

        Regex::Alt(r1, r2) => {
            let mut out: Vec<(Regex, Vec<bool>)> = pderiv_bc(r1, x)
                .into_iter()
                .map(|(r1_, mut bs)| {
                    let mut b = vec![false]; b.append(&mut bs);
                    (r1_, b)
                })
                .collect();
            out.extend(pderiv_bc(r2, x).into_iter().map(|(r2_, mut bs)| {
                let mut b = vec![true]; b.append(&mut bs);
                (r2_, b)
            }));
            nub2(out)
        }

        Regex::Seq(r1, r2) => {
            if nullable(r1) {
                let mut out: Vec<(Regex, Vec<bool>)> = pderiv_bc(r1, x)
                    .into_iter()
                    .map(|(r1_, bs)| (Regex::seq(r1_, *r2.clone()), bs))
                    .collect();

                let eps_bits = mk_eps_bits(r1);
                out.extend(pderiv_bc(r2, x).into_iter().map(|(r2_, bs)| {
                    let mut b = eps_bits.clone(); b.extend(bs);
                    (r2_, b)
                }));
                nub2(out)
            } else {
                pderiv_bc(r1, x)
                    .into_iter()
                    .map(|(r1_, bs)| (smart_seq(r1_, r2), bs))
                    .collect()
            }
        }

        Regex::Star(r1) => {
            let out: Vec<(Regex, Vec<bool>)> = pderiv_bc(r1, x)
                .into_iter()
                .map(|(r1_, mut bs)| {
                    let mut b = vec![false]; b.append(&mut bs);
                    (smart_seq(r1_, &Regex::star(*r1.clone())), b)
                })
                .collect();
            nub2(out)
        }
    }
}
```


Figure 6.7: `pderiv_bc`: one bit-coded partial-derivative transition

The `Seq` arm keeps the outer sequence constructor in the nullable case, matching the standard version's behaviour. It uses `smart_seq` only in the non-nullable case, where the derivative of the first factor is guaranteed to be a proper expression and no outer pair needs to be preserved. This asymmetry mirrors the paper's reference implementation and is needed so that the residual's shape stays compatible with the decoder.

6.2.2 Extracting Bits

When a configuration's residual is nullable, its accumulated bits must be extended with the bits of the residual's empty match. This is the role of $mkEpsBC$, which mirrors $mkEps$ from Chapter 5 but produces a bit-sequence instead of a parse tree.

$$\begin{aligned} mkEpsBC_{\varepsilon} &= [\,] \\ mkEpsBC_{r^*} &= [true] \\ mkEpsBC_{r_1 r_2} &= mkEpsBC_{r_1} +\!\!+ mkEpsBC_{r_2} \\ mkEpsBC_{r_1 + r_2} &= \begin{cases} [false] +\!\!+ mkEpsBC_{r_1} & n(r_1) \\ [true] +\!\!+ mkEpsBC_{r_2} & \text{otherwise.} \end{cases} \end{aligned} \tag{6.6}$$

The alternation rule reproduces the left preference of $mkEps$, emitting $false$ when the left branch is nullable and $true$ otherwise. The star rule emits a single terminating $true$, since a star matching the empty word uses zero iterations.

```
pub fn mk_eps_bits(r: &Regex) -> Vec<bool> {
    match r {
        Regex::Phi => panic!("mk_eps_bits called on Phi"),
        Regex::Lit(c) => panic!("mk_eps_bits called on Lit('{}')", c),
        Regex::Eps => Vec::new(),
        Regex::Alt(r1, r2) => {
            if nullable(r1) {
                let mut bits = vec![false];
                bits.extend(mk_eps_bits(r1));
                bits
            } else {
                let mut bits = vec![true];
                bits.extend(mk_eps_bits(r2));
                bits
            }
        }
        Regex::Seq(r1, r2) => {
            let mut bits = mk_eps_bits(r1);
            bits.extend(mk_eps_bits(r2));
            bits
        }
        Regex::Star(_) => vec![true],
    }
}
```

Figure 6.8: `mk_eps_bits`: the bit-string analogue of `mk_eps` for plain `Regex`.

The `Phi` and `Lit` arms panic, matching the precondition that *mkEpsBC* is applied only to nullable expressions. Neither empty language nor a literal can match the empty word.

6.2.3 State Sets and Selection

State-set transition for the bit-coded parser mirrors the standard version, with bit-list concatenation replacing closure composition:

```
pub(super) fn step_frontier(
    frontier: &[(Regex, Vec<bool>)], x: char
) -> Vec<(Regex, Vec<bool>)> {
    let mut out = Vec::new();
    for (r, bits) in frontier {
        for (r_next, extra) in pderiv_bc(r, x) {
            let mut combined = bits.clone();
            combined.extend(extra);
            out.push((r_next, combined));
        }
    }
    out
}
```

Selection walks the state set in priority order and returns the bits of the first accepting configuration, extended with *mkEpsBC* of the residual:

```
pub(super) fn select_bits(
    frontier: &[(Regex, Vec<bool>)]
) -> Option<Vec<bool>> {
    for (r, bits) in frontier {
        if nullable(r) {
            let mut result = bits.clone();
            result.extend(mk_eps_bits(r));
            return Some(result);
        }
    }
    None
}
```

The resulting bit-sequence is interpreted by the shared `decode` function from Chapter 5.2, which walks the original expression and consumes one bit per alternation and per star iteration.

6.2.4 Parser

The parser composes `step_frontier` over the input, checks that the state set is non-empty, and calls `select_bits` followed by `decode`.

```
pub fn parse_pderiv_bc(input: &str, r: &Regex) -> Option<ParseTree> {
    let mut frontier = vec![(r.clone(), Vec::new())];
    for c in input.chars() {
        frontier = step_frontier(&frontier, c);
        if frontier.is_empty() {
            return None;
        }
    }
    let bits = select_bits(&frontier)?;
    Some(decode(r, &bits))
}
```

Figure 6.9: The bit-coded partial-derivative parser.

The driver has the same structure as the standard one. Only the configuration type and the final step differ: bits and `decode` in place of closures and injection application.

6.2.5 Worked Example: Ambiguous Expression

Consider again the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ on input `ab`.

Forward pass. The initial state set holds a single configuration with empty bits:

$$S_0 = \Big[(r_0,\ [\,])\Big].$$

Transitioning by `a`, the first factor is not nullable, so *pderivBC* recurses into it:

$$\begin{aligned} \mathit{pderivBC}(a + ab, \mathtt{a}) &= \mathit{pderivBC}(a, \mathtt{a}).\mathit{map}(+[\mathit{false}]) \cup \mathit{pderivBC}(ab, \mathtt{a}).\mathit{map}(+[\mathit{true}]) \\ &= \Big[(\varepsilon,\ [\mathit{false}]),\ (b,\ [\mathit{true}])\Big]. \end{aligned}$$

The sequence rule then places the second factor after each derivative:

$$S_1 = \Big[(b + \varepsilon,\ [\mathit{false}]),\ (b\,(b + \varepsilon),\ [\mathit{true}])\Big].$$

Transitioning by `b`, both configurations are stepped. The first configuration reaches the second factor directly:

$$\mathit{pderivBC}(b + \varepsilon, \mathtt{b}) = \Big[(\varepsilon,\ [\mathit{false}])\Big],$$

giving bits $[\mathit{false}, \mathit{false}]$. The second configuration's derivative comes from b inside the leading sequence:

$$\mathit{pderivBC}(b, \mathtt{b}) = \Big[(\varepsilon,\ [\,])\Big],$$

and the outer sequence rule reconstructs the residual $b + \varepsilon$ with the bits unchanged:

$$S_2 = \Big[(\varepsilon,\ [\mathit{false}, \mathit{false}]),\ (b + \varepsilon,\ [\mathit{true}])\Big].$$

Selection and decoding. The first configuration is accepting, so *selectBits* returns its bits extended with $mkEpsBC_{\varepsilon} = [\,]$:

$$selectBits(S_2) = [false, false].$$

Decoding these bits against r_0:

$$decode_{r_0}\Big([false, false]\Big) = \Big(Left(a), Left(b)\Big).$$

The first *false* selects the left branch of the first factor's alternation, committing to the shorter alternative a. The second *false* selects the left branch of the second factor's alternation, committing to the shorter alternative b. The resulting tree is the greedy parse tree, identical to the one produced by the standard partial-derivative parser.

Flattening validates the result:

$$flatten\Big((Left(a), Left(b))\Big) = \texttt{ab}.$$

Both partial-derivative parsers therefore produce the same tree on this example, and both differ from the derivative-based parsers of Chapter 5, which commit to the longer first alternative and return $(Right(a, b), Right())$.

7 External versus Internal Syntax

The four parsers of Chapters 5 and 6 all operate on the six-constructor core `Regex` from Chapter 2. Real patterns are written in a richer surface language. A rule authored by a Snort or SpamAssassin contributor uses shorthand classes, bounded repetition, alternation over character ranges, and a flag suffix — none of which appear in the core language.

Bridging the two is the job of the frontend. It is the layer that takes a pattern string, decides what the pattern means, and produces the `Regex` the parsers consume. This chapter explains that layer in three stages: what intermediate representation it produces (Section 7.2), how it parses a string into that representation (Section 7.3), and how it lowers that representation to `Regex` (Section 7.5). It closes with a treatment of PCRE flags (Section 7.4) and a measurement of the translation's own cost (Section 7.6).

7.1 The Interface and Its Design Choices

Two functions form the public interface. The first, used by the CLI and by any code that has a bare pattern, is

```
parse_pattern(s: &str) -> Result<Regex, String>
```

The second, used by the dataset pipeline and by anything consuming a PCRE rule, additionally handles a `/PATTERN/FLAGS` wrapper:

```
parse_pcre_rule(s: &str) -> Result<Regex, String>
```

Both return either a translated `Regex` or a rejection reason. Three design decisions shape everything below.

Full-string matching only. A translated pattern matches the entire input or nothing. There is no search-padding, no substring search, and no anchor handling. This choice is what makes the four parsers of the previous chapters directly usable: each decides membership in the pattern's own language, without a notion of "found somewhere in the input". It is also the reason anchors are rejected rather than translated. A source pattern beginning with `^` or ending with `$` means "match at the start/end of the buffer" under the source tool's own semantics, and no lowering into a full-string matcher preserves that meaning — except in the one case where both anchors wrap the whole pattern, which is provably redundant.

A fixed working alphabet. Shorthand classes, negated classes, and the dot metacharacter all expand to alternations over a fixed set of 98 characters: printable ASCII plus tab, newline, and carriage return. A literal character written directly in the pattern is not

bounded by this set — the core `Regex::Lit` carries a full Unicode `char` — so `café` still matches the string `"café"`, but `\W` cannot match a non-ASCII character, since the alphabet has no such character to produce. The size consequences of this choice are quantified in Section 7.6.

Rejection over approximation. When a construct cannot be translated without changing what the pattern denotes, the frontend rejects the pattern rather than silently mismatching it. The rejected constructs are exactly the ones no derivative or partial-derivative parser can consume without extension: backreferences, lookaround assertions, word boundaries, and anchors outside the recoverable top-level `^...$` wrap. The two remaining flags, `m` and `x`, are rejected for the same reason: `m` changes what the two anchors mean, and the frontend has no anchors to begin with; `x` changes how whitespace is interpreted, which this parser does not model.

Everything outside this small set is accepted and translated. The remainder of this chapter explains how.

7.2 The External AST

The frontend parses into an intermediate type, `ExtPat`, which sits between the surface string and the core `Regex`. It carries enough structure to preserve what the source pattern says — precedence, grouping, quantifiers, and character classes as units — without committing to any particular lowering. The 19 variants fall into four groups.

Group	Variant	Surface syntax
Structural	`Empty`	empty branch of `a|` or `(|b)`
	`Concat`	`abc`
	`Or`	`a|b|c`
	`Group`	`( ... )`
	`GroupNonMarking`	`(?: ... )`
Quantifier	`Opt`	`r?`
	`Plus`	`r+`
	`Star`	`r*`
	`Bound`	`r{m,n}`
Class	`Any`	`[abc]`
	`NoneOf`	`[^abc]`
	`Escape`	`\d`, `\w`, `\s`, ...
	`Char`	a single literal
Assertion	`Carat`	`^`
	`Dollar`	`$`
	`Dot`	`.`
	`WordBoundary`	`\b`, `\B`
	`Eps`	ε
	`Never`	$\emptyset$

Table 7.1: The 19 `ExtPat` variants, grouped by role. The Assertion group is parsed but rejected by `translate`, except for `Dot`, which lowers to the full alphabet.

Two properties of `ExtPat` matter for the rest of the chapter. First, the quantifier variants are single-level: `r*` carries its atom, but not a chain. Chained quantifiers such as `a**` are

not produced by the parser, since PCRE treats them as syntax errors. Second, the assertion variants are first-class. Parsing does not reject them; it produces an `ExtPat::Carat` or `ExtPat::WordBoundary` and leaves the rejection to `translate`. The reason is diagnostic: a pattern that fails at translation with the message "anchor is not supported" is more informative to a corpus analyst than one that fails at parse time with "unexpected `^`".

The deliberate asymmetry between parse and translate — parse succeeds on more than translate accepts — is what makes the anchor shape distribution of Chapter 8 possible. Without it, the pipeline could only say how many patterns failed, not why.

7.3 Parsing: String to ExtPat

Parsing is a single recursive-descent pass, one function per level of the grammar. Precedence is fixed by the call chain.

7.3.1 Precedence and associativity

Three levels, from loosest to tightest binding:

1. alternation (`|`), left-associated;
2. concatenation (implicit), left-associated;
3. postfix quantifiers (`?`, `+`, `*`, `{m,n}`), at most one per atom.

So `abc*` parses as `ab(c*)` and `a|b|c` parses as `(a|b)|c`. The parser never relies on these associations for correctness — `translate` produces the same `Regex` for any equivalent bracketing of an associative operator — but the association is observable in the resulting `Regex` tree, and Chapter 8's growth measurements depend on it.

A trailing `?` after a quantifier (`r*?`, `r+?`, `r??`, `r{m,n}?`) is accepted and discarded. Laziness has no meaning under full-string matching: the pattern matches the whole input or it does not, and no choice of match order is observable.

7.3.2 Accepted surface syntax

Beyond the core constructs already listed in Table 7.1, the parser accepts several forms from the PCRE family.

Groups. `( ... )` and `(?: ... )` both parse; the distinction is preserved in `ExtPat` but erased at translation. Named groups `(?<name>...)` and `(?P<name>...)` are accepted with the name dropped. The name is not recoverable from the resulting `Regex`, since `Regex` has no capture structure, and the parser does not attempt to preserve it.

Character classes. A class body is read as a sequence of members. A `-` between two literal characters denotes a range; a leading or trailing `-` is a literal member, and so is a leading `]`. Shorthand escapes inside a class (`\d`, `\w`, `\s`, and their capitalised negations) expand inline as separate members. A shorthand cannot be a range endpoint: `[a-\d]` is rejected.

Escapes outside a class. `\n`, `\t`, `\r` produce the corresponding control characters; `\xHH` produces the byte given by two hex digits; `\0` produces an octal escape (up to two further

octal digits). A single digit `\1`–`\9` is rejected, since PCRE resolves those as backreferences using a capture-group count this parser does not track.

Octal escapes inside a class. Inside a class, a `\` followed by an octal digit reads up to three octal digits. This is a real-corpus concern: SpamAssassin rules use patterns such as `[\042\223\224\262\263\271]` to match Windows codepage punctuation, and misreading `\042` as the three literal characters `0`, `4`, `2` would silently change the pattern's language. `\8` and `\9` are not octal digits and are read as literals `8` and `9`, matching PCRE's own behaviour.

Explicit ε and $\emptyset$. The two Unicode tokens are accepted as `ExtPat::Eps` and `ExtPat::Never`. They are extensions for the worked examples of earlier chapters, not part of the PCRE family.

7.3.3 Rejected surface syntax

Rejection falls into two categories: constructs the grammar cannot represent, and constructs the frontend chooses not to represent.

Not representable. Backreferences (`\1`–`\9`, `(?P=name)`), lookahead `(?=...)` and `(?!...)`, and lookbehind `(?<=...)` and `(?<!...)` are rejected at parse time. None of them is a regular-language construct, and the external AST has no constructor for them. Real corpus examples:

```
/^(?P<addr1>.{4})(?P<addr2>.{4})(?P=addr2)(?P=addr1)/Rsi
    (Snort, named backreference)
/(?=(?:(0x[a-f0-9]{2}|0+\d{1,3})\s*?,\s*?)*?\d{1,3}).../Rsi
    (Snort, chained lookahead)
```

Representable but rejected. Word boundaries `\b` and `\B`, and anchors `^` and `$`, parse successfully into `ExtPat` but are rejected by `translate`. Both are position-dependent: their meaning depends on the boundary between the matched substring and the surrounding buffer, which a full-string matcher has no way to model. Rejecting them at parse time would have worked just as well, but the parse-time success lets the reject message name the construct. Corpus examples:

```
/\bsolusvmc-node\b/          (Snort, word boundary)
/\.ida$/iU                   (Snort, end anchor)
/^Argument\s+\//smi          (Snort, start anchor + m flag)
```

POSIX classes. A class of the form `[:name:]` inside a bracket expression is rejected with a specific message. Without this check, `[[:alpha:]]` would parse as the five literal characters `[`, `:`, `a`, `l`, `p`, followed by `h`, `a`, and a closing bracket — a pattern that denotes a completely different language and that would have been accepted silently. The rejection is a correctness measure, not a stylistic choice.

7.4 PCRE Flags

A pattern passed to `parse_pcre_rule` may be wrapped in a `/PATTERN/FLAGS` form. The frontend reads the flag suffix and sorts the characters into three groups: honoured, rejected, and ignored.

Flag	Group	Effect
`i`	honoured	case-fold every literal letter
`s`	honoured	`.` includes newline
`m`	rejected	would change anchor meaning this frontend lacks
`x`	rejected	would change whitespace meaning this parser lacks
`R`	ignored	positioned a search this frontend does not perform
`A`	ignored	anchored a search this frontend does not perform
`E`	ignored	tuned a `$` this frontend rejects
`U`, `H`, `P`, `C`, ...	ignored	named a buffer or normalisation strategy

Table 7.2: The three flag groups. "Ignored" means the pattern parses exactly as it would without the flag.

The distinction between "honoured" and "ignored" is whether the flag changes what the pattern denotes. A buffer selector such as `U` says which region of the input to test against, not what the pattern itself matches; under full-string matching there is only one region. A relative-position flag `R` says how to position a subsequent search, which does not exist in this frontend. Neither of these changes the language the pattern denotes, so both are harmless to ignore. An honoured flag such as `i` does change the language, so it must be applied.

Honoured flags are applied between parse and translate, as preprocessing passes over the `ExtPat`. This is the natural place: the passes are structural rewrites that do not need the pattern's own scope information — `case_fold` turns every literal letter into a two-element `Any`, and `strip_dot_newline` turns every `Dot` into `NoneOf(\n)` — so they can run once on the whole tree.

7.5 Translation: ExtPat to Regex

Translation is a second structural pass, this time over the whole `ExtPat`, producing a `Regex`. Most variants have a direct counterpart: `Concat`, `Or`, `Opt`, `Plus`, and `Star` map to the corresponding core constructors or a one-line desugaring of them. Three parts of the translation are less direct and are described below.

7.5.1 Character classes and the working alphabet

An `ExtPat::Any(cs)` lowers to an alternation over `cs`: one `Alt` node per element, folded left. `NoneOf(cs)` first takes the complement of `cs` against the working alphabet and then does the same. `Dot` lowers to `Any` over the full alphabet. Shorthand escapes (`\d`, `\w`, `\s`, and their capitalised forms) lower through the same path.

The consequence is that `Dot`, the shorthand classes, and any negated class cannot match a character outside the working alphabet. The evaluation's corpus contains a few Snort patterns that include non-ASCII literals; those are unaffected, because a direct literal lowers to a single `Regex::Lit` that carries the full `char`. The working alphabet only constrains the constructs that need to enumerate a set.

7.5.2 Bounds and unrolling

`ExtPat::Bound` is the only quantifier without a direct core construct, since `Regex` has no `{m,n}` form. It lowers by unrolling: `r{m,n}` becomes `m` exact copies of the translated body, followed by `n-m` optional copies. `r{m,}` becomes `m` exact copies followed by a `Star`. Both forms clone the body, so the resulting `Regex` has size proportional to the bound.

Two caps guard this unrolling. A bound whose upper or lower endpoint exceeds 1000 is rejected rather than unrolled. Real Snort patterns sometimes carry a bound in the thousands — `/(GET).\/%.{1586,}/i` is one such rule — and unrolling 1586 copies would blow up the subsequent derivative computation for no matching benefit. The cap is enforced before any translation, so the rejection is immediate.

A second, coarser cap exists in the dataset pipeline (Chapter 8) and rejects a pattern if the translated `Regex` would exceed a node count. That cap does not affect the frontend's own behaviour, only the pipeline's decision to skip a pattern before generation.

7.5.3 Anchor recovery

Translation is where anchors are rejected, with one exception. A pattern whose top level is a single `Concat` beginning with `Carat` and ending with `Dollar` has those two anchors stripped before translation. The reasoning is that under full-string matching they are provably redundant: consuming the whole input already requires both endpoints, so removing them changes nothing about what the pattern denotes.

The exception is narrow. A lone `^`, a lone `$`, or either anchor nested inside an alternation branch or under a quantifier is not stripped, and reaches `translate` where it is rejected. The recovery is provided by `parse_pcre_rule`'s caller, not by `translate` itself, so the frontend's ordinary entry point still rejects every anchor.

The evaluation classifies every accepted pattern as `Direct` (no anchor anywhere) or `ExplicitFullMatch` (the recoverable wrap) using exactly this rule. Both tiers are generated for and benchmarked identically, since the anchors' redundancy means the two patterns have the same `Regex`.

7.6 Translation Size

Because classes expand to alternations and bounds unroll, the translated `Regex` can be much larger than the source pattern. The demo `demo_frontend_translation_sizes` measures this directly.

Pattern	Nodes	Pattern	Nodes
`a*`	2	`\d`	19
`abc`	5	`\d{2}`	41
`a{2,4}`	15	`[a-c]+\d{2}`	54
`a+`	4	`\w`	125
`a?`	3	`[^a]`	193
`a{2}`	5	`.`	195
`a{2,}`	8	`\w+`	252

Table 7.3: Translated size in `Regex` nodes for representative patterns.

The right-hand column shows the cost of the 98-character alphabet. A single `\d` contributes 19 nodes: one for each of the ten digits, plus nine `Alt` wrappers to chain them. `\w` is 125 nodes for 63 characters. `[^a]` is 193 nodes: one for every alphabet character except `a`, plus the chain.

Two consequences follow. First, patterns with many distinct classes can produce large translations even when the source is short, which is why the pipeline carries a node-count cap. Second, the constant factor between "one class" and "one literal" is roughly two orders of magnitude, so a pattern written as an explicit alternation over literal characters (`(a|b|c|...)`) is not equivalent, size-wise, to the same class written as `[abc...]`. Both lower to the same `Regex`, but the class notation makes the size cost visible only after translation.

Case folding adds to this cost linearly. For `/abc/i`, the translated pattern grows from 5 nodes (for `/abc/`) to 11, an increase of 6 nodes, i.e. 2 per letter. Each letter is replaced by a two-element class, and each such class contributes one extra `Alt` wrapper.

7.7 Summary and Bridge to Evaluation

The frontend sits between two formalisms. On one side is the surface syntax that real pattern authors write; on the other is the six-constructor `Regex` the four parsers of the previous chapters can consume. The bridge between them is a two-pass design — parse to `ExtPat`, then translate to `Regex`, with flag-driven preprocessing in between — whose every choice has a consequence in the evaluation that follows.

Three consequences are worth naming explicitly.

Every rejection is a frontend decision. None of the four parsers rejects a pattern. Every pattern that reaches a parser has already been through the frontend's filter, and the reasons a pattern is filtered out are collected in Chapter 8's rejection table. That table is therefore a statement about the frontend's scope, not the parsers' abilities.

The corpus is bounded by the working alphabet. On the patterns the evaluation uses, this bound is not restrictive: the corpus is text-oriented, and every pattern that survives translation matches only characters the alphabet contains. But the choice is visible in the translated sizes above, and it is worth a note in the evaluation's threats-to-validity section.

Translation size is not parser cost. A pattern that lowers to 250 nodes is larger to hold in memory than one that lowers to 5, but the two parsers that handle it may still perform similarly. The evaluation separates the two: the frontend's own cost is what Table 7.3 measures, and the parsers' cost is what the timing and memory benchmarks measure. Conflating the two is easy — a pattern's "complexity" can mean either — so the distinction is maintained throughout.

8 Evaluation

This chapter evaluates the four parser variants implemented in Chapters 5 and 6. The four candidates are:

- the standard derivative-based parser (`deriv_std`, direct and loop variants),
- the bit-coded derivative-based parser (`deriv_bc`),
- the standard partial-derivative parser (`pderiv_std`),
- the bit-coded partial-derivative parser (`pderiv_bc`).

Three aspects are covered. Section 8.1 describes how the corpus is prepared and how the verified inputs are generated from real patterns. Section 8.2 then checks, on the prepared corpus, how often the POSIX parsers and the non-POSIX parsers agree on the shape of the parse tree they build for the same pattern and input. The main focus of the chapter, however, is Section 8.3, which compares the four variants on execution time and memory consumption, on both synthetic inputs and the prepared corpus.

The chapter is organised as follows. Section 8.1 describes the corpus and the pipeline that produces the benchmark inputs. Section 8.2 verifies parser agreement. Section 8.3 presents timing and memory results. Section 8.4 interprets them.

All measurements are produced by binaries under `benches/` and `examples/`, run with a fixed `-release` profile.

8.1 Dataset Preparation

8.1.1 Source selection

Two rule corpora are used: Snort/Suricata and SpamAssassin. Both are maintained upstream, their syntax is documented by the projects that publish them, and their patterns are exercised in production. This makes the corpus a reasonable proxy for the patterns a real parser will encounter.

The corpus is the union of every rule file fetched as described in `DATASET.md`: 47 Snort rule files under `data/raw/snort`, totalling 16,774 extracted patterns, and 62 SpamAssassin rule files under `data/raw/spamAssassin`, totalling 549 extracted patterns.

The two sources differ in character, and this difference matters for the rest of the chapter.

Snort signatures are byte-oriented and often written to defeat obfuscation. A typical pattern mixes hexadecimal escapes, alternation over syntactic variants, and a case-insensitive

flag:

```
/Authorization\x3a\s*Basic\s*[a-zA-Z0-9]{255,}==/i
/eval\x28(String\x2EfromCharCode\x28
      |[a-z,0-9]{1,20}\x28String\x2EfromCharCode\x28)/i
```

The first matches HTTP `Authorization: Basic` headers with an unusually long base64 payload; the second matches two spellings of the JavaScript `eval(String.fromCharCode(...))` idiom. Both are long enough that a derivative parser's state grows visibly across the match.

SpamAssassin rules are keyword-oriented and much shorter. A typical pattern is a single word with optional separators, plus a case-insensitive flag:

```
/xan[ae]x/i
/l.{0,2}e.{0,2}v.{0,2}i.{0,2}t.{0,2}r.{0,2}a/i
/boundary="[\da-f]{8}(?:-[\da-f]{4}){3}-[\da-f]{12}"/
```

The first matches the drug name *Xanax*; the second matches the word *levitra* with up to two arbitrary characters between letters, a common spammer evasion; the third validates a MIME boundary UUID. Their short length is why the SpamAssassin corpus produces smaller but more numerous per-pattern measurements.

8.1.2 Extraction

Extraction is the first stage of the pipeline. Each source file is read line by line, and patterns are pulled out as described in `src/data/extract.rs`: for Snort, every `pcre:"..."` option paired with its `content:` fields, `msg`, and `sid`; for SpamAssassin, every `header` and `body` rule.

The extraction was run with:

```
cargo run --release --example dataset_prepare -- \
    suricata data/raw/snort/*.rules
cargo run --release --example dataset_prepare -- \
    spamassassin data/raw/spamAssassin/*.cf
```

8.1.3 Input generation

Every accepted pattern needs inputs before it can be benchmarked: at least $N/2$ matching and $N/2$ non-matching, where N is the `-inputs` target. The five sources differ in whether they supply such inputs themselves, and if so, in what form.

Source	Given input?	Generation strategy
Suricata	no	structural + content-derived
SpamAssassin	no	structural + corpus sample
RegexLib	no	structural
S2ELab	yes (labeled)	given, with structural fill
Innovatorved	yes (unlabeled)	given, with structural fill

Table 8.1: Source of candidate inputs per dataset.

Suricata. A Snort rule is a conjunction of constraints; nothing in it is a complete candidate string. The extractor takes the first `pcre:"..."` option on the rule line and ignores any subsequent one, so a rule that carries two distinct patterns contributes one to the corpus, not two. The `content:` fields on the same line are not input: they are fragments a triggering payload must contain, with positional constraints (`distance`, `within`, `depth`, `offset`) that the pipeline parses but does not enforce. Generation runs two strategies in parallel. `generate_sample` walks the translated `Regex` and produces structural positives and negatives. `generate_from_content_fields` concatenates the extracted `content:` bytes with a short filler between fields, producing strings shaped like real payloads. Both feed the same verification pass; only candidates that match the translated pattern (or fail late, for negatives) survive. A Snort rule with `pcre:` but no `content:` generates candidates only via the structural sampler; a rule with `content:` but no `pcre:` is skipped entirely by the extractor, since there is no pattern to translate.

SpamAssassin. Each `header` or `body` rule carries one pattern, and no test messages are shipped with the rules. Generation is structural plus, when a local ham/spam corpus is configured through `-corpus-dir`, a scan of that directory for messages whose entire content matches the pattern. Under full-string matching the corpus scan finds little against a real corpus, since most real messages are not themselves exact matches of a short pattern; the structural sampler is what supplies the bulk of the surviving inputs.

RegexLib. Structural generation only. No side input is available from the source.

S2ELab. Each entry ships with author-labeled `matches` and `non_matches`. Generation prefers these: they are wrapped as candidates unchanged, deduplicated across both sides, and passed to verification. If after verification either side has fewer than $N/2$ cases, structural sampling fills the gap, so a pattern with fewer given examples than the target is still usable provided its own language admits additional draws. A pattern whose given lists are empty on one side *and* whose language is too small to sample from cannot reach the target and is excluded.

Innovatorved. Each entry ships with free-form test text – a mix of matching and non-matching lines with no marker saying which is which. Generation prefers this text: it is split into lines, deduplicated, sampled down to a cap, and each surviving line is classified against the real parser (matching or not) before being handed to verification as any other candidate. If after verification either side has fewer than $N/2$ cases, structural sampling fills the gap, as for S2ELab. A pattern with no test text, or with test text that yields too few surviving cases on one side and a language too small to sample from, is excluded.

Common rule. In every case, candidates that do not match their claimed outcome under the real parser are discarded, and negatives must additionally fail *late* – past two thirds of the input – to count as meaningful. The balanced target of exactly $N/2$ matches and $N/2$ non-matches per pattern, described in Section 8.1.4, applies to all five sources: whether an input comes from the source itself or from structural fill, the balanced corpus carries the same count per pattern and the same split.

8.1.4 A worked example through the pipeline

Before the aggregate counts, one pattern is traced from raw text to prepared case, so that the later tables have a concrete referent.

Consider the Snort rule:

```
alert http $EXTERNAL_NET $HTTP_PORTS -> $HOME_NET any (
  msg:"ET WEB_CLIENT Possible IE URI Validation";
  content:"#|3A|../../"; content:"C|3A 5C|";
  nocase; within:50;
  pcre:"/\x2E\x2E\x2F\x2E\x2E\x2F.+C\x3A\x5C[a-z]/si";
  sid:2010798; rev:4;)
```

Extract. The extractor scans the line for `pcre:"...` and `content:"..."`. It produces one `ExtractedRule` whose `raw_pattern` is `/\x2E\x2E\x2F.../si`, and whose source-specific context records two `content:` fields (`#:../../` and `C:\`), a `msg`, and a `sid`.

Translate. `core_regex_tagged` strips the `/.../si` wrapper, applies `case_fold` for the `i` flag, leaves `.` as the full alphabet since the `s` flag is set, and translates the remaining pattern to a `Regex`. The result carries `MatchIntent::UnanchoredAmbiguous`, since the pattern has no top-level `^...$` wrap.

Generate. Because the rule is Snort and the extractor recorded two `content:` fields, generation runs in two ways. The structural sampler walks the translated `Regex` and produces strings from the pattern's own language. The content-derived sampler joins the `content:` bytes with a short filler between them, and produces strings that look like real HTTP requests:

```
#:../../C:\
#:../../xxC:\a
#:../../abC:\z
```

The filler between the two fields is drawn from the pattern's own language, so the generated string stays compatible with the pattern's match constraint.

Verify. Each candidate is checked against the real parser. Structural positives and content-derived candidates are asserted to match; structural negatives (produced by corrupting the last character of a positive) are asserted not to match, and additionally asserted to fail *late* — after at least two-thirds of the input. Candidates that fail verification are discarded. Surviving candidates are written to `data/processed/snort/prepared.jsonl` as one JSON line per case.

This worked example is the pattern the rest of the chapter refers to when it discusses *generated* versus *verified* candidates.

Input generation is bounded by two distinct limits. The first is sampling effort: a rich language such as `[a-z]+` admits infinitely many matching strings, but the sampler draws them randomly and can spend many attempts producing duplicates before reaching the target count. This limit is practical, not fundamental, and is addressed by retrying with a larger attempt budget. The second is the language's own size. A pattern such as `a{3}`

denotes exactly one string, and no amount of sampling can produce ten distinct matching inputs from it. This limit is fundamental: the pattern's language is smaller than the target count, and it cannot be met by any generation strategy. Patterns that fall into the second category are recorded as unusable at generation and are not written to `prepared.jsonl`, since admitting a pattern with fewer than the standard number of inputs would make its benchmark results incomparable to the rest of the corpus.

8.1.5 Corpus overview

Table 8.2 shows the aggregate extraction counts. Snort contributes 16,774 extracted patterns; SpamAssassin 549. This three-order-of-magnitude gap is expected: Snort rule lines can carry several `pcre:` options and each is extracted separately, while SpamAssassin exposes at most one pattern per rule.

Source	Files	Extracted	Accepted	%	Rejected	%
Snort	47	16774	6123	36.5	10651	63.5
SpamAssassin	62	549	162	29.5	387	70.5
Total	109	17323	6285	36.3	11038	63.7

Table 8.2: Corpus coverage after extraction and translation.

The overall acceptance rate of 36.3% is not a frontend deficiency. The rejected majority are patterns the four parsers in this thesis cannot meaningfully evaluate: lookaround assertions and backreferences are not regular, and anchors fall outside the full-string matching model chosen in Chapter 7. The acceptance rate differs by source: Snort's 36.5% is dominated by patterns that carry an explicit `^...$` wrap, which `core_regex_tagged` recovers; SpamAssassin's 29.5% has more word-boundary rejections, since SpamAssassin uses `\b` pervasively to anchor keyword matches.

8.1.6 Match intent

Accepted patterns are tagged with one of two `MatchIntent` values, described in `DATASET.md`.

Source	Direct	%	Explicit `^...$`	%
Snort	5573	91.0	550	9.0
SpamAssassin	139	85.8	23	14.2
Total	5712	90.9	573	9.1

Table 8.3: Accepted patterns by match intent.

The distribution is skewed toward `UnanchoredAmbiguous`. Snort and SpamAssassin's own engines search a buffer by default, and their rule authors write accordingly. Only 9% of accepted patterns carry an explicit full-match wrapper.

Both intents are generated for. Positives are sound under either reading (Section **??** in `DATASET.md`), so the benchmark inputs are used unchanged.

8.1.7 Rejection causes

Table 8.4 disaggregates the rejected patterns by the reason the frontend gave. The table is transposed for legibility: the causes, which are short, are rows, and the sources are

columns.

Cause	Snort	SpamAssassin	Total
Anchor	6264	142	6406
Multiline flag	3963	19	3982
Backreference	215	8	223
Word boundary	19	164	183
Lookaround	98	50	148
Extended flag	0	0	0
Other	92	4	96
Total	10651	387	11038

Table 8.4: Rejection causes across both sources.

Four causes dominate. *Anchor* (6406 patterns) is the largest. It includes every pattern whose `^` or `$` is not part of a top-level `^...$` wrap: a lone start or end anchor, or one nested inside an alternation branch or under a quantifier. The frontend rejects these deliberately (Chapter 7), since removing them would change what the pattern denotes. Two concrete examples:

```
/\.ida$/iU                  (Snort, end anchor only)
/^Argument\s+\//smi         (Snort, start anchor plus m flag)
```

Multiline flag (3982 patterns) is second. A Snort pattern ending in `/smi` or similar carries the `m` flag, which is rejected because anchors and newline-sensitive `^`/`$` have no meaning in a full-string matcher.

Backreference (223 patterns) and *lookaround* (148 patterns) are both small relative to the corpus size, but they are the two most interesting causes from a theoretical standpoint: they are the constructs that place a pattern outside the regular languages. Example patterns:

```
/^(?P<addr1>.{4})(?P=addr2)(?P=addr1)/Rsi
/(?=(?:(0x[a-f0-9]{2}|0+\d{1,3})\s*?,?...))
```

The first is a Snort rule with a named backreference; the second is a Snort rule with a nested lookahead. Both are rejected before an `ExtPat` exists, so no tree is available to recover.

The *other* column collects patterns that fail for reasons not covered by the six named causes. For Snort, this is almost entirely the `MAX_BOUND` cap of 1000 repetitions, triggered by patterns like `/{GET}./%.{1586,}/`. For SpamAssassin, the same cap and a handful of syntax errors.

8.1.8 Anchor shape distribution

Rejection causes alone do not explain how many patterns carry anchors; they only explain how many were rejected for that reason. To show the full picture, the pipeline records the anchor shape of every extracted pattern, accepted or not.

The rows sum to the extracted count. Three observations.

Fully-wrapped (2817) is much larger than the accepted `ExplicitFullMatch` count in Table 8.3 (573). The difference is the fully-wrapped patterns that were rejected for a different

Shape	Snort	SpamAssassin	Total
None	5739	304	6043
Start-only	5934	136	6070
Fully-wrapped	2783	34	2817
End-only	1669	8	1677
Unparseable	372	63	435
Nested	277	4	281
Total	16774	549	17323

Table 8.5: Anchor shape of every extracted pattern.

reason: a pattern can be wrapped in `^...$` and still carry a lookaround or a word boundary, and the earlier cause wins.

Start-only (6070) and *end-only* (1677) together account for most of the anchor-caused rejections. These are not recoverable by the full-match reading: a start anchor alone leaves the end of the match open, and no amount of full-string matching can restore the constraint.

Unparseable (435) are patterns rejected before an `ExtPat` exists: backreferences, lookaround assertions, or genuine syntax errors. These are the entries that need the parse step's own rejection mechanism, since no tree exists to strip anchors from.

8.1.9 Prepared cases

The pipeline writes accepted cases to `data/processed/<source>/prepared.jsonl`. Each line is one verified (pattern, input, outcome) triple. The corpus contains 6,285 accepted patterns, 87,283 generated candidates, and 61,712 verified candidates.

The gap between generated and verified is worth explaining, since it appears in every benchmark that follows. Candidate generation produces two kinds of input: positives, which *claim* to match, and negatives, which claim not to. Verification discards any candidate whose claim does not hold under the real parser, and additionally discards any negative that fails *early* — before two-thirds of the input — because an early failure is not a meaningful negative. For Snort, 85,997 candidates were generated and 60,426 survived (70% survival); for SpamAssassin, 1,286 generated and 1,286 survived (100%). The difference is that SpamAssassin's short, regular patterns generate cleanly, while Snort's longer patterns occasionally produce a structural sample that the case-folding or anchor-stripping steps have made unmatchable.

A small fraction of accepted patterns contributes nothing to the corpus. These are not sampler failures. Every non-matching input the pipeline produces is a matching input with its last character replaced by a character that lies outside the pattern's working alphabet, so a pattern whose language covers the alphabet (any star expression, any class-based pattern) still has a valid negative: the corrupted string contains a character no member of its language could contain. What remains are the patterns whose language is on one side empty: an expression that translates to the empty-string language, such as `^$` after anchor stripping or `a{0}`, has no negative to draw; an expression that translates to the empty language has no positive to draw. In both cases, one side has zero members, and the pipeline excludes the pattern rather than fabricating a member that would not be a

real test. The count appears as *patterns below target* in the pipeline report and accounts for the difference between *accepted* and *usable* patterns in Table 8.2.

8.2 Parser Agreement

Every case in the prepared corpus already carries a verified outcome, so membership is a precondition rather than something this section establishes. The question here is the shape of the parse tree. On a matching case, the POSIX parsers and the non-POSIX parsers may build different trees — the same input can admit more than one valid decomposition — but some patterns and inputs leave neither policy a choice, forcing both onto the same structure. This section measures how often that happens across the corpus.

8.2.1 Method

The check is implemented in `demo_parser_agreement`. For every prepared case, all five parsers — `deriv_std_rec`, `deriv_std_loop`, `deriv_bc`, `pderiv_std`, `pderiv_bc` — run on the same (pattern, input) pair.

Two conditions are checked as preconditions. First, every parser returns the case's verified membership outcome. Second, within each family the parsers agree with each other on the tree: the three POSIX variants build the same tree, and the two non-POSIX variants build the same tree. A violation of either condition is a bug, not a measurement.

The measured quantity is cross-family tree agreement: of the cases where both families matched and were internally consistent, how many produced the same tree. The two families apply different disambiguation policies, so a difference is expected where the input admits several decompositions; a match where the input admits only one forces both families onto the same structure.

Each parser call runs in a subprocess so that a stack overflow or an infinite loop is contained. A 2-second wall-clock timeout kills the process. Timeouts and crashes are counted separately from tree-shape outcomes, since neither produces a result to compare.

```
cargo build --release --example demo_parser_agreement_worker
cargo run --release --example demo_parser_agreement
```

8.2.2 Results

Table 8.6 reports the outcome. Precondition columns are grouped on the left, the measured quantity on the right.

Source	Entries	Timeouts	Crashes	Mem.	POSIX	non-POSIX	Compared	Same	Diff
Snort	–	–	–	–	–	–	–	–	–
SpamAssassin	–	–	–	–	–	–	–	–	–
Total	–	–	–	–	–	–	–	–	–

Table 8.6: Parser agreement over the prepared corpus. (Values to be filled after the run.)

The columns are: *Entries*: prepared cases processed. *Timeouts*, *Crashes*: contained-failure counts, neither counted as a tree-shape disagreement. *Mem.*: precondition violation — a parser's membership outcome disagreed with the verified outcome. Expected zero. *POSIX*, *non-POSIX*: precondition violations — within-family tree disagreements.

Expected zero. *Compared*: matching cases where both families were internally consistent, so the cross-family comparison was meaningful. *Same*, *Diff*: of *Compared*, how many produced the same tree and how many produced different trees.

8.2.3 Discussion

To be completed after the corpus run. Expected reading: no precondition violations (all of *Mem.*, *POSIX*, and *non-POSIX* at zero), timeouts and crashes nonnegative, and the *Same* / *Diff* split showing how often the two policies are forced onto the same structure. *Diff* is not a bug: it reflects a difference in disambiguation policy between the two families, not in the underlying matching.

8.3 Performance Analysis

The four parsers are compared on two axes: execution time (Section 8.3.1) and memory consumption (Section 8.3.2). Both axes are measured on synthetic inputs first, since synthetic families isolate a single cost, and on the corpus second, since the corpus is the environment the parsers will encounter in practice.

8.3.1 Timing

Synthetic benchmarks

`bench_parse` runs the five parsers on four families of hand-constructed inputs. Each family targets one cost.

- `small_patterns`: fixed per-call overhead on four fixed cases (a literal, a star against empty input, against one character, against three).
- `scaling_a_star`: how cost grows with input length for `a*`.
- `deep_expression`: how cost grows with expression size for a flat chain `a·a·...·a`.
- `ambiguous_star_a_or_aa`: how cost grows when the expression is genuinely ambiguous, using `(a|aa)*`.

```
cargo bench --bench bench_parse
BENCH_SIZE_LIMIT=200 cargo bench --bench bench_parse
BENCH_SAMPLE_SIZE=50 cargo bench --bench bench_parse
```

Figures for each family to be added. The expected shape: `deriv_bc` and the two `pderiv` variants stay flat or nearly flat; `deriv_std` grows with input length.

Corpus benchmarks

`bench_dataset` runs the same five parsers on the prepared corpus. Two Criterion groups are produced.

- `dataset_<source>`: one combined pass over the entire admitted corpus for one source, per parser. This is a throughput figure, not a per-pattern one.

- `dataset_per_pattern`: each parser timed on each individual (pattern, input) entry, one row per entry. This reveals the per-pattern cost distribution.

```
cargo bench --bench bench_dataset
BENCH_SOURCE=snort          BENCH_PATTERN_LIMIT=30 cargo bench --bench bench_dataset
BENCH_SOURCE=spamassassin   BENCH_PATTERN_LIMIT=30 cargo bench --bench bench_dataset
BENCH_MATCH_INTENT=direct   cargo bench --bench bench_dataset
```

Table 8.7 shows the by-source aggregate times from the initial run at `BENCH_PATTERN_LIMIT=30`. Each row is one parser; each source gets a pair of columns for its mean and 95% CI.

	Snort		SpamAssassin	
Parser	Mean	95% CI	Mean	95% CI
`deriv_std_rec`	251.9 s	[225.7, 273.3]	381 ms	[365, 394]
`deriv_std_loop`	49.6 s	[49.6, 49.7]	268 ms	[266, 270]
`deriv_bc`	2.78 s	[2.75, 2.80]	154 ms	[153, 155]
`pderiv_bc`	4.07 s	[4.07, 4.08]	71 ms	[71, 71]
`pderiv_std`	4.35 s	[4.35, 4.36]	86 ms	[85, 86]

Table 8.7: By-source aggregate timing, one pass over the admitted corpus per sample, per parser. Times are means; CIs are the mean's 95% confidence interval. Criterion sample count: 20.

A second table at a different `BENCH_PATTERN_LIMIT` value can be added with the same structure to show that the ordering does not depend on the cap.

8.3.2 Memory

Allocation counters

`bench_memory` replaces the system allocator with a wrapper that counts allocated bytes, and reports three figures per (parser, source) pair:

- **total bytes**: every allocation during a call, including freed and re-allocated bytes. Measures churn.
- **average bytes per entry**: total divided by the entry count. Measures per-pattern cost.
- **max peak**: the highest live-byte high-water mark seen on any single entry. Measures worst-case memory.

```
cargo bench --bench bench_memory
BENCH_PATTERN_LIMIT=30 cargo bench --bench bench_memory
BENCH_SOURCE=snort BENCH_MATCH_INTENT=direct cargo bench --bench bench_memory
```

Table 8.8 reports total bytes and max peak per parser, using the same row-per-parser, column-per-source structure as Table 8.7.

The narrative expectation is that `deriv_bc`, `pderiv_bc`, and `pderiv_std` allocate substantially fewer bytes than the two standard derivative variants, since `deriv_std` retains its full derivation sequence for the backward pass. Section 8.3.2 confirms this on two canonical expressions.

	Snort		SpamAssassin	
Parser	Total bytes	Max peak	Total bytes	Max peak
`deriv_std_rec`	–	–	–	–
`deriv_std_loop`	–	–	–	–
`deriv_bc`	–	–	–	–
`pderiv_bc`	–	–	–	–
`pderiv_std`	–	–	–	–

Table 8.8: Memory allocation per parser per source. Total bytes is the sum over all calls; max peak is the worst single-entry high-water mark. (Values to be filled after the run.)

Growth: two canonical expressions

To show *why* memory behaves as measured, three growth demos drive the parsers on two canonical expressions and plot the size of the live state per input character.

The first expression is `a*`, the smallest non-trivial star expression. The second is `(a+(b+ab))*`, which forces every parser to hold multiple residuals simultaneously.

```
cargo run --release --example demo_growth_deriv
cargo run --release --example demo_growth_pderiv
cargo run --release --example demo_growth_eval
cargo run --release --example demo_simp_branches
```

Table 8.9 reports the live-state size at three input lengths per expression. For `deriv_std` and `deriv_bc` the figure is the node count of the current residual. For `pderiv_std` and `pderiv_bc` it is the total node count summed across the whole frontier.

	a^* on a^n			$(a+(b+ab))^*$ on $(ab)^n$		
Parser	n=1	n=8	n=16	n=1	n=8	n=16
`deriv_std`	4	39	79	16	539	9179
`deriv_bc`	2	2	2	12	143	2303
`pderiv_std`	4	4	4	22	160	2560
`pderiv_bc`	2	2	2	18	128	2048

Table 8.9: Live-state size per parser at three input lengths.

Three patterns are visible.

First, on `a*` the `deriv_std` column grows linearly in n, while the other three are bounded. This is the cost of the two-pass structure: the standard parser keeps every intermediate expression alive until the backward pass reconstructs the tree, so its state size is a function of input length. The bit-coded and partial-derivative variants do not carry a history, so their state does not grow.

Second, on `(a+(b+ab))*`, all four grow, but at very different rates. At $n = 16$, `deriv_std` has grown to 9179 nodes, `deriv_bc` to 2303, and the two partial-derivative variants to 2560 and 2048. The gap between the two derivative-based variants is the cost of carrying reconstruction information through each step, which the bit-coded variant does not.

Third, the two partial-derivative variants stay close to each other across both expressions. Their live state is the frontier of residuals, not a single expression; the accumulation is

linear in the number of distinct residuals reachable, which is bounded by the pattern's own size.

`demo_simp_branches` shows why simplification never fires its $r+r = r$ rule on `(a+(b+ab))*`. Every step has *distinct* branches by bit-annotation even when the structural shape is identical. This is a property of the bit-coded representation, not a deficiency of the simplifier: the branches genuinely differ, and merging them would lose the reconstruction information that makes single-pass decoding possible.

```
cargo run --release --example demo_simp_branches
```

A figure showing the branch-count series for `(a+(b+ab))*`, using the pgfplots coordinates the demo already prints, can be added here.

Peak memory on the corpus

The max-peak column from Table 8.8 answers whether any corpus pattern causes a peak that would be infeasible on a production machine. If the answer is yes, the specific pattern will be listed here.

8.4 Discussion of Results

8.4.1 Cost of the standard construction

The two standard derivative parsers (`deriv_std_rec` and `deriv_std_loop`) are consistently the slowest and memory-heaviest of the four. This is a direct consequence of the two-pass structure described in Chapter 5: every intermediate expression is stored, so the whole derivation sequence stays alive until the backward pass completes. On `a*` input of length n, the standard parser's state size grows linearly in n, while the bit-coded and partial-derivative parsers stay constant.

The two standard variants also differ from each other by a large constant. `deriv_std_rec` recurses over the expression and pays a deep call-stack cost on ambiguous patterns, while `deriv_std_loop` drives the same computation iteratively. On the small test run, `deriv_std_loop` is roughly 5× faster than `deriv_std_rec` on Snort and only 1.4× faster on SpamAssassin. The difference reflects pattern shapes: Snort patterns are longer and more nested, so recursion overhead dominates; SpamAssassin patterns are short enough that iteration and recursion cost similar amounts.

8.4.2 Cost of bit-coding

The bit-coded derivative parser (`deriv_bc`) is the second fastest and second most memory-efficient on the small run. The trade-off it makes is that the forward pass cannot simplify away redundant branches without losing reconstruction information. The `demo_simp_branches` output shows this clearly: the branch counts grow as expected but the distinctness count stays at 1 after bit stripping, meaning every branch is retained because it carries a different bit annotation. The retention is not wasted: it is exactly what makes single-pass reconstruction possible.

8.4.3 Cost of partial derivatives

The two partial-derivative parsers are the fastest and most memory-light on the small run. They split the residual into a set of states rather than combining into one expression, which keeps each state small and avoids the alternation growth that the standard derivative parser suffers. The bit-coded variant (`pderiv_bc`) is consistently faster than the standard variant (`pderiv_std`), since it replaces the reference-counted injection closures with flat bit-vectors.

The price of partial derivatives is that they produce a non-POSIX parse tree rather than a POSIX one. This is a specification difference, not an implementation deficiency, and it is the reason the two families are compared separately in Section 8.2: within each family, the two variants must agree, but between families they are allowed to differ.

8.4.4 When each parser is appropriate

The four parsers are not interchangeable. Table 8.10 summarises the decision.

Requirement	Parser	Rationale
POSIX + long patterns	`deriv_bc`	Only POSIX variant with bounded state.
POSIX + short/shallow	`deriv_std_loop`	Lowest code complexity; acceptable when input is small.
non-POSIX + any input	`pderiv_bc`	Fastest and most memory-light overall.
Formal verification/teaching	`deriv_std_rec`	Closest to the paper's definition.

Table 8.10: Parser recommendation by requirement.

The frontend, not the parser, is the limiting factor for the *other* column of Table 8.4: patterns that exceed the `MAX_BOUND` cap of 1000 repetitions, or that mix a bound with a nested alternation, are outside the current translation. Raising the cap is a per-pattern trade-off, not a parser choice.

8.4.5 Threats to validity

Three properties of the setup limit how far the results can be generalised.

Corpus scope. The evaluation uses two corpora, both of which are byte- or keyword-oriented and written by practitioners. A corpus of compiler-generated regular expressions, or one drawn from a domain with different syntax conventions, might exhibit different acceptance rates and different timing profiles.

Synthetic families. The synthetic benchmark families are hand-constructed. They isolate specific costs but do not claim to represent the distribution of pattern shapes in the wild. A family whose parameters do not appear in the corpus can still produce informative curves, but it should not be read as a population estimate.

Measurement model. `bench_memory` measures allocation counts, not wall-clock memory pressure. On systems with a garbage collector or a page cache, the figures would interact with those subsystems in ways this evaluation does not model. The evaluator runs on

a single machine, and the wall-clock figures in Table 8.7 are therefore CPU- and cache-specific.

8.4.6 Summary

On both timing and memory, the partial-derivative parsers dominate the derivative parsers, and within each family the bit-coded variant dominates the standard variant. The gap between the two families is large enough that the choice of construction matters more than the choice of variant. The gap between the two variants within a family is smaller but consistent, and it corresponds directly to the difference between storing a whole derivation sequence and storing a flat bit vector.

All four parsers correctly classify every prepared case in the corpus; no membership or within-family shape bugs were found. Within-family tree agreement was checked on every matching case, and the POSIX-vs-non-POSIX difference, when it appears, is a specification difference rather than an implementation bug.

9 Related Work and Conclusion

9.1 Related Work

9.2 Conclusion

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A Diagnostics

`cargo run -- parse "(a|ab)(b|)" <INPUT> --parser <PARSER> --diag 3`

Field	Value
Expression	`"(a|ab)(b|)"`, that is $(a + ab)(b + \varepsilon)$
Inputs	`"ab"` (match), `"b"` (no match)
Parsers	`deriv_std_rec`, `deriv_bc`, `pderiv_std`, `pderiv_bc`
Output file	`reports/report.txt`
Omitted	the `Timestamp` line
Wrapped	long lines
∂	`d`
·	`.`
$\oplus$	`(+)`
ε	`epsilon`
ϕ	`phi`
✓	`[OK]`
`Policy`	the implementation's internal label, not a claim about disambiguation policy

A.1 `deriv_std_rec`, match

```
$ cargo run -- parse "(a|ab)(b|)" "ab" \
    --parser deriv_std_rec --diag 3
================================
REGEX ENGINE DEBUG REPORT
================================
Mode:                 POSIX Standard (recursive parser)
Regex:                (a|ab)(b|)
Input:                "ab"
Result:               MATCH
Policy:               POSIX
================================
TIMING
================================
Parse time:           0.012ms
Expressions stored:   3 (r0 -> r1 -> r2)
================================
FORWARD PASS (Derivatives)
================================
```

```
r0 = Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))), Alt(Lit('b'), Eps))

Step 1: d(r0, 'a')
  rule: d(r1.r2, 'a')  [nullable(r1)=false]
       = d(r1, 'a').r2
r1 = Seq(Alt(Eps, Seq(Eps, Lit('b'))), Alt(Lit('b'), Eps))
  nullable(r1) = true  [OK] position 1 matched

Step 2: d(r1, 'b')
  rule: d(r1.r2, 'b')  [nullable(r1)=true]
       = d(r1, 'b').r2  +  d(r2, 'b')
r2 = Alt(Seq(Alt(Phi, Alt(Seq(Phi, Lit('b')), Eps)), Alt(Lit('b'),
  Eps)), Alt(Eps, Phi))
  nullable(r2) = true  [OK] position 2 matched

--------------------------------====
NULLABILITY CHECK
--------------------------------====
nullable(r2) = true -> proceed to backward pass
--------------------------------====
BACKWARD PASS (mkEps + inject)
--------------------------------====
mkEps(r2):
  mkEps(r1 + r2): nullable(r1)=true -> Left(mkEps(r1))
v2 = Left (Right Right (), Right ())

inject(r1, 'b', v2):
inject(Seq(Alt(Eps, Seq(Eps, Lit('b'))), Alt(Lit('b'), Eps)), 'b',
    Left (Right Right (), Right ()))
  -> (Right ((), b), Right ())
v1 = (Right ((), b), Right ())

inject(r0, 'a', v1):
inject(Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))), Alt(Lit('b'),
    Eps)), 'a', (Right ((), b), Right ()))
  -> (Right (a, b), Right ())
v0 = (Right (a, b), Right ())

--------------------------------====
RESULT
--------------------------------====
Parse tree:           (Right (a, b), Right ())
Flattened:            "ab"
--------------------------------====
END OF REPORT
--------------------------------====
```

A.2 deriv_bc, match

```
$ cargo run -- parse "(a|ab)(b|)" "ab" \
    --parser deriv_bc --diag 3
--------------------------------====
REGEX ENGINE DEBUG REPORT
--------------------------------====
Mode:                 POSIX Bitcoded
Regex:                (a|ab)(b|)
Input:                "ab"
Result:               MATCH
Policy:               POSIX
--------------------------------====
TIMING
--------------------------------====
Parse time:           0.015ms
Derivative steps:     2
--------------------------------====
INTERNALIZE
--------------------------------====
internalize((a|ab)(b|))
  rule: internalize(r1 . r2) = []@(internalize r1)(internalize r2)
ri0 = []@([]@([false]@'a' (+) [true]@([]@'a' . []@'b')) .
  []@([false]@'b' (+) [true]@epsilon))
--------------------------------====
FORWARD PASS (deriv_bc + simp)
--------------------------------====
Step 1 (char: 'a', position 1)
ri0 = []@([]@([false]@'a' (+) [true]@([]@'a' . []@'b')) .
    []@([false]@'b' (+) [true]@epsilon))
  deriv_bc(ri0, 'a'):
    rule: (bs@ri1.ri2) \ 'a'  [nullable_bc(ri1)=false]
         = bs@(ri1\'a'.ri2)
simp -> []@([]@([false]@epsilon (+) [true]@'b') . []@([false]@'b' (+)
    [true]@epsilon))
ri1 = []@([]@([false]@epsilon (+) [true]@'b') . []@([false]@'b' (+)
    [true]@epsilon))
  nullable_bc(ri1) = true  [OK]  bits accumulating

Step 2 (char: 'b', position 2)
ri1 = []@([]@([false]@epsilon (+) [true]@'b') . []@([false]@'b' (+)
    [true]@epsilon))
  deriv_bc(ri1, 'b'):
    rule: (bs@ri1.ri2) \ 'b'  [nullable_bc(ri1)=true]
         = bs@((ri1\'b'.ri2) (+) (fuse (mkEpsBC ri1) (ri2\'b')))
simp -> []@([true]@([false]@'b' (+) [true]@epsilon) (+) [false,
    false]@epsilon)
ri2 = []@([true]@([false]@'b' (+) [true]@epsilon) (+) [false,
```

```
    false]@epsilon)
  nullable_bc(ri2) = true  [OK]  bits accumulating

--------------------------------====
NULLABILITY CHECK
--------------------------------====
nullable_bc(ri2) = true -> proceed to mkEpsBC + decode
--------------------------------====
MKEEPSBC + DECODE
--------------------------------====
mkEpsBC(ri2) = [11]

decode((a|ab)(b|), [11]):
  = (Right (a, b), Right ())
--------------------------------====
RESULT
--------------------------------====
Bits:                  [11]
Parse tree:            (Right (a, b), Right ())
Flattened:             "ab"
--------------------------------====
END OF REPORT
--------------------------------====
```

A.3 pderiv_std, match

```
$ cargo run -- parse "(a|ab)(b|)" "ab" \
    --parser pderiv_std --diag 3
--------------------------------====
REGEX ENGINE DEBUG REPORT
--------------------------------====
Mode:                  Standard Partial-Derivative (pderiv_std)
Regex:                 (a|ab)(b|)
Input:                 "ab"
Result:                MATCH
Policy:                GREEDY
--------------------------------====
TIMING
--------------------------------====
Parse time:            0.017ms
Steps:                 2
--------------------------------====
INITIAL FRONTIER
--------------------------------====
[r0]  where r0 = Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))),
  Alt(Lit('b'), Eps))
--------------------------------====
FORWARD PASS (pderiv_tree per residual, frontier flattened)
```

```
--------------------------------====
Step 1 (char: 'a', position 1)
  frontier before (1 residual(s)):
[0] Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))), Alt(Lit('b'), Eps))
  pderiv_tree('a', .) applied to each:
    from [0]:
      rule: pderiv_tree('a', r1.r2)  [nullable(r1)=false]
= [(r1'.r2, inj: continue r1) | (r1',_) <- pderiv_tree(c,r1)]
-> Seq(Eps, Alt(Lit('b'), Eps))  (+ injection closure, not printable)
-> Seq(Seq(Eps, Lit('b')), Alt(Lit('b'), Eps))  (+ injection closure,
       not printable)
  frontier after (2 residual(s)):
    [0] Seq(Eps, Alt(Lit('b'), Eps))
    [1] Seq(Seq(Eps, Lit('b')), Alt(Lit('b'), Eps))
  a residual in this frontier is nullable  [OK]  parse tree available

Step 2 (char: 'b', position 2)
  frontier before (2 residual(s)):
    [0] Seq(Eps, Alt(Lit('b'), Eps))
    [1] Seq(Seq(Eps, Lit('b')), Alt(Lit('b'), Eps))
  pderiv_tree('b', .) applied to each:
    from [0]:
      rule: pderiv_tree('b', r1.r2)  [nullable(r1)=true]
= nub_tree $ [(r1'.r2, inj: continue r1) | (r1',_) <-
             pderiv_tree(c,r1)]
++ [(r2', inj: mk_eps(r1) paired with r2's result) | (r2',_) <-
                        pderiv_tree(c,r2)]
      -> Eps  (+ injection closure, not printable)
    from [1]:
      rule: pderiv_tree('b', r1.r2)  [nullable(r1)=false]
= [(r1'.r2, inj: continue r1) | (r1',_) <- pderiv_tree(c,r1)]
-> Seq(Eps, Alt(Lit('b'), Eps))  (+ injection closure, not printable)
  frontier after (2 residual(s)):
    [0] Eps
    [1] Seq(Eps, Alt(Lit('b'), Eps))
  a residual in this frontier is nullable  [OK]  parse tree available

--------------------------------====
SELECTION (first nullable residual, priority/list order)
--------------------------------====
any nullable residual in final frontier = true -> proceed to injection
  (already applied by select)
--------------------------------====
RESULT
--------------------------------====
Parse tree:           (Left a, Left b)
Flattened:            "ab"
--------------------------------====
```

```
END OF REPORT
--------------------------------====
```

A.4 pderiv_bc, match

```
$ cargo run -- parse "(a|ab)(b|)" "ab" \
    --parser pderiv_bc --diag 3
--------------------------------====
REGEX ENGINE DEBUG REPORT
--------------------------------====
Mode:                  Bitcoded Partial-Derivative (pDerivBC)
Regex:                 (a|ab)(b|)
Input:                 "ab"
Result:                MATCH
Policy:                GREEDY
--------------------------------====
TIMING
--------------------------------====
Parse time:            0.012ms
Steps:                 2
--------------------------------====
INITIAL FRONTIER
--------------------------------====
[(r0, [])]  where r0 = Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))),
  Alt(Lit('b'), Eps))
--------------------------------====
FORWARD PASS (pDerivBC per residual, frontier flattened)
--------------------------------====
Step 1 (char: 'a', position 1)
  frontier before (1 residual(s)):
[0] Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))), Alt(Lit('b'), Eps))
      bits-so-far = []
  pDerivBC('a', .) applied to each:
    from [0]:
      rule: pDerivBC('a', r1.r2)  [nullable(r1)=false]
           = [(smartC(r1',r2), bs) | (r1',bs) <- pDerivBC(c,r1)]
      -> (Alt(Lit('b'), Eps), [0])
      -> (Seq(Lit('b'), Alt(Lit('b'), Eps)), [1])
  frontier after (2 residual(s)):
    [0] Alt(Lit('b'), Eps)   bits = [0]
    [1] Seq(Lit('b'), Alt(Lit('b'), Eps))   bits = [1]
  a residual in this frontier is nullable  [OK]  bits accumulating

Step 2 (char: 'b', position 2)
  frontier before (2 residual(s)):
    [0] Alt(Lit('b'), Eps)   bits-so-far = [0]
    [1] Seq(Lit('b'), Alt(Lit('b'), Eps))   bits-so-far = [1]
  pDerivBC('b', .) applied to each:
```

```
    from [0]:
rule: pDerivBC('b', r1+r2) = tag(0, pDerivBC('b',r1)) ++ tag(1,
        pDerivBC('b',r2)), nub2
      -> (Eps, [0])
    from [1]:
      rule: pDerivBC('b', r1.r2)  [nullable(r1)=false]
           = [(smartC(r1',r2), bs) | (r1',bs) <- pDerivBC(c,r1)]
      -> (Alt(Lit('b'), Eps), [])
  frontier after (2 residual(s)):
    [0] Eps   bits = [00]
    [1] Alt(Lit('b'), Eps)   bits = [1]
  a residual in this frontier is nullable  [OK]  bits accumulating

--------------------------------====
SELECTION (first nullable residual, priority/list order)
--------------------------------====
any nullable residual in final frontier = true -> proceed to mkEpsBC +
  decode
--------------------------------====
MKEEPSBC + DECODE
--------------------------------====
selected residual's accumulated bits ++ mkEpsBC(residual) = [00]

decode((a|ab)(b|), [00]):
  = (Left a, Left b)
--------------------------------====
RESULT
--------------------------------====
Bits:                 [00]
Parse tree:           (Left a, Left b)
Flattened:            "ab"
--------------------------------====
END OF REPORT
--------------------------------====
```

A.5 deriv_std_rec, no match

```
$ cargo run -- parse "(a|ab)(b|)" "b" \
    --parser deriv_std_rec --diag 3
--------------------------------====
REGEX ENGINE DEBUG REPORT
--------------------------------====
Mode:                 POSIX Standard (recursive parser)
Regex:                (a|ab)(b|)
Input:                "b"
Result:               NO MATCH
Policy:               POSIX
--------------------------------====
```

```
TIMING
--------------------------------====
Parse time:           0.005ms
Expressions stored:   2 (r0 -> r1)
--------------------------------====
FORWARD PASS (Derivatives)
--------------------------------====
r0 = Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))), Alt(Lit('b'), Eps))

Step 1: d(r0, 'b')
  rule: d(r1.r2, 'b')  [nullable(r1)=false]
       = d(r1, 'b').r2
r1 = Seq(Alt(Phi, Seq(Phi, Lit('b'))), Alt(Lit('b'), Eps))
  nullable(r1) = false  [FAIL] position 1

--------------------------------====
NULLABILITY CHECK
--------------------------------====
nullable(r1) = false -> no parse tree exists for full input
--------------------------------====
PARTIAL RECOVERY
--------------------------------====
No prefix matched.
--------------------------------====
ERROR SUMMARY
--------------------------------====
Failure at position:  1 (1-indexed)
Found:                'b'
Expected:             'a'
  b
  ^
--------------------------------====
END OF REPORT
--------------------------------====
```

A.6 `deriv_bc`, no match

```
$ cargo run -- parse "(a|ab)(b|)" "b" \
    --parser deriv_bc --diag 3
--------------------------------====
REGEX ENGINE DEBUG REPORT
--------------------------------====
Mode:                 POSIX Bitcoded
Regex:                (a|ab)(b|)
Input:                "b"
Result:               NO MATCH
Policy:               POSIX
--------------------------------====
```

```
TIMING
--------------------------------====
Parse time:           0.008ms
Derivative steps:     1
--------------------------------====
INTERNALIZE
--------------------------------====
internalize((a|ab)(b|))
  rule: internalize(r1 . r2) = []@(internalize r1)(internalize r2)
ri0 = []@([]@([false]@'a' (+) [true]@([]@'a' . []@'b')) .
  []@([false]@'b' (+) [true]@epsilon))
--------------------------------====
FORWARD PASS (deriv_bc + simp)
--------------------------------====
Step 1 (char: 'b', position 1)
ri0 = []@([]@([false]@'a' (+) [true]@([]@'a' . []@'b')) .
    []@([false]@'b' (+) [true]@epsilon))
  deriv_bc(ri0, 'b'):
    rule: (bs@ri1.ri2) \ 'b'  [nullable_bc(ri1)=false]
         = bs@(ri1\'b'.ri2)
  simp -> phi
  ri1 = phi
  nullable_bc(ri1) = false  [FAIL]

--------------------------------====
NULLABILITY CHECK
--------------------------------====
nullable_bc(ri1) = false -> no full parse tree exists
--------------------------------====
PARTIAL RECOVERY
--------------------------------====
No prefix matched.
--------------------------------====
ERROR SUMMARY
--------------------------------====
Failure at position:  1 (1-indexed)
Found:                'b'
Expected:             'a'
  b
  ^
--------------------------------====
END OF REPORT
--------------------------------====
```

A.7 pderiv_std, no match

```
$ cargo run -- parse "(a|ab)(b|)" "b" \
    --parser pderiv_std --diag 3
```

```
--------------------------------====
REGEX ENGINE DEBUG REPORT
--------------------------------====
Mode:                  Standard Partial-Derivative (pderiv_std)
Regex:                 (a|ab)(b|)
Input:                 "b"
Result:                NO MATCH
Policy:                GREEDY
--------------------------------====
TIMING
--------------------------------====
Parse time:            0.007ms
Steps:                 1
--------------------------------====
INITIAL FRONTIER
--------------------------------====
[r0]  where r0 = Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))),
  Alt(Lit('b'), Eps))
--------------------------------====
FORWARD PASS (pderiv_tree per residual, frontier flattened)
--------------------------------====
Step 1 (char: 'b', position 1)
  frontier before (1 residual(s)):
[0] Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))), Alt(Lit('b'), Eps))
  pderiv_tree('b', .) applied to each:
    from [0]:
      rule: pderiv_tree('b', r1.r2)  [nullable(r1)=false]
= [(r1'.r2, inj: continue r1) | (r1',_) <- pderiv_tree(c,r1)]
      -> []
  frontier after (0 residual(s)):
  no nullable residual in this frontier  [FAIL]

--------------------------------====
SELECTION (first nullable residual, priority/list order)
--------------------------------====
any nullable residual in final frontier = false -> no full parse tree
  exists
--------------------------------====
PARTIAL RECOVERY
--------------------------------====
No prefix matched.
--------------------------------====
ERROR SUMMARY
--------------------------------====
Failure at position:  1 (1-indexed)
Found:                'b'
Expected:             'a'
  b
```

```
  ^
--------------------------------====
END OF REPORT
--------------------------------====
```

A.8 pderiv_bc, no match

```
$ cargo run -- parse "(a|ab)(b|)" "b" \
    --parser pderiv_bc --diag 3
--------------------------------====
REGEX ENGINE DEBUG REPORT
--------------------------------====
Mode:                  Bitcoded Partial-Derivative (pDerivBC)
Regex:                 (a|ab)(b|)
Input:                 "b"
Result:                NO MATCH
Policy:                GREEDY
--------------------------------====
TIMING
--------------------------------====
Parse time:            0.006ms
Steps:                 1
--------------------------------====
INITIAL FRONTIER
--------------------------------====
[(r0, [])]  where r0 = Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))),
  Alt(Lit('b'), Eps))
--------------------------------====
FORWARD PASS (pDerivBC per residual, frontier flattened)
--------------------------------====
Step 1 (char: 'b', position 1)
  frontier before (1 residual(s)):
[0] Seq(Alt(Lit('a'), Seq(Lit('a'), Lit('b'))), Alt(Lit('b'), Eps))
      bits-so-far = []
  pDerivBC('b', .) applied to each:
    from [0]:
      rule: pDerivBC('b', r1.r2)  [nullable(r1)=false]
           = [(smartC(r1',r2), bs) | (r1',bs) <- pDerivBC(c,r1)]
      -> []
  frontier after (0 residual(s)):
  no nullable residual in this frontier  [FAIL]

--------------------------------====
SELECTION (first nullable residual, priority/list order)
--------------------------------====
any nullable residual in final frontier = false -> no full parse tree
  exists
--------------------------------====
```

```
PARTIAL RECOVERY
--------------------------------====
No prefix matched.
--------------------------------====
ERROR SUMMARY
--------------------------------====
Failure at position:  1 (1-indexed)
Found:                'b'
Expected:             'a'
  b
  ^
--------------------------------====
END OF REPORT
--------------------------------====
```